

ABSTRACT

The nucleon electromagnetic form factors G_E and G_M describe how charge and magnetization are distributed inside the proton and neutron, and extending their measurement to the highest four-momentum transfers Q^2 yet reached is the central goal of the Super Bigbite Spectrometer (SBS) program in Jefferson Lab’s Hall A. Because three of the SBS experiments extract these form factors from polarization observables, precise knowledge of the longitudinal electron beam polarization is critical. At the extremely high Q^2 targeted by SBS, the uncertainty of this beam polarization can easily become the limiting systematic error for the entire measurement.

This dissertation presents the Hall A Møller polarimetry program developed to provide these essential, calibrated beam polarizations. The analysis encompasses robust period-by-period corrections (charge normalization, deadtime, accidentals) and a comprehensive systematic uncertainty budget evaluating target polarization, analyzing power, bleedthrough, false-asymmetry, and spin-precession effects. For production periods lacking direct measurements, beam polarizations are instead inferred using Wien-angle offsets and spin precession models.

Across all thirteen SBS production periods (2022–2025), the delivered longitudinal beam polarizations spanned from 80.5% to 87.6%. In the eight periods with direct Møller measurements, systematic uncertainties were strictly controlled between 0.88% and 2.3%, with statistical uncertainties $\leq 0.21\%$. For the five inferred periods, total systematic uncertainties ranged from 1.4% to 4.6%. Validated by overlapping Hall C measurements that confirm consistent spin-transport behavior, these fully calibrated polarizations and their rigorously defined uncertainties are successfully delivered to the SBS physics analyses.

**MØLLER POLARIMETRY FOR NUCLEON FORM FACTOR
EXPERIMENTS IN THE SUPER BIGBITE SPECTROMETER
(SBS) PROGRAM**

by

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CHAPTER 1

Introduction and Physics Motivation

Nucleons—the protons and neutrons that make up the atomic nucleus and carry nearly all of an atom’s mass—are not elementary, but composite objects built from quarks and gluons bound by the strong interaction. Their internal structure, including quantities as basic as the charge radius, the magnetic moment, and the mass, is encoded in the electromagnetic form factors G_E and G_M , which describe how charge and magnetization are distributed inside the nucleon. These form factors are probed in elastic electron scattering, and measuring them as a function of the squared four-momentum transfer Q^2 maps the nucleon’s structure across distance scales [1, 2].

The most revealing measurements lie at high Q^2 , where the elastic cross sections are smallest and precise data have long been sparse. Extending the proton and neutron form factors to the highest Q^2 yet reached is the goal of the Super Bigbite Spectrometer (SBS) program in Hall A at Jefferson Lab. Three of its four experiments determine a form factor—or a ratio of form factors—from a polarization observable, and each one depends on precise knowledge of the electron beam polarization: the quantity that Møller polarimetry, the subject of this thesis, is built to measure [3, 4, 5].

1.1 Nucleon Electromagnetic Form Factors

The experimental study of nucleon structure began with the elastic electron–scattering measurements carried out by Hofstadter and collaborators at the Stanford High Energy Physics Laboratory in the mid-1950s. These experiments scattered electrons of up to a few hundred MeV from hydrogen. By observing how the electrons deflected compared to the theoretical baseline for a simple, zero-dimensional charge, the team provided the first direct evidence that protons and neutrons actually possess a finite physical size and an internal distribution of charge and magnetization [6, 7]. The elastic form factors are functions of the momentum transfer. They describe how the nucleon’s electromagnetic response differs from that of a simple point particle. It was for this body of work that Hofstadter shared the 1961 Nobel Prize in Physics.

The Sachs convention uses two functions of the squared four-momentum transfer, Q^2 , to describe the nucleon’s electromagnetic structure. These are the electric form factor $G_E(Q^2)$ and the magnetic form factor $G_M(Q^2)$ [8]. These form factors describe how charge and magnetization are distributed within the nucleon. At $Q^2 = 0$, they equal the particle’s static charge and magnetic moment: $G_E^p(0) = 1$, $G_M^p(0) = \mu_p$, $G_E^n(0) = 0$, and $G_M^n(0) = \mu_n$. A simple formula called the dipole form, $G_D(Q^2) = \left[1 + Q^2/0.71 \text{ (GeV}/c)^2\right]^{-2}$, approximately describes the proton’s form factors and the neutron’s magnetic form factor over a wide range of momentum transfers. This dipole form acts as a baseline; comparing actual measurements against it reveals finer details of the nucleon’s structure [9].

Two complementary classes of measurement give access to these form factors. The traditional method is a Rosenbluth, or longitudinal–transverse, separation of the unpolarized elastic cross section. In the one-photon-exchange approximation, the unpolarized elastic cross section is described by the Rosenbluth formula [10] in terms of the Sachs electric and

magnetic form factors,

$$\frac{d\sigma}{d\Omega} = \sigma_{\text{Mott}} \frac{\varepsilon G_E^2(Q^2) + \tau G_M^2(Q^2)}{\varepsilon(1 + \tau)}, \quad (1.1)$$

with $\tau = Q^2/(4M^2)$ and $\varepsilon = [1 + 2(1 + \tau) \tan^2(\theta_e/2)]^{-1}$, where σ_{Mott} is the point-charge Mott cross section,

$$\sigma_{\text{Mott}} = \left(\frac{\alpha \cos(\theta_e/2)}{2E_e \sin^2(\theta_e/2)} \right)^2 \frac{E'_e}{E_e}, \quad (1.2)$$

with α the fine-structure constant, E_e (E'_e) the incident (scattered) electron energy, and θ_e the electron scattering angle. Isolating G_E^2 from G_M^2 requires measuring the cross section at several values of ε — that is, at several combinations of beam energy and scattering angle — for each fixed Q^2 . The electric term is multiplied by ε , while the magnetic term is multiplied by τ . As Q^2 increases, τ becomes much larger than 1. At these high values, G_M completely dominates the cross section, making the electric contribution very difficult to extract [9].

The second class of measurement uses polarization observables. The polarization-sensitive experiments exploit the Akhiezer–Rekalo relation [11, 12], which gives the form-factor ratio directly from the transverse (P_t) and longitudinal (P_ℓ) components of the recoil nucleon polarization,

$$\frac{G_E}{G_M} = -\frac{P_t}{P_\ell} \frac{E_e + E'_e}{2M} \tan\left(\frac{\theta_e}{2}\right), \quad (1.3)$$

and the polarized-target double-spin asymmetry approach used in GEN-II relies on the Donnelly–Raskin formalism [13] detailed in Sec. 1.4.2. These methods extract the ratio G_E/G_M by comparing two polarization components measured at the same time. This simultaneous measurement allows overall uncertainties — such as the exact beam polarization — to cancel out. As a result, polarization techniques are much less vulnerable to the systematic errors that ruin Rosenbluth separations at high Q^2 [9].

The decisive importance of the polarization approach became apparent around the turn

of the century. Jefferson Lab experiments by Jones et al. [14] and Gayou et al. [15] applied this polarization technique to the proton. They discovered that the ratio $\mu_p G_E^p / G_M^p$ dropped steadily as Q^2 increased. This sharply contradicted earlier Rosenbluth measurements, which had suggested the ratio stayed roughly constant near 1 [14, 15, 16]. By $Q^2 \approx 5.6 (\text{GeV}/c)^2$ the polarization measurements gave a ratio of roughly ~ 0.3 , while Rosenbluth extractions remained near unity [15]. Subsequent experiments proved this discrepancy was real, not an error. Physicists now believe it is caused by two-photon-exchange effects. These effects strongly distort Rosenbluth measurements but naturally cancel out in polarization ratios [9, 17]. This result established polarization-based measurements as the precision tool of choice for the form-factor ratio at high Q^2 , and it is the methodological foundation on which the SBS program is built.

1.2 The Role of Beam Polarization

The polarization-based form-factor extractions (GEn-II, GEn-RP, GEp-V) require a precise measurement of the longitudinal beam polarization P_{beam} via Møller polarimetry; the role of P_{beam} in each is described in the respective experiment section. The beam polarization, P_{beam} , is a critical ingredient in these measurements. It directly factors into every observable used to calculate a form factor. At the extremely high Q^2 values targeted by SBS, our uncertainty in P_{beam} becomes the limiting factor for the asymmetry measurement. How it enters, however, differs qualitatively between the two kinds of polarization measurement used in the program.

In the GEn-II experiment's asymmetry method, the raw measured asymmetry is smaller than the true physics asymmetry. It gets diluted by several factors (Sec. 1.4.2): $A_{meas} = P_{beam} P_{target} P_n f A_{phys}$. To find the neutron electric form factor, scientists must work back-

wards from this equation. Under these specific conditions, the asymmetry essentially isolates the G_E^n form factor. Because the relationship is direct and linear, any percentage error in the polarization measurements adds directly to the error of the final G_E^n result:

$$\left(\frac{\delta G_E^n}{G_E^n}\right)^2 \simeq \left(\frac{\delta A_{\text{meas}}}{A_{\text{meas}}}\right)^2 + \left(\frac{\delta P_{\text{beam}}}{P_{\text{beam}}}\right)^2 + \left(\frac{\delta P_{\text{target}}}{P_{\text{target}}}\right)^2 + \left(\frac{\delta P_n}{P_n}\right)^2 + \left(\frac{\delta f}{f}\right)^2. \quad (1.4)$$

This means beam-polarization errors transfer almost one-to-one. For example, a 2% error in P_{beam} causes roughly a 2% error in G_E^n . Because the asymmetry is very small at high Q^2 , even modest beam-polarization uncertainties can consume a substantial fraction of the total allowed error budget. No single external threshold is imposed; the operative requirement is that the beam-polarization systematic remain subdominant to the other multiplicative factors in the asymmetry — the target polarization, the effective neutron polarization P_n , and the dilution factor — each of which contributes at the few-percent level. This sets a working target at the $\sim 1\%$ level, consistent with the demonstrated Hall A Møller performance.

The GEN-RP and GEp-V experiments use a different method. They calculate the form-factor ratio by comparing two recoil-polarization components measured at the exact same time. By taking a ratio, the beam polarization P_{beam} largely cancels out mathematically (Secs. 1.4.3 and 1.4.4). This cancellation is the principal advantage of the ratio method and is why recoil-polarization measurements can achieve higher precision than target-asymmetry techniques [9]. There is therefore no formal systematic budget that the polarimetry must satisfy for these two experiments. However, accurate polarimetry is still necessary. P_{beam} is still needed to properly calibrate the individual components [18]. Furthermore, if the beam polarization shifts or drifts during the measurement process, it will introduce new errors into the results.

Fortunately, Møller polarimetry can meet these strict requirements. The Møller polarimeter in Hall A measures P_{beam} by scattering beam electrons off a magnetized foil. This device has consistently delivered high-precision results (around 1% uncertainty) for two decades [18, 19]. In its most rigorous recent runs (the PREX-2 and CREX experiments), it achieved an impressive 0.89% and 0.85% precision. The main remaining limitation was our knowledge of the foil’s exact magnetization [18]. This sub-percent performance comfortably meets the $\sim 1\%$ working target of GEN-II — the most demanding of the three measurements — and far exceeds what the ratio-based GEN-RP and GEp extractions require, where P_{beam} cancels in the form-factor ratio to leading order. Delivering and validating this measurement for the SBS experiments is the central concern of this thesis.

1.3 Overview of the SBS Program

The Super Bigbite Spectrometer (SBS) collaboration is a working group within the Jefferson Lab Hall A collaboration whose program is a series of high-precision measurements of the elastic nucleon electromagnetic form factors using the 12 GeV CEBAF beam, comprising four experiments: G_M^n (E12-09-019), G_E^n via polarized ^3He (E12-09-016), G_E^n/G_M^n via recoil polarimetry (E12-17-004), and $\mu_p G_E^p/G_M^p$ via polarization transfer (E12-07-109) [20, 21]. What sets SBS apart from the previous generation of experiments is that each of these measurements is extended to the highest Q^2 ever achieved for the relevant quantity: G_M^n to $13.6 (\text{GeV}/c)^2$ (previous high-precision limit $\sim 4.8 (\text{GeV}/c)^2$ from Lachniet et al. [22]); G_E^n to $\sim 10 (\text{GeV}/c)^2$ (previous limit $3.4 (\text{GeV}/c)^2$ from Riordan et al. [23]); and $\mu_p G_E^p/G_M^p$ to $\sim 12 (\text{GeV}/c)^2$ (previous limit $8.54 (\text{GeV}/c)^2$ from Puckett et al. [24]). The remainder of this section describes the shared apparatus around which these measurements are built; the individual experiments and the quantities each one extracts are presented in Sec. 1.4.

1.3.1. Super Bigbite Spectrometer

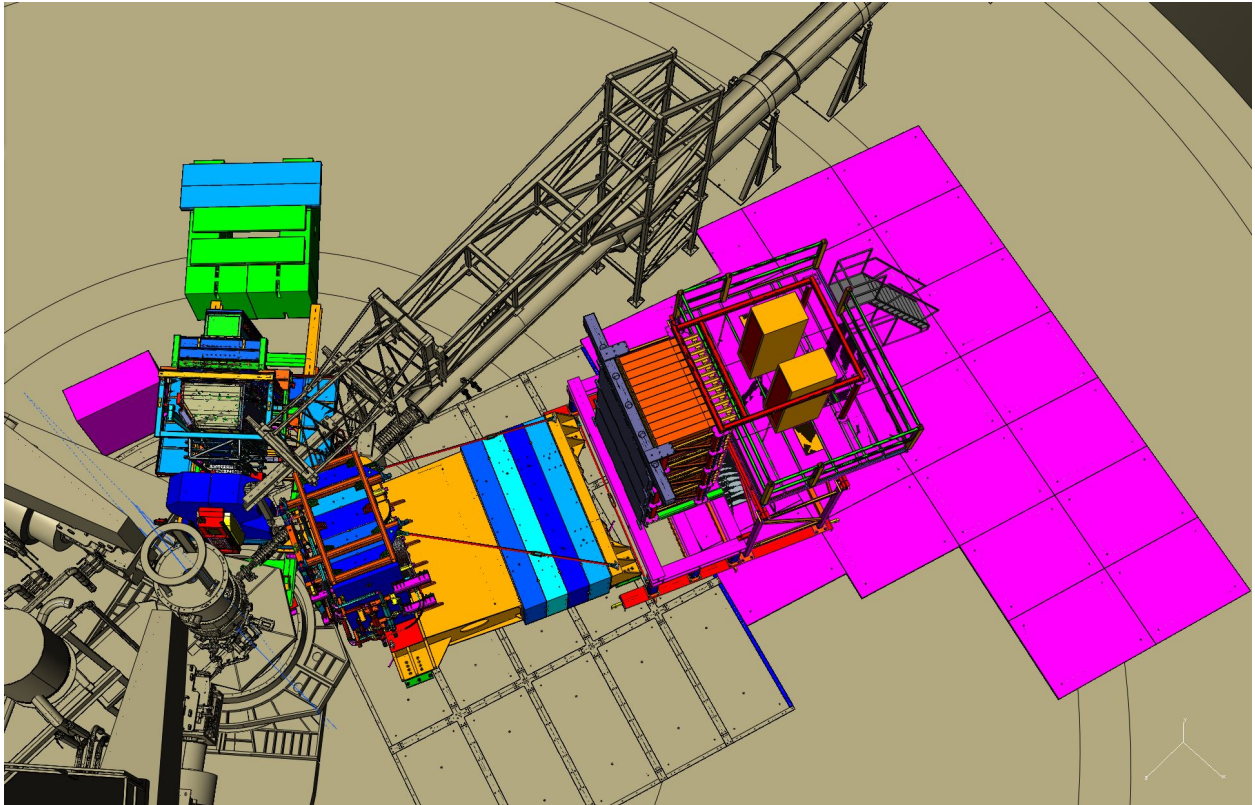


Figure 1: Experimental layout used for the G_M^n and G_E^n measurements in the SBS program.

The SBS apparatus is built around a reconfigured 100-ton BNL 48D48 warm-coil dipole magnet, with a horizontal slit through the yoke that allows the beamline to pass through the magnet; this short target-to-magnet distance enables a ≈ 70 msr solid-angle acceptance [21]. The hadron arm uses Gas Electron Multiplier (GEM) detectors — with $\approx 70 \mu\text{m}$ spatial resolution and counting-rate capability above 500 kHz/cm^2 [25] — for charged-particle tracking; for the neutron experiments only the Front Tracker (positioned between the target and the SBS dipole) is required to give the recoil trajectory, while the GEp polarimeter additionally uses a Back Tracker downstream of the dipole (composed of the Second and Third Trackers) to measure secondary scattering angles [25]. The downstream

Hadron Calorimeter (HCAL), built new for the SBS program, is a sampling calorimeter of 288 modules arranged in 24 rows $\times$ 12 columns; each module consists of 40 alternating layers of iron and plastic scintillator and provides ≈ 0.5 ns timing resolution, 3–4 cm position resolution, and near-identical detection efficiency for protons and neutrons [26, 27]. The electron arm in the three neutron experiments is the BigBite Spectrometer (BBS), a single-dipole magnet instrumented with five layers of GEM detectors, the Gas Ring-Imaging Cherenkov detector (GRINCH) for pion rejection, a timing hodoscope, and the BigBite Calorimeter (BBCal) — a lead-glass preshower + shower stack with $\approx 6.2\%$ energy resolution, 1.2 cm position resolution, and 0.5 ns timing that provides the primary electron trigger [20]. For the proton form-factor experiment (GEp-V), the BBS electron arm is replaced by a newly built lead-glass Electromagnetic Calorimeter (ECAL) on beam-left, together with a 2352-bar scintillator Coordinate Detector (CDET) for vertical hit determination and photon-background rejection [28]; the SBS hadron arm is reconfigured as a proton focal-plane polarimeter, with CH_2 analyzer blocks placed between the SBS dipole and the Back Tracker, where secondary scattering provides the proton polarization signal [25].

1.4 The SBS Experiments

1.4.1. Neutron Magnetic Form Factor (GM_n)

The SBS program designates Experiment E12-09-019 as GMn; its physics objective is to extract the neutron magnetic form factor G_M^n from quasi-elastic deuterium yields. It ran from October 2021 to February 2022 using a 15 cm liquid-deuterium (LD_2) target together with a reference liquid-hydrogen (LH_2) cell, covering six kinematic settings across five distinct Q^2 values up to beam energies of 9.9 GeV [20]. G_M^n is extracted via the Durand ratio method [22, 29, 30], in which the quasi-elastic neutron and proton yields from deuterium

form the ratio

$$R = \frac{\sigma[d(e, e'n)]}{\sigma[d(e, e'p)]} = a(Q^2, \theta_e) \frac{(G_E^n)^2 + (\tau/\varepsilon) (G_M^n)^2}{(G_E^p)^2 + (\tau/\varepsilon) (G_M^p)^2}, \quad (1.5)$$

where $a(Q^2, \theta_e)$ is a kinematic prefactor. The measured ratio is related to Eq. (1.5) schematically by $R_{\text{meas}} \approx R \cdot (\epsilon_n/\epsilon_p)$ at leading order, where ϵ_n and ϵ_p are the neutron and proton detection efficiencies in HCAL; the full extraction additionally accounts for radiative corrections, nuclear corrections (final-state interactions, meson-exchange currents), and backgrounds. Once the small G_E^n contribution and the proton form factors are taken from independent measurements, G_M^n follows from R . Since R is built from unpolarized cross sections, no beam polarization is required and Møller polarimetry is not used. The experiment covers $3.0 \leq Q^2 \leq 13.6 \text{ (GeV}/c)^2$ using BBS for electron detection and the SBS dipole + HCAL for nucleon detection; the SBS dipole deflects the proton vertically while leaving the neutron undeflected, providing spatial separation of n and p on the HCAL face. The primary experimental challenge is equalizing the n and p detection efficiencies in HCAL and controlling inelastic π^- contamination, which would otherwise dominate the systematic uncertainty at the highest Q^2 points.

1.4.2. Neutron Electric Form Factor ($GE_n - II$)

Known as GEN-II, Experiment E12-09-016 measures the neutron electric form factor G_E^n through beam-target double-spin asymmetries on polarized ^3He —the observable that imposes the strictest beam-polarization requirements in the program. It ran from October 2022 to October 2023 using a polarized ^3He target — a double-chambered glass cell at ≈ 7 atm filled via Rb/K hybrid spin-exchange optical pumping (SEOP), with a 60 cm target chamber — operating at beam currents up to $60 \mu\text{A}$, achieving a polarization-weighted luminosity approximately three times higher than any previous polarized- ^3He electron-scattering experiment [20, 31]. In the Donnelly–Raskin formalism [13], the beam-target double-spin asymmetry for $^3\vec{H}e(\vec{e}, e'n)$ depends on the polar and azimuthal angles (θ^*, φ^*) of the target

spin relative to $\vec{q}$:

$$A_{\text{phys}}(\theta^*, \varphi^*) = -\frac{v_{T'} (G_M^n)^2 \cos \theta^* + 2 v_{TL'} G_E^n G_M^n \sin \theta^* \cos \varphi^*}{v_L (G_E^n)^2 + v_T (G_M^n)^2}, \quad (1.6)$$

where v_L , v_T , $v_{T'}$, and $v_{TL'}$ are kinematic structure functions defined in Ref. [13]. The experiment orients the target spin perpendicular to $\vec{q}$ ($\theta^* = 90^\circ$, $\varphi^* = 0$), which zeroes the $(G_M^n)^2 \cos \theta^*$ longitudinal term and isolates the transverse $G_E^n G_M^n$ interference term that maximizes sensitivity to G_E^n . The measured asymmetry is related to the physics asymmetry by

$$A_{\text{meas}} = P_{\text{beam}} \cdot P_{\text{target}} \cdot P_n \cdot f \cdot A_{\text{phys}}, \quad (1.7)$$

where $P_n \approx 0.86$ is the effective neutron polarization of polarized ^3He [32], f is the dilution factor accounting for unpolarized background contributions (quasi-elastic scattering from the two protons in ^3He , the N_2 buffer gas used to mitigate wall depolarization, and the glass cell walls), and P_{beam} is the longitudinal beam polarization measured by the Hall A Møller polarimeter [18, 19]. The spectrometer configuration is identical to GMn (BBS + SBS + HCAL) with the addition of the polarized ^3He target ladder, Helmholtz holding-field coils, NMR and EPR polarimetry systems, and magnetic shielding protecting the target from the SBS and BBS fringe fields. Main challenges include maintaining high ^3He polarization at elevated beam current, accurately cross-calibrating NMR and EPR target polarimetry, and controlling final-state interaction and meson-exchange-current corrections at high Q^2 .

1.4.3. Neutron Electric Form Factor by Recoil Polarimetry ($GE_n - RP$)

GEN-RP (Experiment E12-17-004) determines the neutron form-factor ratio G_E^n/G_M^n by comparing longitudinal and transverse components of the recoil-neutron polarization. It ran in April–May 2024 at a single kinematic point with $Q^2 \sim 4.5 (\text{GeV}/c)^2$ using a 15 cm

LD₂ target [20, 33]. Applying Eq. (1.3) to the recoiling neutron yields

$$\frac{G_E^n}{G_M^n} = -\frac{P_{S'}}{P_{L'}} \frac{E_e + E'_e}{2M_n} \tan\left(\frac{\theta_e}{2}\right), \quad (1.8)$$

where $P_{L'}$ and $P_{S'}$ are the longitudinal and sideways (in-plane transverse) polarization components of the recoil neutron. For a longitudinally polarized electron beam of helicity $h = \pm 1$ and polarization P_{beam} , the measured components are

$$P_{L'}^{\text{meas}} = h P_{beam} P_{L'}, \quad P_{S'}^{\text{meas}} = h P_{beam} P_{S'}, \quad (1.9)$$

so that although P_{beam} cancels in the ratio $P_{S'}/P_{L'}$ to leading order, it is needed for the absolute calibration of each component and for the analyzing-power determination [18]. The SBS arm is reconfigured as a neutron focal-plane polarimeter consisting of a front GEM tracker, a passive analyzer (iron for the novel charge-exchange channel $np \rightarrow pn$, or CH₂/HDPE for the conventional $np \rightarrow np$ elastic channel), a rear GEM tracker that provides the angular reconstruction of the secondary scattered charged particle, and HCAL as trigger; the SBS dipole induces spin precession of the charge-exchange-produced proton between the front and rear trackers. The two polarimetry channels run simultaneously, allowing the novel charge-exchange analyzing power to be validated against the established np elastic channel. The principal challenge is measuring and validating the charge-exchange analyzing power, which must be well understood before the method can be scaled to higher Q^2 in future experiments.

1.4.4. Proton Electric Form Factor ($GEp - V$)

GEp-V (Experiment E12-07-109) extends the proton polarization-transfer method to extract $\mu_p G_E^p / G_M^p$ at the highest Q^2 yet attempted in this class of measurement. It ran from 8 April to 25 August 2025 across three running periods, at beam energies of 6.4 GeV and

10.7 GeV, using a 30 cm liquid-hydrogen (LH₂) target [34, 35]. The experiment extends the proton polarization-transfer program — pioneered by Jones et al. [14] and most recently pushed to $Q^2 = 8.54 (\text{GeV}/c)^2$ by Puckett et al. [24] — up to $Q^2 \approx 12 (\text{GeV}/c)^2$. The form-factor ratio is extracted via Eq. (1.3) applied to the recoil proton:

$$\mu_p \frac{G_E^p}{G_M^p} = -\mu_p \frac{P_t}{P_\ell} \frac{E_e + E'_e}{2M_p} \tan\left(\frac{\theta_e}{2}\right). \quad (1.10)$$

For a longitudinally polarized beam, the individual transferred polarization components are [12, 14]

$$h P_{beam} P_\ell = h P_{beam} \frac{E_e + E'_e}{M_p} \sqrt{\tau(1+\tau)} \tan^2(\theta_e/2) \frac{(G_M^p)^2}{I_0}, \quad (1.11)$$

$$h P_{beam} P_t = -h P_{beam} 2\sqrt{\tau(1+\tau)} \tan(\theta_e/2) \frac{G_E^p G_M^p}{I_0}, \quad (1.12)$$

with $I_0 = (G_E^p)^2 + (\tau/\varepsilon)(G_M^p)^2$, so that P_{beam} enters every individually measured component and the Møller polarimeter is required for the absolute calibration of each [18], even though it cancels in the ratio P_t/P_ℓ to leading order. The electron arm consists of a newly built lead-glass ECAL on beam-left, together with the CDET coordinate detector [28]; the SBS proton polarimeter on beam-right consists of the Front Tracker upstream of the SBS dipole, CH₂ analyzer blocks downstream of the dipole, the Back Tracker (Second + Third Trackers) behind the analyzer to reconstruct the secondary scattering angles, and HCAL as trigger, with the SBS dipole serving simultaneously as momentum analyzer and spin-precession element. Key experimental challenges include the steep falloff of the figure-of-merit at high Q^2 , the declining proton-CH₂ analyzing power A_y , photon backgrounds in ECAL, and managing GEM occupancy in the proton arm under the high-luminosity conditions required to accumulate adequate statistics.

CHAPTER 2

Thomas Jefferson National Accelerator Facility

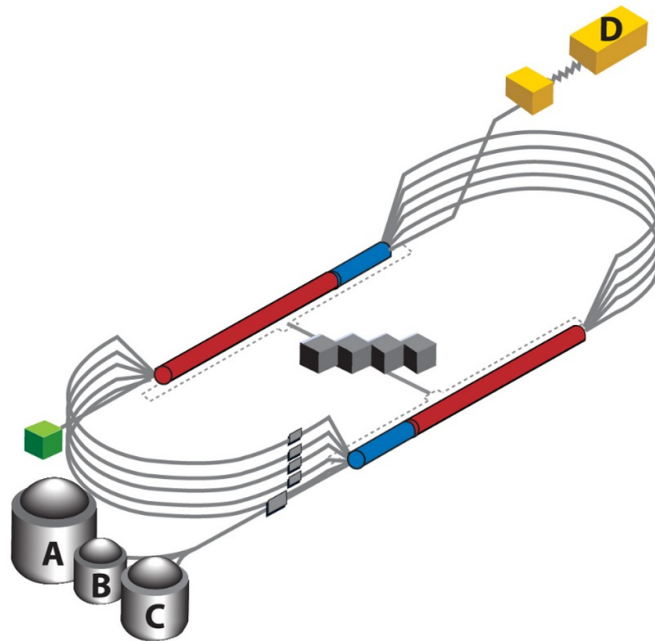


Figure 2: Schematic overview of the Jefferson Lab Continuous Electron Beam Accelerator Facility (CEBAF), showing the injector, recirculating linacs/arcs, and experimental hall beam delivery paths.

2.1 Overview of Continuous Electron Beam Accelerator Facility (CEBAF)

This chapter describes the physical infrastructure at the Thomas Jefferson National Accelerator Facility that underlies the measurements presented in this thesis, framed in terms of what Hall A Møller polarimetry requires from the rest of the machine. The Continuous Electron Beam Accelerator Facility (CEBAF) produces polarized electrons at the injector, accelerates them through recirculating superconducting linacs, and delivers them to the experimental halls. The Super Bigbite Spectrometer (SBS) program is the immediate context for this capability; its requirement for absolute beam polarization motivates the Møller measurements developed in the chapters that follow [36].

The sections below follow that beam from the photoinjector through acceleration, arc transport, hall delivery, and the beam-monitoring systems used in the halls, before summarizing the spectrometer installations at Jefferson Lab. The photoinjector system produces polarized electrons from a strained-superlattice GaAs photocathode illuminated by four gain-switched diode lasers at 249.5 MHz inside a 200 keV DC photogun [37]. Fast helicity reversal with a Pockels cell and slower polarization sign changes with the insertable half-wave plate (iHWP) set the spin state at the source [38], and a pair of orthogonal Wien filters rotate the spin to the precalculated angle required for longitudinal polarization in each hall after the precession accumulated in the arcs, with the Wien-angle setting anchored by the “spin dance” calibration that underlies Hall A Møller polarimetry.

Once launched, the beam is accelerated by two antiparallel superconducting RF linacs operating at 1.497 GHz and steered through recirculation arcs that provide up to 5-pass delivery to 11 GeV for Halls A, B, and C and 5.5-pass delivery to 12 GeV for Hall D. RF sep-

arator cavities and septum magnets multiplex the interleaved bunches into the assigned halls; the $\pm 37.5^\circ$ bends into Halls A and C introduce additional spin precession that the Wien settings must compensate. Beam monitoring and instrumentation—including RF-cavity beam current monitors, stripline beam position monitors, and a fast raster system—supply the intensity and trajectory information needed for normalization in the halls [39, 40], though Hall A Møller running at $\approx 1 \mu\text{A}$ operates without the raster. Polarized-beam characterization along this path—from the injector Mott polarimeter to the hall Compton and Møller devices—is treated in Chapter 3; the Hall A Møller polarimeter that is the central focus of this thesis is described there in detail.

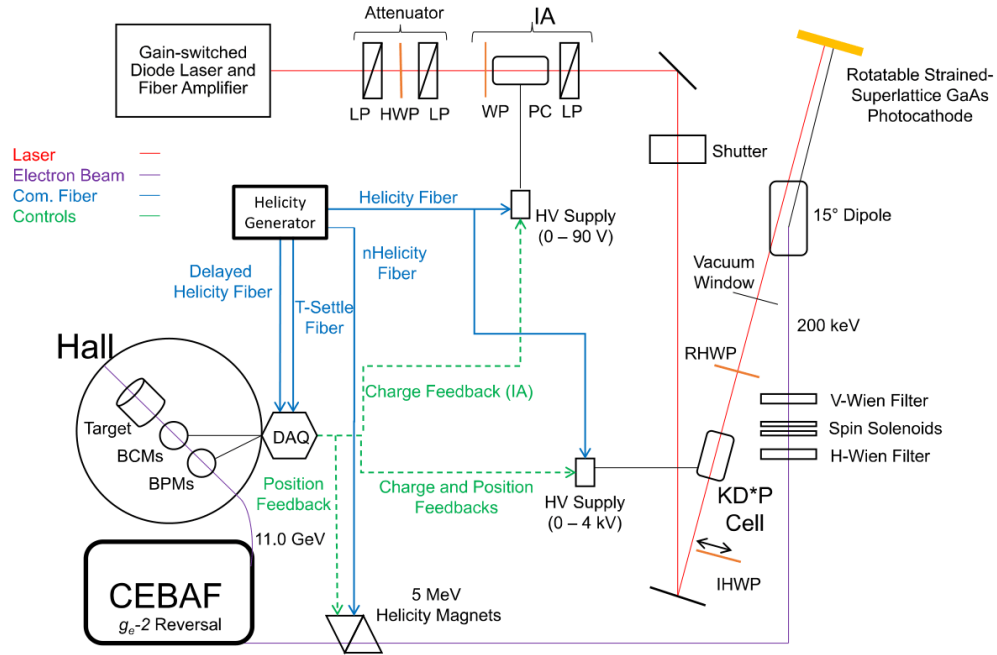


Figure 3: Overview of the various stages of beam generation and manipulation at the CEBAF injector, including the feedback mechanisms used to suppress helicity-correlated beam asymmetries [41].

2.2 The Photoinjector System

Laser Source

The photoinjector at CEBAF is built around a high-voltage DC photo-gun operating at 200 keV [42, 43, 44], which employs a GaAs photocathode. The system has undergone several iterative developments, drawing inspiration from research facilities such as SLAC [45], and in its current form serves as CEBAF's polarized beam source [46]. Four independent lasers pulse sequentially, each at 249.5 MHz, in order to deliver a simultaneous continuous wave (CW) electron beam to four separate experimental halls [47, 48, 49]. Following the 12 GeV upgrade, the injector system is capable of supplying up to approximately $200 \mu\text{A}$ of combined beam current to all experimental halls at a beam energy of 12 GeV [41].

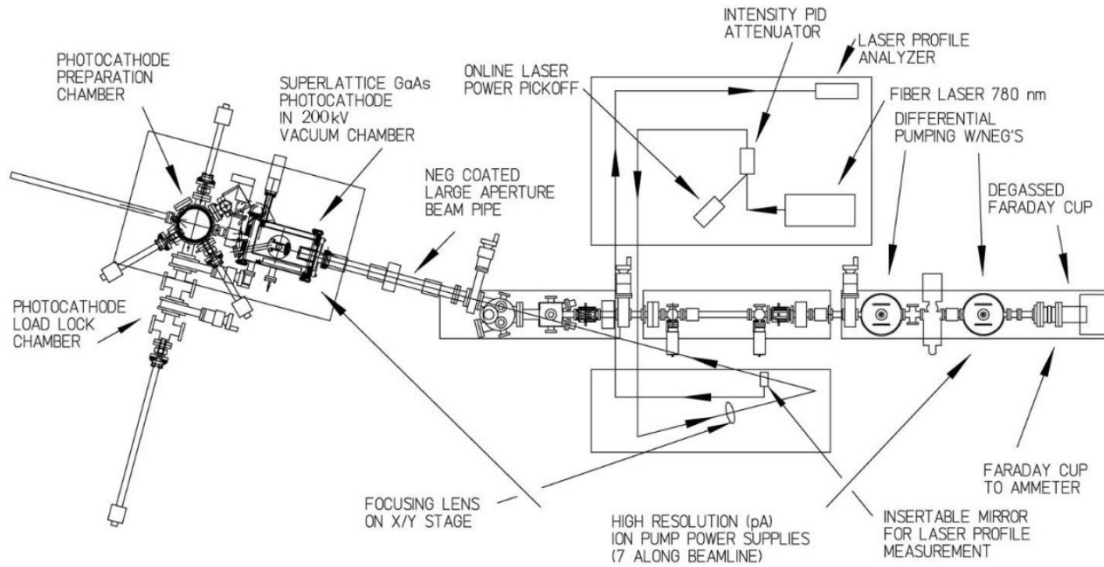


Figure 4: Schematic diagram of the CEBAF photoinjector system [42].

Photocathode

The strained superlattice gallium arsenide (GaAs) photocathodes used at CEBAF are capable of producing electron beams with up to 92% beam polarization at approximately 1% quantum efficiency (QE) [50, 51, 52]. The specific photocathode employed at CEBAF consists of 14 paired layers of GaAs/GaAsP topped with a final highly doped GaAs surface layer, which delivers electrons with over 85% spin-polarization to the experimental halls [53, 54]. The quantum efficiency also imposes constraints on the permissible laser power and the wavelength of the laser light [53]. Under continuous operation, the QE of the photocathode degrades over time, which in turn reduces the operational lifetime of the photo-gun. To mitigate this, the QE of the photocathode — also referred to as the photocathode yield — is measured on a daily basis. Once the yield has dropped sufficiently, the photocathode is reheated, the necessary chemical treatments are reapplied to reactivate its surface, and the yield is recovered. The laser spot position is then repositioned [37, 55].

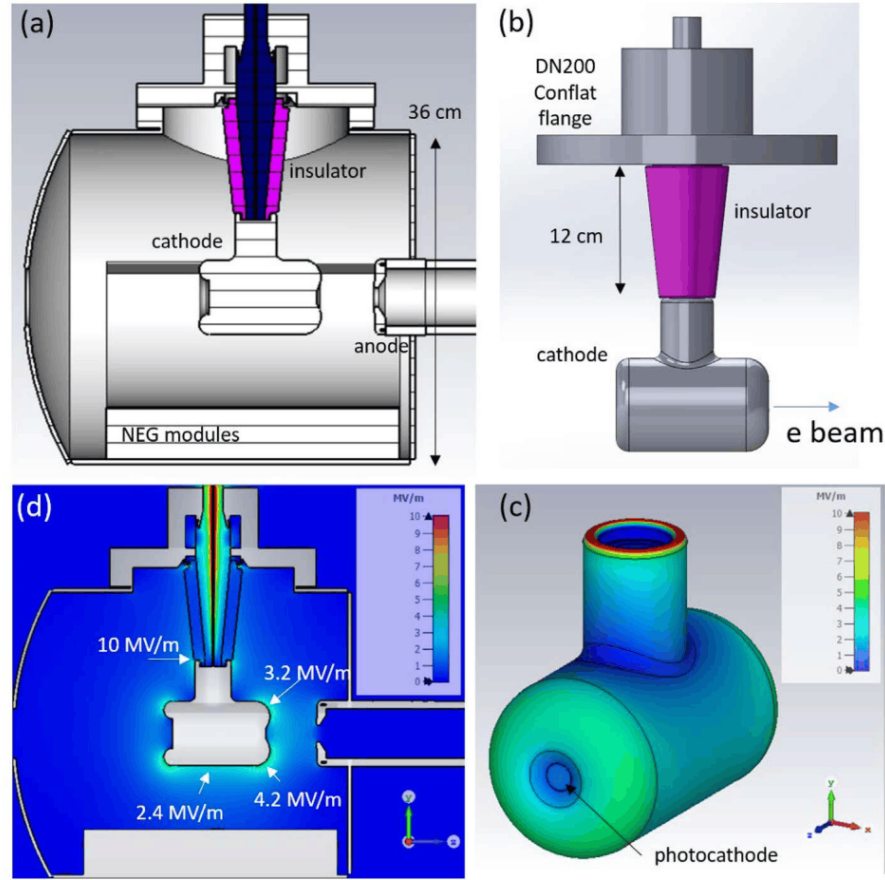


Figure 5: Illustrations of the CEBAF DC high-voltage photogun and cathode electrode. Shown are: (a) a side view of the photogun assembly featuring an inverted insulator geometry with key components identified; (b) a close-up view of the tee-shaped stainless steel cathode electrode; (c) the electric field strength distribution along the surface of the tee-shaped cathode electrode at a bias voltage of 100 kV; and (d) an electrostatic field map of the complete photogun assembly [41].

2.3 Fast Helicity Reversal — Pockels Cell

The quarter-wave Pockels cell at CEBAF is an electro-optic device that, when a voltage is applied across it, rotates the polarization of the laser light and converts the linearly polarized input into the circularly polarized output that ultimately produces the electron beam

[56]. It serves as the fast helicity reversal device in the system. Depending on the experimental requirements, the Pockels cell generates a specific helicity pattern designed to reduce the sensitivity of the helicity-correlated asymmetry to low-frequency accelerator noise [57]. The pattern employed during the Super Bigbite Spectrometer (SBS) experimental program was a quartet pattern, shown in Figure 6, which depicts the relationship between the Pockels cell voltage and the helicity state of the beam electrons [58].

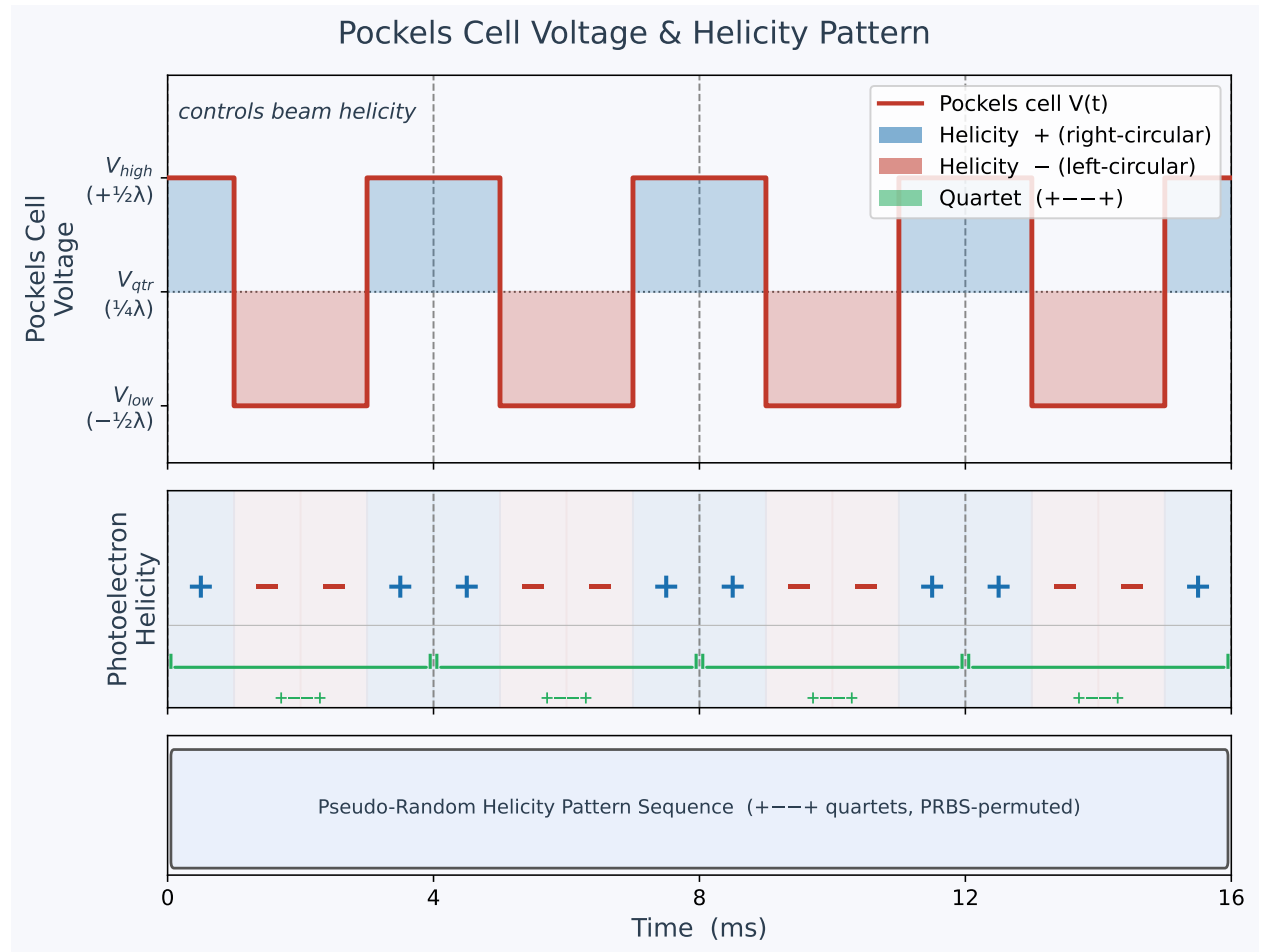


Figure 6: The quartet helicity pattern used during the SBS experimental program, shown alongside the corresponding Pockels cell voltage.

2.4 Spin Manipulation at Injection

Half-Wave Plates

Two half-wave plates are installed at the injector: the insertable half-wave plate (iHWP) and the rotatable half-wave plate (RHWP). The iHWP is a passive optical component that serves as a slow helicity reversal method, rotating the polarization direction of the linearly polarized laser light by 90° when inserted [59], which switches the Pockels cell output from a right circularly polarized state to a left circularly polarized one [60]. The iHWP has two states, IN and OUT. Insertion of the iHWP preserves all instrument settings, voltages, and optical configurations while flipping only the electron beam polarization sign; averaging measurements taken in the two states therefore helps constrain false asymmetries originating from the laser source and from electronic cross-talk [38]. Parity-violation experiments may require frequent iHWP state changes in order to precisely measure the beam polarization and minimize the resulting false asymmetries [61]. The rotatable half-wave plate, in contrast, is used to rotate any residual linear polarization present after the Pockels cell, thereby minimizing its impact on the helicity-correlated beam asymmetry downstream.

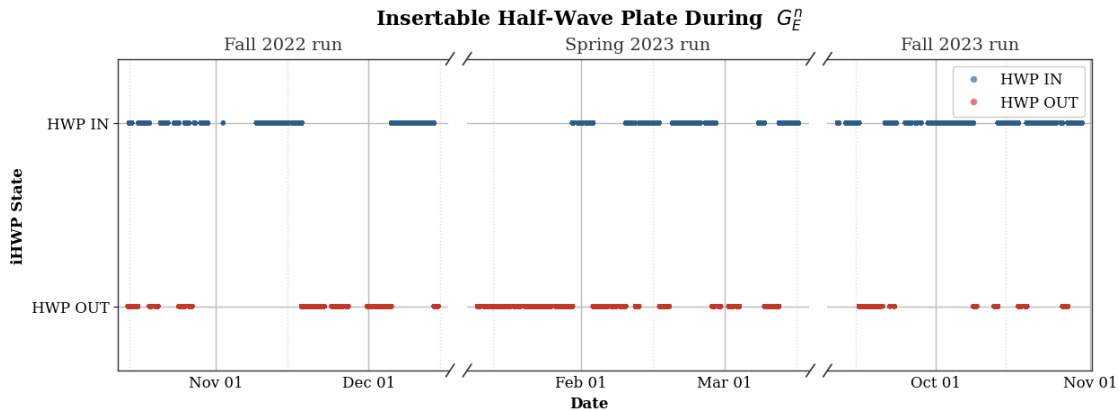


Figure 7: The insertable half-wave plate (iHWP) state recorded across the Fall 2022, Spring 2023, and Fall 2023 run periods of the G_E^n experiment. Blue and red markers indicate the HWP IN and HWP OUT states, respectively.

Wien Filters

Multiple experimental halls at CEBAF may request a polarized beam simultaneously [62]. Because the electrons undergo spin precession as they propagate through the accelerator (discussed in more detail in Section 4.5), they would otherwise arrive at the halls with their spins pointing in undesired directions, even though many experiments require a longitudinally polarized beam [63]. To address this, the electron beam is passed through the Wien filter — a spin rotator — immediately after generation, which manipulates the spin orientation by a precalculated angle chosen to maximize the longitudinal polarization delivered to each experimental hall. The Wien filter applies mutually orthogonal static electric and magnetic fields to the beam in such a way that the Lorentz force on the beam trajectory vanishes, while the spin precesses by an angle

$$\theta_W = \frac{eBL}{mc\beta\gamma^2} \left(\frac{g}{2} \right) \approx \frac{eBL}{mc\beta\gamma^2}, \quad (2.1)$$

where e is the electron charge, B is the magnetic field strength, L is the effective length of the filter, m is the electron mass, c is the speed of light, β and γ are the usual relativis-

tic factors, and g is the electron gyromagnetic ratio [64]. CEBAF employs two orthogonal Wien filters — one applying a horizontal and the other a vertical magnetic field — separated by two solenoids that provide an in-plane rotation [65]. To calibrate the Wien filters and establish absolute knowledge of the spin orientation, a procedure known as a “spin dance” is performed, which is the standard practice for Hall A Møller polarimetry. The procedure involves stepping the Wien angle across a range of values and extracting the cosine response of the various polarimeters at both the hall of interest and the injector [66, 67].

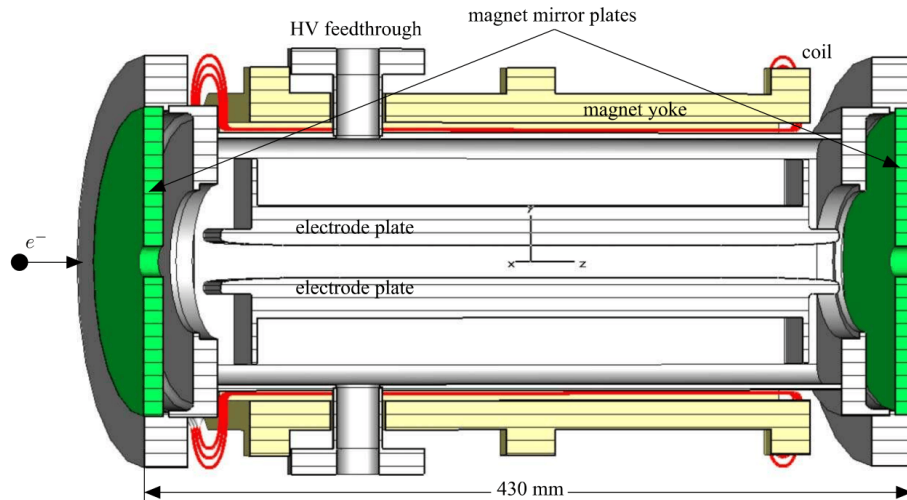


Figure 8: Cross section of the Wien filter installed at Jefferson Lab [68].

2.5 Acceleration

The electron beam is accelerated at Jefferson Lab by two antiparallel superconducting radio-frequency (SRF) linear accelerators (Linacs), referred to as the North Linac and the South Linac, which are connected by magnetic recirculation arcs. Both Linacs operate at the fundamental CEBAF frequency of 1497 MHz. Inside the Linacs, standing electromagnetic waves are established in niobium cavities; the ultra-relativistic electron

bunches are precisely timed to ride the crest of the longitudinal electric field, thereby progressively increasing the energy of the electron beam to the level required by each experimental hall. The facility currently supports up to 5-pass operation through the recirculation arcs, providing a maximum beam energy of 11 GeV for Halls A, B, and C, and a 5.5-pass operation delivering up to 12 GeV for Hall D [69, 70]. CEBAF houses approximately 50 cryomodules, which serve as the fundamental building blocks of the accelerator. Each cryomodule contains a string of the superconducting resonant cavities responsible for boosting the energy of the electrons as they traverse the Linacs. The energy gain provided by each cryomodule varies depending on its cavity design and when it was refurbished or upgraded, and on average they currently contribute approximately 1.1 GeV of energy per Linac per pass [71]. The cryogenic operation of these cavities is critical to the machine's performance; maintaining the niobium structures at approximately 2 K nearly eliminates resistive Joule heating, thereby allowing the accelerator to run in a highly efficient continuous-wave (CW) mode rather than in a pulsed mode. This ultra-low temperature environment is sustained by a continuous flow of liquid helium supplied by the Central Helium Liquefier (CHL) at the laboratory [72, 73].



Figure 9: A cryomodule for the CEBAF 12 GeV upgrade, which houses the superconducting radio frequency cavities used to accelerate the electron beam [74].

2.6 The Recirculation Arcs

The recirculation arcs, also referred to as the bending arcs, are located at the two ends of the accelerator system and serve to redirect the beam back to the opposite Linac for continued acceleration, effectively multiplying the total energy gain without requiring a correspondingly long linear footprint. Because the beam in the Linacs consists of interleaved electron bunches at several different discrete energies simultaneously, the east and west ends together contain a total of 10 distinct arcs, which are stacked vertically. A dedicated "spreader" system of vertical dipoles and focusing quadrupoles is used to separate

the bunches by momentum and transport the beam vertically from the Linacs to the specific arc selected for each pass, while a corresponding "recombiner" merges them before the next Linac [75]. The principal horizontal bending of the beam is achieved through a series of precision dipole magnets distributed along each arc. To maintain the exact same geometric bending radius for all passes, the magnetic fields must scale with the momentum. The lowest-energy arcs each contain 16 dipoles, while the higher-energy arcs each contain 32 dipole magnets to accommodate the significantly greater magnetic rigidity of the beam at higher energies, preventing the need for impossibly strong individual magnetic fields [76]. Two of the main physical considerations associated with transporting high-energy electrons through these arcs are the emission of synchrotron radiation and the spin precession of the electrons. Synchrotron radiation results in undesirable beam energy loss and an increase in energy spread (emittance growth) that scales strongly with the beam energy. Meanwhile, due to the electron's anomalous magnetic moment, its spin precesses relative to its momentum vector as it traverses the transverse magnetic fields of the dipoles, a dynamic that constitutes an important source of systematic uncertainty for polarization-dependent experiments at the experimental halls. Both of these phenomena will be discussed in greater detail in the next chapter [77, 78].



Figure 10: A bending arc at CEBAF, where a series of dipole magnets steer the electron beam between successive passes through the Linacs [79].

2.7 RF Separators and Hall Bending Magnets

At the downstream end of the South Linac, the simultaneous delivery of continuous-wave beam to multiple experimental halls is achieved using normal-conducting TEM-mode RF separator cavities. These cavities operate at a subharmonic of the fundamental CEBAF frequency (typically 499 MHz) to provide a precisely timed, transverse electromagnetic kick that selectively deflects the interleaved electron bunches based on their destination [80, 81]. For Halls A, B, and C, the spatially separated beamlets are subsequently intercepted by DC current-sheet septum magnets. The septum design is critical, as it produces a strong

dipole field to further bend the kicked beam while maintaining a sharp boundary to a nearly field-free region, ensuring that adjacent, undeflected beam bunches pass undisturbed. These septum magnets operate with high current densities in the range of 28–48 A/mm² and have lengths between 1 and 3 m [82]. The operation of Hall D is decoupled from this RF separation scheme. Because Hall D exclusively requires the highest-energy (12 GeV) beam from a final 5.5-pass acceleration through the North Linac, it is fed via direct magnetic extraction immediately after Arc 10 rather than through the shared South Linac switchyard [72]. Following separation, the lower-pass beams are transported through the beam switchyard to their respective targets. For Halls A and C, significant additional magnetic bending is required to steer the beam by 37.5° and −37.5°, respectively. This large geometric deflection dictates a massive array of dipoles and introduces substantial electron spin precession that must be precisely compensated for at the injector, whereas Hall B is positioned straight ahead along the linac axis and requires no horizontal bending [83].

2.8 Beam Monitoring and Instrumentation

Beam Monitors

Continuous, non-destructive monitoring of the electron beam’s properties is essential for normalizing scattering rates and ensuring precise kinematic reconstruction. A Beam Current Monitor (BCM) is an RF resonant cavity used to measure the intensity of the electron beam, providing both the instantaneous beam current and the total accumulated charge delivered to the experimental hall [84]. As the periodic electron bunches pass through the cylindrical cavity, they excite a fundamental electromagnetic mode (typically tuned to the 1497 MHz CEBAF frequency) whose output amplitude is directly proportional to the

beam current. The measured analog RF output of the BCM is processed into a frequency signal that must be calibrated against an absolute reference. Because the RF cavities are highly stable over short periods but lack absolute zero-point calibration, this reference is provided by a parametric DC current transformer; in the case of Hall A, this absolute reference is the Unser Monitor. The calibrated BCM frequency is then continuously converted into a precision beam current value. A Beam Position Monitor (BPM) is a cylindrical device through which the beam passes, typically equipped with four symmetric wire stripline antenna electrodes that pick up the induced RF signals from the passing beam. While the periodic beam structure provides the fundamental frequency, the precise transverse position of the beam centroid within the beam pipe is extracted by calculating the normalized difference-over-sum of the induced voltage amplitudes measured by opposing electrode pairs [85]. Two or more BPMs are typically placed several meters apart in order to mathematically reconstruct the full beam trajectory—yielding both absolute spatial position and incident angle—particularly in the critical regions close to the main experimental setup [86]. To complement the non-destructive BCMs and BPMs, harp scanners are utilized to provide a high-precision, absolute measurement of the beam’s transverse profile and position [87]. These devices consist of thin wires mounted on a frame that a stepper motor drives directly across the beam path [87]. As the wires intercept the beam, the resulting secondary electron emission or induced radiation shower is measured to construct a highly accurate beam intensity profile. Because the wire physically obstructs the beam, harp scans are an invasive, destructive diagnostic and are typically performed only during dedicated beam tuning or calibration intervals rather than during continuous data acquisition [87].

Rasterization

Because the intrinsic transverse profile of the continuous wave (CW) electron beam delivered by the accelerator is exceptionally small—typically on the order of $100\ \mu\text{m}$ —a GeV-scale beam delivered at currents greater than or equal to $20\ \mu\text{A}$ represents an extreme localized power density. If left unmitigated, this concentrated energy deposition would rapidly exceed the thermal limits of most physics targets at Jefferson Lab, causing unacceptable localized boiling and density fluctuations in cryogenic targets or outright melting and structural failure in solid targets. To prevent catastrophic damage to these experimental targets and to maintain a stable target density for accurate scattering rate normalization, the cross-sectional area covered by the beam must be artificially expanded to distribute the thermal load over a much larger footprint [88]. This is accomplished by passing the beam through an upstream fast raster system, typically consisting of pairs of orthogonal AC dipole magnets. By driving these magnets with rapidly varying currents modulated at high, non-resonant harmonic frequencies (often in the kilohertz range), the unperturbed beam spot is swiftly swept across a few millimeters in both transverse directions, effectively creating a uniform two-dimensional spatial distribution averaged over the thermal relaxation time of the target [19]. During the demanding, high-luminosity running of the SBS program in Hall A, several distinct raster patterns—such as squares or circles generated by careful selection of the horizontal and vertical sweeping waveforms—were employed across the various experiments in order to optimally match the specific size and geometry of each target cell, maximizing the usable interaction volume while steering clear of the target walls. Møller polarimetry, by contrast, does not require any raster pattern, as it is strictly operated at very low, diagnostic beam currents, typically around $1\ \mu\text{A}$. At these significantly reduced currents, the modest thermal load is easily dissipated through natural conduction in the iron target foil [89].

2.9 Spectrometers

A magnetic spectrometer is an essential device used to measure the properties of charged particles produced in nuclear reactions. It uses static magnetic fields to bend the paths of particles, allowing an array of downstream detectors to record their trajectories, momenta, and identities. The underlying physics relies on the Lorentz force, $F = qv \times B$. When a charged particle with momentum p and charge q enters a uniform magnetic field B (created by a dipole magnet), it is forced to traverse a circular arc of radius $\rho = p/(qB)$. Because the bend angle is fixed by the layout of the detectors, where a particle lands on the focal plane directly maps to its momentum. To improve the system, quadrupole magnets—which act like optical lenses whose field strength increases away from the center, illustrated in Figure 23—are used to focus the particle beam. This enlarges the solid angle of particles the spectrometer can accept without losing them.

When evaluating these instruments, physicists look at two primary performance metrics that fundamentally trade off against each other: momentum resolution ($\Delta p/p$) and solid-angle acceptance ($\Delta\Omega$). Momentum resolution measures how precisely the device can determine a particle's momentum, with modern electron-scattering systems achieving $\Delta p/p \lesssim 10^{-4}$ [19, 90]. Solid-angle acceptance measures the physical "field of view" required to capture rare reactions. Because any focusing system can only image a finite volume of particles onto a detector, improving one metric usually degrades the other. For instance, the High Resolution Spectrometer (HRS) pair in Hall A at Jefferson Lab offers incredible resolution ($\Delta p/p < 2 \times 10^{-4}$) but captures a very narrow window of particles (about 6 msr) [19]. Conversely, the BigBite single-dipole spectrometer sacrifices precision for volume, achieving a massive 96 msr acceptance at a much looser resolution of roughly 8×10^{-3} [91].

This modern landscape of complementary spectrometers evolved directly from early β -ray spectroscopy. In the 1940s and 1950s, Kai Siegbahn pioneered highly precise iron-free spectrometers to study radioactive sources, establishing the foundational concepts of magnetic focusing and dispersion [92]. As particle accelerators grew more powerful in the 1960s, spectrometers had to scale up. At SLAC, massive devices using multiple magnets were used to measure proton form factors, ultimately leading to the discovery of quarks [93, 94]. Over the decades, laboratories worldwide experimented with various magnet combinations, such as the QDD and QDQ designs at Saclay and NIKHEF [91, 95], and the QQDD facilities at MIT-Bates and Mainz MAMI [90]. This progression led to advanced setups like the QQDQ arrangement (two superconducting quadrupoles, a dipole, and a third quadrupole) used in Jefferson Lab’s Hall A [19].

Today, precise electron-scattering experiments require these varied configurations to simultaneously identify scattered electrons and resulting hadronic particles. High-quality spectrometers are essential to reconstruct the missing mass and energy of collisions, allowing scientists to separate specific elastic and quasi-elastic events from the inelastic background. Because no single system can do it all, modern physics facilities rely on a complementary mix of tools. They pair QQDQ designs for pinpoint resolution with single-dipole BigBite-style designs for massive acceptance and high luminosities [91, 96], alongside toroidal designs like CLAS that cover almost 4π in azimuth at a moderate resolution [97]. By carefully selecting the right magnet configuration, physicists can successfully explore the full spectrum of nuclear physics.

2.10 The Experimental Halls

CEBAF delivers a continuous-wave electron beam, carrying up to 11 GeV across three of its four 5-pass arcs to Halls A, B, and C, and delivering a 12 GeV fifth pass to Hall D [72]. Rather than a single shared hall, CEBAF was built with four separate experimental areas because no one configuration can simultaneously maximize momentum resolution, solid-angle acceptance, momentum reach, and high-flux polarized photon delivery; each hall is therefore optimized for a different part of this trade-off space.

Experimental Hall A — from high-resolution focusing spectrometry to a large-installation hall: form factors, parity violation, and the SBS program

Hall A is the largest of Jefferson Lab’s experimental halls [19]. Historically, it operated as a focusing-spectrometer hall utilizing two identical High Resolution Spectrometers (HRS). Each HRS featured a QQDQ sequence of three superconducting quadrupoles and a 6.6-m dipole that bent charged particles vertically by 45° . This setup achieved a momentum resolution better than 2×10^{-4} , a horizontal angular resolution under 2 mrad, a central-momentum range of 0.3–4 GeV/c, and a solid angle of a few msr per arm [19]. Pairing this extreme resolution with luminosities exceeding $10^{38} \text{ cm}^{-2} \text{ s}^{-1}$ made the HRS the perfect workhorse for high-precision measurements. This included the HAPPEX strange-form-factor measurements, the PREX-2 determination of the ^{208}Pb neutron skin [98], and the CREX measurement on ^{48}Ca [99].

For the 12 GeV era, Hall A transitioned into a long-term installation hall hosting the MOLLER experiment, the future SoLID program, and the Super BigBite Spectrometer (SBS) [100]. The hall’s current physics program is anchored by the SBS alongside the original BigBite spectrometer. BigBite provides a massive solid-angle acceptance of nearly 100 msr, an angular resolution of 4 mrad, and a momentum resolution of roughly 5×10^{-3} [101]. SBS

itself uses a 48D48 dipole magnet instrumented with GEM trackers designed to handle CEBAF’s highest available luminosities (around $10^{39} \text{ cm}^{-2} \text{ s}^{-1}$) with solid angles typically between 50–100 msr [96, 101, 102]. This combination of high luminosity and large acceptance allows SBS to push neutron magnetic form factor measurements (GMn) to extreme momentum transfers, reaching $Q^2 \approx 13.6 \text{ (GeV/c)}^2$ [103]. To support these demanding measurements, the hall relies on a Møller polarimeter to determine beam polarization, recently achieving a precision of 0.89% for PREX-2 and 0.85% for CREX [18].

Experimental Hall B — large acceptance, low luminosity, multi-particle final states

Hall B is home to the CEBAF Large Acceptance Spectrometer (CLAS) family, designed around a philosophy opposite to that of Hall A: trading luminosity for maximum acceptance. The original CLAS (1997-2012) utilized a six-coil superconducting toroidal magnet to achieve momentum resolutions of roughly 0.5% at forward angles and operating luminosities up to $10^{34} \text{ nucleon cm}^{-2} \text{ s}^{-1}$ [97]. For the 12 GeV era, the system was rebuilt as CLAS12 [104]. This upgrade added a 5T superconducting solenoid around the target for large-angle central tracks while retaining a 3.6T peak-field toroid for forward tracks [105, 106]. This two-magnet geometry provides CLAS12 with quasi- 4π hadronic coverage, allowing physicists to detect complete multi-particle final states in a single event.

Experimental Hall C — high momentum, high luminosity, precision Rosenbluth and L/T separations

Hall C houses two highly-focusing, heavily-shielded magnetic spectrometers: the High Momentum Spectrometer (HMS) and the newer Super High Momentum Spectrometer (SHMS) [107]. The HMS utilizes a QQQD magnet configuration reaching a maximum central momentum of 7.5 GeV/c, an 18% momentum bite, a momentum resolution better than 0.1%, a solid angle greater than 6 msr, and a minimum central scattering angle of

12.5° [108]. The SHMS extends this central momentum coverage to 11 GeV/c, offering a momentum acceptance of $-10\% < \delta < +22\%$, a momentum resolution of 0.03–0.08%, a solid angle of roughly 5 msr, and a scattering angle reach down to 5.5° [107]. By combining excellent systematic control with high momentum reach and very high luminosity [109, 110], Hall C is the optimal facility for precision Rosenbluth and longitudinal–transverse cross-section separations [107].

Experimental Hall D — linearly polarised photon beam, meson spectroscopy

Unlike the other halls, Hall D was built specifically for the 12 GeV upgrade to operate with a photon beam rather than an electron beam. A 12 GeV electron beam hits a thin 20 μm diamond wafer to produce a tagged, linearly polarized photon beam via coherent bremsstrahlung. This beam features a coherent peak between 8.4 GeV and 9.1 GeV with 40% linear polarization [111, 112]. The photons are directed into the GlueX detector, built around a refurbished SLAC LASS solenoid that runs at 1350 A to provide an axial field of roughly 2 T around a liquid-hydrogen target [113]. Combined with precision pair spectrometers and polarimeters [112, 113], this hermetic solenoidal setup makes Hall D the world’s leading facility for hunting exotic hybrid mesons (states with $J^{PC} = 1^{-+}, 0^{+-}$, etc., which are forbidden in simple $q\bar{q}$ models) and mapping the gluonic excitation spectrum.

CHAPTER 3

Beam Polarimetry at Jefferson Lab

3.1 Polarimetry Concepts — Longitudinal vs. Transverse

Polarized electrons at CEBAF are produced by photoemission from a strained-superlattice GaAs photocathode and emerge from the source longitudinally polarized, with their spin oriented parallel or antiparallel to the beam momentum [114, 115]. The nuclear physics program generally requires *longitudinal* polarization at the target inside each experimental hall, and the spin direction does not remain aligned with the momentum during transport. To compensate for the spin precession of the beam described in detail in Section 4.5, one or more Wien filters in the injector rotate the spin at low energy so that, after the cumulative precession, the polarization arrives longitudinal at the target [115]. Because the experimental halls are served simultaneously at different energies and pass numbers, delivering maximal longitudinal polarization to every hall at once is a non-trivial optimization of the Wien-filter orientation and the linac energy sharing [115].

These spin-transport properties dictate where, and how, the polarization is measured. A Mott polarimeter located in the low-energy (~ 5 MeV) injector is sensitive to the *transverse* component of the polarization, since the Mott left–right asymmetry vanishes for spin aligned with the momentum; it is therefore used to characterize and optimize the polarized source and to help establish the launch spin orientation before acceleration [114, 116]. The polarimeters in the experimental halls—Compton and Møller—instead measure the

longitudinal polarization that the experiments actually use.

No single polarimetry technique is ideal across all running conditions, which is why CE-BAF relies on several complementary devices [117]. Møller polarimetry provides a fast, absolute measurement but is invasive: it uses a ferromagnetic foil target and can operate only at low beam current, interrupting the physics program [117]. Compton polarimetry is non-invasive and runs concurrently with production, but its analyzing power varies rapidly with beam energy—even changing sign across the scattered-particle spectrum—which complicates operation over a broad energy range [117]. Mott polarimetry, in turn, operates only at the injector, before acceleration. Using these independent techniques together, based on three distinct scattering processes, allows their results to be cross-checked against one another. This redundancy has proven essential for identifying and controlling systematic uncertainties, and for verifying working assumptions such as the independence of the beam polarization from beam current [117, 118].

3.2 The Mott Polarimeter

The Mott polarimeter is located at the injector site and measures the transverse polarization of the electron beam, serving as a crucial diagnostic tool to verify the beam polarization before it is sent to the accelerator. The fundamental operating principle of the Mott polarimeter is based on the elastic Coulomb scattering of polarized electrons off a high- Z target nucleus, typically gold [114, 119]. During this interaction, the strong electric field of the nucleus interacts with the magnetic moment of the incident electron via spin-orbit coupling, which results in a spin-dependent scattering probability. This spin-orbit interaction produces an azimuthal spatial asymmetry in the scattered electron distribution. By utilizing detectors positioned symmetrically (e.g., left and right) relative to the beamline, an

experimental left-right scattering asymmetry, A , is measured. The transverse polarization of the beam, P_y , is then extracted using the following relationship [116, 120]:

$$P_y = \frac{A}{S_{\text{eff}}(\theta)} \quad (3.1)$$

Here, $S_{\text{eff}}(\theta)$ is the effective Sherman function, which characterizes the analyzing power of the target at a given scattering angle θ . While the theoretical Sherman function $S(\theta)$ assumes a perfectly point-like interaction, real experimental targets have a finite thickness. This finite thickness introduces multiple Coulomb scattering events that depolarize the electrons and dilute the analyzing power. Consequently, $S_{\text{eff}}(\theta)$ is carefully determined via detailed Monte Carlo simulations to correct for these multiple scattering and detector geometry effects [121]. An elevation view of the Mott Polarimeter at Jefferson Lab is shown in Figure 11.

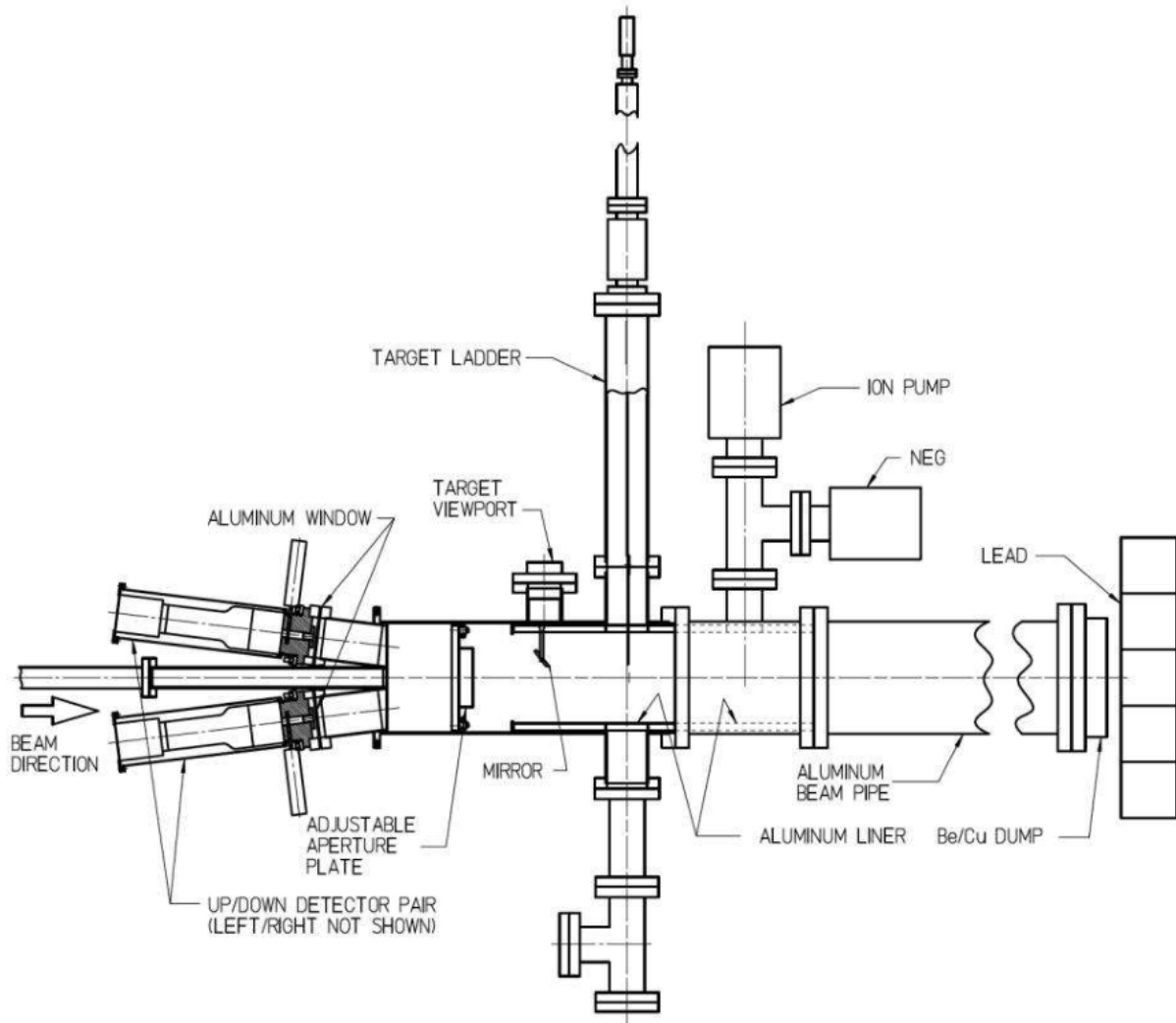


Figure 11: Elevation view of the Mott polarimeter at Jefferson Lab [116].

3.3 Compton Polarimeters

Jefferson Lab operates two Compton polarimeters, one in Hall A and one in Hall C. Unlike Møller polarimetry, Compton polarimetry is considered a non-invasive polarimetry technique, since it can operate continuously and in parallel with the main experimental program without disrupting data taking [38, 122]. Both the Hall A and Hall C Compton

polarimeters rely on the Compton backscattering of an incident photon off an electron in the beam: a high-power laser is directed into the electron beam, and the resulting scattered photons and recoil electrons are recorded by a photon detector and an electron detector, either of which can independently determine the electron beam polarization [117, 123, 124]. The Hall A Compton polarimeter apparatus consists of a four-dipole magnetic chicane, a high-finesse external Fabry-Perot cavity that serves as a photon amplifier, a 532 nm laser system used as the photon source, a photon detector, an electron detector, collimators, and beam position monitors [88, 125, 126]. During the SBS program, the Hall A Compton polarimeter was not operational. Compton polarimetry is performed by exploiting the following relationship [121, 127]:

$$A_{\text{measured}} = P_{\text{beam}} P_{\gamma} A_{\text{th}} \quad (3.2)$$

where P_{beam} is the polarization of the electron beam, P_{γ} is the circular polarization of the incident laser photon, and A_{th} is the theoretical longitudinal analyzing power.

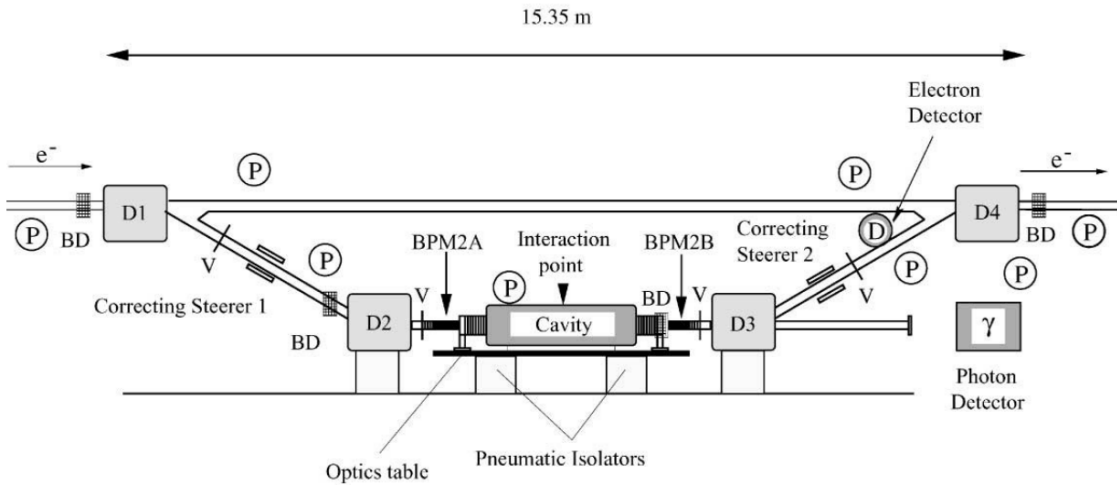


Figure 12: Schematic of the Hall A Compton polarimeter at Jefferson Lab [123].

3.4 The Møller Polarimeters

Møller polarimetry determines the longitudinal polarization of the electron beam from the spin-dependent cross section of elastic electron–electron (Møller) scattering between the polarized beam electrons and the polarized atomic electrons of a ferromagnetic foil target [128, 129]. The measured helicity asymmetry factorizes as

$$A_{\text{measured}} = -P_{\text{beam}} P_{\text{target}} \langle A_{zz} \rangle, \quad (3.3)$$

where P_{beam} and P_{target} are the longitudinal polarizations of the beam and target electrons, and $\langle A_{zz} \rangle$ is the longitudinal analyzing power averaged over the spectrometer acceptance. The analyzing power is fixed by QED and depends only on the center-of-mass scattering angle θ ; in general it is a tensor (A_{zz}, A_{xx}, A_{yy}) reflecting the relative orientation of the beam and target spins. For purely longitudinal spins its magnitude is maximal, $|A_{zz}| = 7/9$, at $\theta = 90^\circ$, where it is a factor of seven larger than the transverse analyzing power, so all three polarimeters tune their acceptance around this kinematic point [114, 128].

The three polarimeters share the same essential features. In every case the target electron polarization is only of order 8%, because only the electrons in the $3d$ shell of the iron are spin-polarized while the inner-shell electrons are not [128, 129]. The measurement is consequently invasive: the foil is inserted into the beam and the current is reduced to a few microamperes to avoid depolarization from beam heating, and the scattered pair is detected in coincidence to suppress the dominant background from Mott (electron–nucleus) scattering in the foil [117, 128]. The polarimeters differ chiefly in how the target is magnetized. The older approach magnetizes a high-permeability alloy foil in-plane with a modest holding field of order 100 G, tilting the foil so that the magnetization has a longitudinal projection along the beam; this is experimentally simple, but the target polarization

is known only to a few percent [114]. The newer approach, pioneered for Hall C, instead drives a pure iron foil to out-of-plane saturation with a strong ($\approx 3\text{--}4\text{ T}$) field along the beam, which makes the target polarization precisely calculable and reduces the dominant systematic to the sub-percent level [114, 128].

Each experimental hall is equipped with its own Møller polarimeter because each hall receives independently configured beam and runs its own program, so each requires a dedicated, local polarization measurement. Operating several Møller polarimeters—alongside the Mott and Compton devices—also provides redundancy and a means of cross-calibration: the effective analyzing powers of the different polarimeters have been compared directly by measuring the same beam, which tests for unrecognized errors in any single device [117, 130].

Hall A Møller Polarimeter

The Hall A Møller polarimeter is the primary focus of this thesis and is covered extensively throughout this document. Conceptually it follows the high-precision design used in Hall C: a pure iron foil magnetized to out-of-plane saturation by a high-field superconducting magnet, with a quadrupole-and-dipole spectrometer that focuses the Møller pair near the 90° center-of-mass maximum and detects the two electrons in coincidence [129]. Its detailed layout, procedures, and systematic analysis are deferred to Chapters 5, 6, 7, respectively.

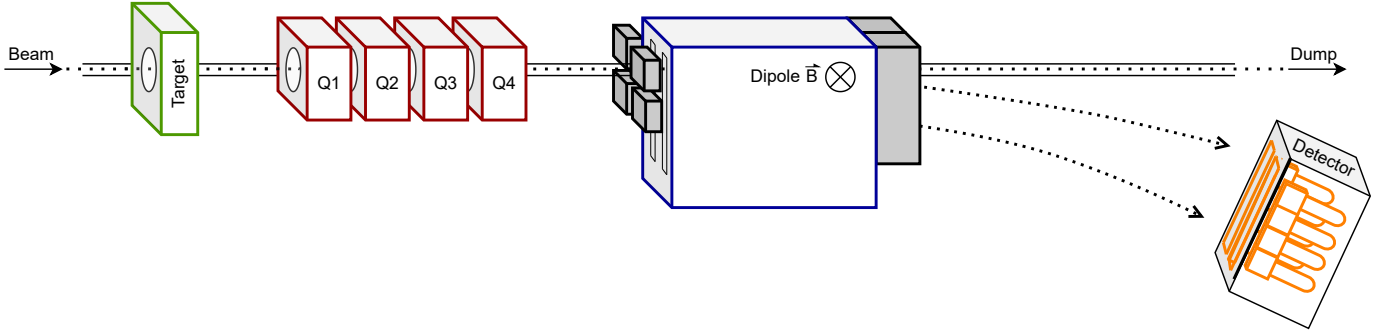


Figure 13: Simple diagram of the Hall A Møller polarimeter. Refer to Figure 20 for more details.

Hall B Møller Polarimeter

This is the primary device used by experimental Hall B to measure the beam polarization. It adopts the older, in-plane magnetization scheme: a ferromagnetic alloy (Permendur) foil tilted with respect to the beam and magnetized by a low Helmholtz field, two quadrupole magnets, and two lead/scintillator total-absorption detectors operated in coincidence [97, 104]. Because the target polarization of an in-plane, tilted foil is less precisely known, the precision achieved by Hall B Møller polarimetry is at the level of roughly 1.5–3%, coarser than the out-of-plane polarimeters of Halls A and C.

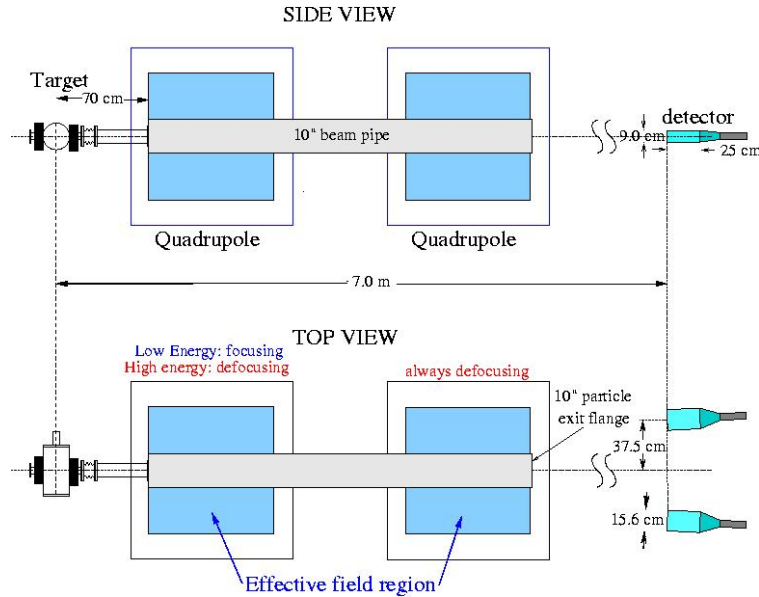


Figure 14: The Hall B Møller polarimeter at Jefferson Lab.

Hall C Møller Polarimeter

The Hall C Møller polarimeter was the first to use the “brute-force” out-of-plane saturation technique now shared with Hall A [117, 128]. A pure iron foil is oriented perpendicular to the beam and driven to magnetic saturation by a superconducting solenoid producing a field of $\approx 3\text{--}4$ T along the beam, well above the ~ 2.2 T required to saturate iron; because the foil magnetization is then at its precisely known maximum, the target electron polarization (of roughly 8%) is determined with small systematic uncertainty. The scattered Møller pairs are separated from the primary beam and from the Mott background by two quadrupole magnets working together with movable collimators, and are detected in coincidence between opposing beam-helicity states. With this approach the Hall C polarimeter provides absolute longitudinal polarization measurements with an accuracy better than 0.5% [118, 131].

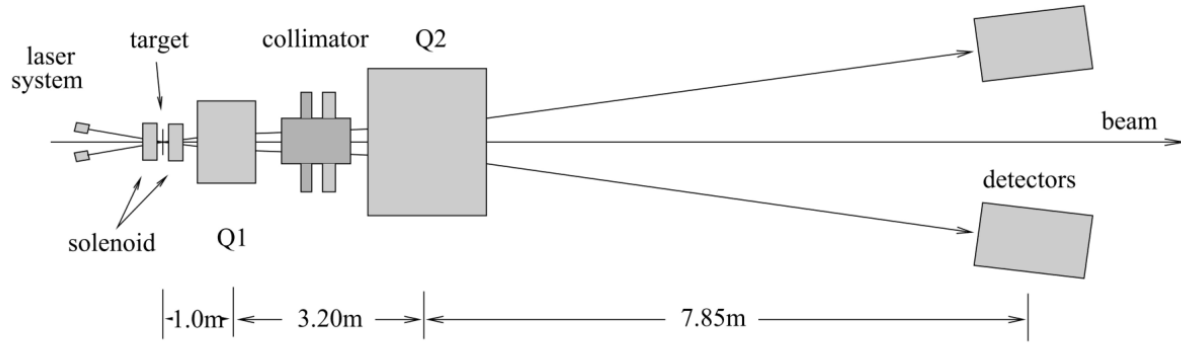


Figure 15: Schematic of the Hall C Møller polarimeter at Jefferson Lab [128].

CHAPTER 4

Theory of Møller Polarimetry

4.1 Møller Scattering

$$e^-(p_1) + e^-(p_2) \rightarrow e^-(p_3) + e^-(p_4) \tag{4.1}$$

Møller scattering plays a central role in the history and development of quantum electrodynamics (QED) [132], beginning with Møller's 1930s relativistic analysis of electron-electron scattering, which first established the importance of scattering between identical fermions and the associated exchange symmetry of the amplitude. The theoretical framework was expanded to include spin and helicity, enabling calculations of polarized cross sections in QED. Møller scattering served as a rigorous test of QED and as a valuable experimental tool with polarized beams. [133]

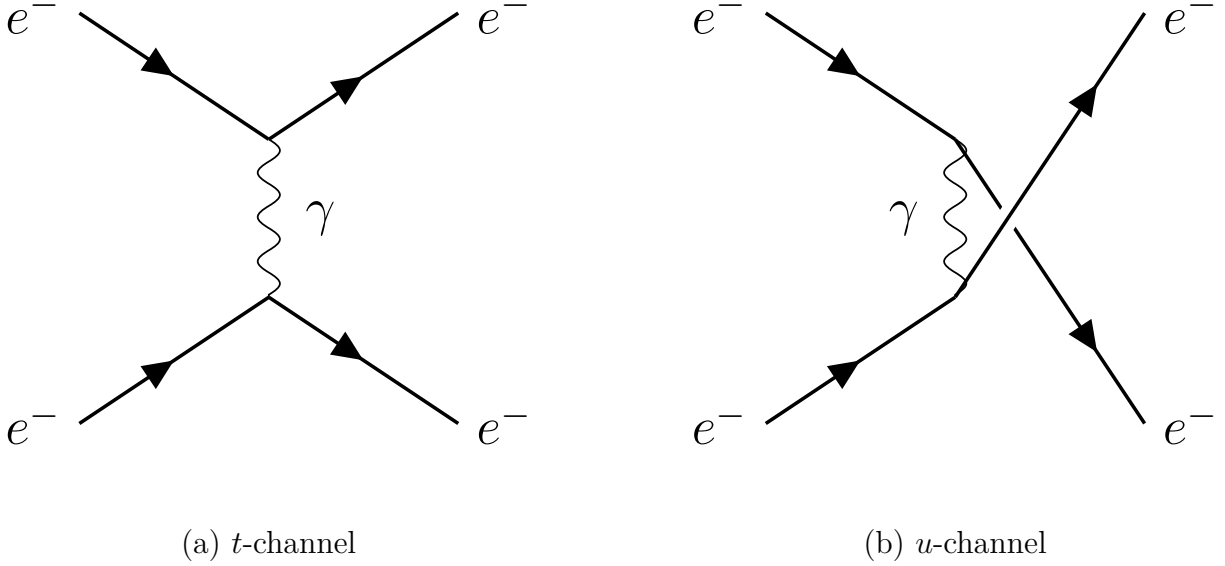


Figure 16: Lowest-order Feynman diagrams representing electron–electron (Møller) scattering: (a) the t -channel diagram, in which the electrons exchange a virtual photon without crossing, corresponding to the assignment $p_1 \rightarrow p_3$ and $p_2 \rightarrow p_4$; and (b) the u -channel diagram, in which the outgoing electrons are exchanged, reflecting the indistinguishability of the final-state fermions and enforcing the antisymmetry of the total scattering amplitude. Both diagrams contribute to the full calculation of the Møller scattering cross-section.

The Mandelstam variables are defined as

$$s = (p_1 + p_2)^2 = (p_3 + p_4)^2 \quad (4.2)$$

$$t = (p_1 - p_3)^2 = (p_4 - p_2)^2 \quad (4.3)$$

$$u = (p_1 - p_4)^2 = (p_3 - p_2)^2 \quad (4.4)$$

Equation 4.1 illustrates the assignment of the four-momenta p_i . In the simple case of an incoming electron beam incident on a fixed target, the elastic nature of Møller scattering implies that both the incoming and outgoing particles are electrons with identical mass,

and the four-momenta can be written as

$$p_1 = (0, 0, p_b, p_b) \quad (4.5)$$

$$p_2 = (0, 0, 0, m) \quad (4.6)$$

$$p_3 = (0, p' \sin \theta, p' \cos \theta, p') \quad (4.7)$$

$$p_4 = (0, -p' \sin \theta, p' \cos \theta, p') \quad (4.8)$$

where p_b is the beam momentum, p' the momentum of the scattered electron, and m is the electron mass. The square of the center-of-mass energy is defined as

$$s = (p_1 + p_2)^2 = m^2 + m^2 + 2mE_b \quad (4.9)$$

which reduces to

$$s_0 = 2mp_b \quad (4.10)$$

in the ultra-relativistic limit $E_b \approx p_b \gg m$. Using the tree-level QED scattering amplitude and standard trace techniques, one obtains the well-known differential cross section for unpolarized Møller scattering [83]

$$\frac{d\sigma_0^u}{d\Omega}(s) = \frac{\alpha^2 (3 + \cos^2 \hat{\theta})^2}{s \sin^4 \hat{\theta}} \quad (4.11)$$

Here, $\hat{\theta}$ denotes the center-of-mass scattering angle and α is the fine-structure constant. A correction term accounts for polarized Møller scattering by introducing spin-dependent interference terms in the cross section. These terms arise from helicity-dependent amplitudes and will be discussed further in Sections 4.2 and 7.7. The general polarized cross-section

can be written as

$$\frac{d\sigma_0}{d\Omega}(s) = \frac{\alpha^2 (3 + \cos^2 \hat{\theta})^2}{s \sin^4 \hat{\theta}} \left\{ 1 - P_z^B P_z^T A_{zz}(\hat{\theta}) - P_t^B P_t^T A_{tt}(\hat{\theta}) \cos(2\phi - \phi_B - \phi_T) \right\} \quad (4.12)$$

where P_z^B and P_z^T denote the longitudinal polarizations of the beam and target electrons, P_t^B and P_t^T represent their transverse polarizations. The quantities A_{zz} and A_{tt} are the longitudinal and transverse analyzing powers. In this discussion, the azimuthal angle ϕ is taken to be zero. The analyzing powers are directly calculable within QED and depend only on the scattering kinematics, which makes polarized Møller scattering particularly suitable for absolute beam polarimetry [129, 134].

To determine the kinematic relations between the electron mass and momenta, we apply energy-momentum conservation. This allows the laboratory scattering angle to be expressed in terms of the center-of-mass angle and the beam energy. Following the treatment of Møller polarimetry in [134], one finds that, in the small-angle approximation, the expression is

$$\theta_{lab}^2 = \frac{1}{p_b p'} \cdot \frac{s_0}{2} (1 - \cos \hat{\theta}) = 2m_e \left(\frac{1}{p'} - \frac{1}{p_b} \right) \quad (4.13)$$

By using the four-momentum transfer squared, $Q^2 = -t$, together with the previously derived relations, one can connect the scattered-electron momentum in the center-of-mass frame to laboratory observables, arriving at the following relationship

$$p' = \frac{p_b}{2} (1 \pm \cos \hat{\theta}) \quad (4.14)$$

Levchuk Effect

All previous expressions assume a stationary target electron. In reality, target electrons in a magnetized iron foil are bound to atomic sites and have intrinsic momenta resulting

from their atomic wave functions. This phenomenon, known as the Levchuk effect, was analyzed in detail by L. G. Levchuk [135]. The Fermi motion of bound electrons smears the effective center-of-mass energy and reduces the observed analyzing power, thereby adding systematic uncertainty to beam polarization measurements[136]. Consequently, the initial four-momentum must be modified from $p_2 = (0, 0, 0, m)$ to $p_2 = (0, 0, p_t, m)$, where p_t is the target-electron momentum component along the beam direction. The modified center-of-mass energy squared becomes

$$s = s_0 \left(1 - \frac{\vec{p}_t \cdot \hat{n}}{m} \right) \quad (4.15)$$

where $\hat{n}$ is the beam direction, and since $\vec{p}_t \cdot \hat{n}$ varies between $|p_t|$ and $-|p_t|$, the factor in parentheses ranges from $1 - \frac{|p_t|}{m}$ to $1 + \frac{|p_t|}{m}$, depending on the projection of the target momentum[137].

Radiative corrections

In addition to the Levchuk effect, radiative corrections must be considered [58]. These are higher-order corrections that modify both the effective scattering angles and momenta of the incoming and outgoing electrons. Comprehensive treatments of QED radiative corrections to polarized Møller scattering have been presented by [138] and applied in precision parity-violating experiments such as [139]. These corrections are essential for achieving percent-level accuracy in polarization measurements and electroweak observables. [140, 141]

The modified kinematic relations can be expressed in terms of fractional energy losses $(1 - x_i)$ associated with each four-momentum p_i ,

$$p' = \frac{p_b x_1 x_3}{2} (1 + \cos \hat{\theta}) \quad (4.16)$$

$$\theta_{lab}^2 = 2m_e x_2 \left(\frac{x_3}{p'} - \frac{1}{p_b x_1} \right) \cdot \left(1 - \frac{\vec{p}_t \cdot \hat{n}}{m_e} \right) \quad (4.17)$$

Including these terms ensures that the reconstructed scattering angles and momenta correctly account for energy degradation prior to and after the hard scattering event, which is particularly important in high-precision polarimetry applications [134].

4.2 Møller Polarimetry

Møller polarimetry leverages the fact that the spin-dependent part of the electron-electron scattering cross section is both well understood within quantum electrodynamics (QED) and maximized under specific kinematic conditions[142, 143]. For longitudinally polarized beam electrons scattering off polarized atomic electrons in a magnetized target, the differential cross section can be written as the sum of an unpolarized term and a spin-dependent interference term proportional to the product of the beam and target polarizations[144]. Experimentally, the measurable quantity is a counting asymmetry calculated from the number of detected Møller scattering events for opposite beam helicity states [145, 146].

$$A_{\text{meas}} = \frac{N_+ - N_-}{N_+ + N_-} \quad (4.18)$$

where N^+ and N^- denote the event yields for positive and negative beam helicity, respectively. This measured asymmetry relates to the underlying physics through

$$A_{\text{meas}} = -P_{\text{beam}} P_{\text{target}} \langle A_{zz} \rangle \quad (4.19)$$

where P_{beam} and P_{target} are the beam and target electron polarizations, and A_{zz} is the longitudinal analyzing power derived directly from the polarized Møller scattering matrix element[132, 147]. Theoretical expressions for A_{zz} exhibit a strong angular dependence as

shown in Equations 4.20 and 4.21, reaching a maximum magnitude at a center-of-mass scattering angle of 90° , which motivates the fixed-angle acceptance design adopted by modern Møller polarimeters. Full forms of the longitudinal and transverse analyzing powers are presented by Equations 4.20 and 4.21.

Unlike hadronic polarimeters, the analyzing power in Møller polarimetry is purely electromagnetic and can be calculated to sub-percent precision using perturbative QED. Radiative corrections are well understood theoretically and separate cleanly in helicity space, so the dominant systematic uncertainties are experimental in origin, primarily from factors such as target polarization.

A polarimetry-specific factor is that the target electrons are bound in atomic orbitals rather than being free particles. In magnetized iron targets, the polarized electrons that contribute to the Møller asymmetry mainly come from the 3d valence shell, whose spin alignment causes the overall target polarization. Scattering from these polarized electrons is parity conserving [38, 148]. The measured asymmetry results entirely from QED spin-spin correlations and does not depend on parity-violating weak interactions.

Additionally, the finite momentum of bound electrons causes small kinematic smearing effects, known as the Levchuk effect, which mainly result from scattering off unpolarized inner-shell electrons. These electrons have significant intrinsic momentum due to atomic binding, which broadens the effective center-of-mass kinematics and slightly reduces the observed asymmetry. Scattering from these inner shells also conserves parity and only adds as background to the polarized signal. The separation between polarized valence electrons and unpolarized core electrons, along with the lack of parity violation in Møller scattering, differentiates this technique from parity-violating electron scattering methods used in beam diagnostics and electroweak studies [38].

4.3 Analyzing Power

The analyzing power is the proportionality factor relating the measured spin-dependent asymmetry to the product of beam and target polarizations, and quantifies the inherent sensitivity of the scattering cross section to the relative spin alignment of the two electrons. Any uncertainty in the analyzing power propagates directly into the extracted beam polarization and all downstream physics observables. The longitudinal analyzing power $A_{zz}(\hat{\theta})$ is the spin–spin correlation coefficient appearing in the polarized cross section, Eq. 4.12; it depends only on the center-of-mass scattering angle. Equivalently, it is the asymmetry between antiparallel and parallel pure spin states, $A_{zz} = (\sigma_{\uparrow\downarrow} - \sigma_{\uparrow\uparrow})/(\sigma_{\uparrow\downarrow} + \sigma_{\uparrow\uparrow})$. The experimentally measured asymmetry is this analyzing power diluted by the beam and target polarizations, $A_{\text{meas}} = -P_{\text{beam}}P_{\text{target}}\langle A_{zz} \rangle$ (Eq. 4.19) [136]. The expressions for the longitudinal and transverse analyzing powers in terms of the center-of-mass scattering angle $\hat{\theta}$ are

$$A_{zz}(\hat{\theta}) = \frac{(7 + \cos^2 \hat{\theta}) \sin^2 \hat{\theta}}{(3 + \cos^2 \hat{\theta})^2} \quad (4.20)$$

$$A_{tt}(\hat{\theta}) = \frac{\sin^4 \hat{\theta}}{(3 + \cos^2 \hat{\theta})^2} \quad (4.21)$$

Differentiating these expressions shows that the maximum values for the longitudinal and transverse analyzing powers are 7/9 and 1/9, respectively, both occurring at a center-of-mass scattering angle of 90° . The relationship between scattering angle and analyzing power is illustrated in Figure 17.

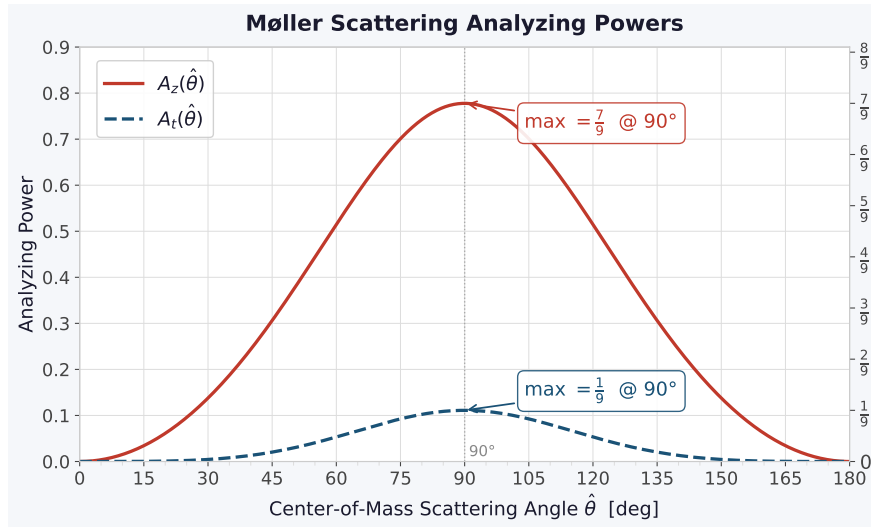


Figure 17: Longitudinal (A_{zz}) and transverse (A_{tt}) analyzing powers as a function of the center-of-mass scattering angle $\hat{\theta}$.

In practical experiments, the effective analyzing power is obtained by averaging the theoretical analyzing power over the detector acceptance, including effects from spectrometer optics, finite angular and momentum resolution, and target electron momentum distributions (e.g., the Levchuk effect), with additional corrections from radiative effects. As a result, accurate calculation of the analyzing power depends on detailed Monte Carlo simulations that model the full experimental geometry and response. Since experiments can operate at various beam energies, the Møller magnet settings differ between configurations, so the analyzing power must be computed separately for each. The list of analyzing powers used during the SBS nucleon form factor experiments is given in Section 4.6.

4.4 Target Polarization

Historically, early implementations of Møller polarimetry used weak magnetic fields applied to ferromagnetic targets positioned at an angle relative to the beam direction [137].

This relied on partial magnetization and geometric projections of the target polarization to define the analyzing power [149]. Although sufficient for early beam-polarization measurements, this approach limited the achievable precision because the degree of target electron polarization was not well constrained and was sensitive to field strength, alignment, and material hysteresis. Modern Møller polarimeters have thus shifted to using strong magnetic fields of about 4 T applied to thin iron foils oriented perpendicular to the beam, ensuring magnetic saturation and a well-defined longitudinal component of the target electron polarization [137, 150]. In this setup, the main systematic uncertainty in the measured beam polarization comes from the knowledge of the target electron polarization, which depends on the relationship between the material’s saturation magnetization and the fraction of magnetization carried by electron spin [129, 151, 152]. More recent research has revisited the fundamental spin polarization of magnetized iron and nickel using updated magnetic data and improved theoretical models, greatly enhancing the accuracy of the effective target polarization used in precision measurements [18, 129, 134, 153].

4.5 Spin Precession and the Spin Dance

Spin Precession

The beam electrons undergo spin precession as they traverse the curved transport arcs of CEBAF. The magnitude of this precession depends on the electron energy and the total bend angle accumulated along the beamline [154, 155]. For relativistic electrons in magnetic fields, the spin motion is governed by the Thomas-BMT equation [156, 157], which describes the precession of the spin vector relative to the momentum vector. In the case of planar bending in CEBAF, this reduces to a simple proportionality between the spin

precession angle and the orbital bend angle. The spin precession can be expressed as [115]:

$$\Delta\phi = \frac{E}{440.65 \text{ MeV}} \times \Delta\theta \quad (4.22)$$

where $\Delta\phi$ is the spin precession angle, E is the beam energy, and $\Delta\theta$ is the total bend angle. The accumulated spin precession grows proportionally with energy, making precise control of beam energy essential for polarization delivery. In the current laboratory setup, each experimental hall may receive a beam at a different energy. Since multiple halls can require polarized beams at the same time, the accelerator must optimize the Wien filter setting to maximize the longitudinal polarization delivered to each hall. The total spin precession accumulates gradually as the beam passes through successive magnetic bends in the recirculating arcs. Both horizontal and vertical bends contribute to the overall bend angle and must be included when calculating spin transport. Additionally, synchrotron radiation emitted in the bending magnets causes small but significant energy losses, which slightly alter the accumulated precession [158].

The measured longitudinal polarization at a given hall depends directly on the total spin precession, since only the longitudinal component contributes to parity-sensitive observables and Møller polarimetry. The relationship between the delivered longitudinal polarization P_L and the Wien filter angle at injection θ_W can be written schematically as

$$P_{\text{measured}} = P_0 \cos(\theta_{\text{Wien}} + \Delta\phi_{\text{tot}}) \quad (4.23)$$

where P_0 is the source polarization, and $\Delta\phi_{\text{tot}}$ is the total accumulated spin precession from the injector to the experimental hall [159]. This sinusoidal dependence arises from projecting the spin vector onto the beam momentum direction after transport through the accelerator lattice.

Using a Python script [160] that computes the spin precession for a given beam energy and experimental hall, I generated a diagram illustrating the relationship between Wien angle and beam polarization for Halls A, B, and C. The output exhibits a sinusoidal behavior, consistent with the cosine dependence shown above. The script also incorporates energy losses due to synchrotron radiation in the recirculating arcs, ensuring that the calculated precession angles accurately reflect the beam energy at each stage of acceleration.

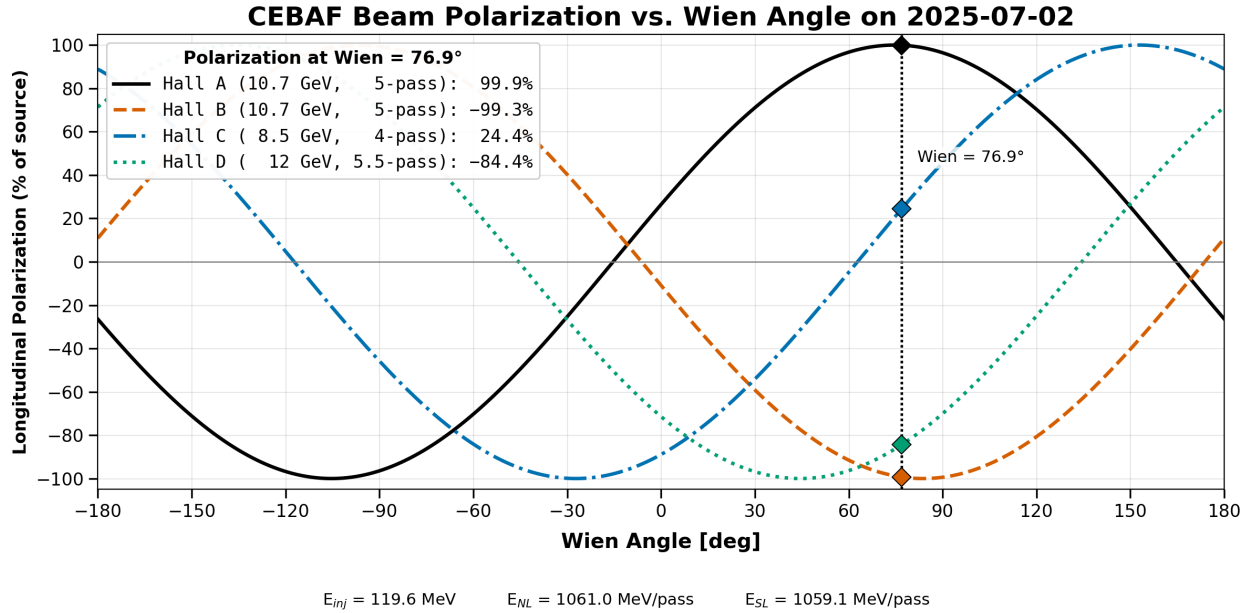


Figure 18: Beam polarization as a function of Wien angle for Halls A, B, C, and D. During this period, only Halls A, B and D required a polarized electron beam.

The result, Figure 18, demonstrates that each hall requires a different Wien angle to achieve fully longitudinal polarization. This difference arises because each hall extracts the beam after a different number of passes through the accelerator, resulting in distinct total bend angles and accumulated spin precession. Consequently, a single Wien setting cannot simultaneously maximize longitudinal polarization for all halls, requiring an operational compromise based on experimental priorities.

The Spin Dance

A “spin dance” is a coordinated calibration procedure used at accelerator facilities like CEBAF to map and optimize the spin direction (polarization) of an electron beam delivered to multiple experimental halls simultaneously [121]. As operators incrementally rotate the beam’s initial spin angle at the source using a Wien filter, the measured polarization at each downstream hall rises and falls in a smooth cosine wave [130]. Because the beam travels through a different number of magnetic bends and at different total energies to reach each hall, these polarization waves are out of phase with one another, creating a synchronized but offset “dance” of data. This procedure is practically essential for multi-hall operations: it allows the accelerator to find the best compromise spin angle so all halls get highly polarized beam, and it provides a direct way to cross-calibrate the different halls’ polarimeters against the exact same beam source. The underlying physics relies on the Thomas-BMT equation, which dictates that an electron’s spin precesses (rotates) relative to its momentum based on a strict fundamental constant of roughly 440.65 MeV, derived from the electron’s mass and magnetic properties. Because this relationship strictly ties the spin rotation to the beam’s bend angle and energy [161], physicists can use the spin dance data to calculate the accelerator’s energy with extreme precision (better than 10^{-4}), making it a powerful complement to traditional magnetic energy measurements. While dedicated spin dances were frequently performed and published during CEBAF’s older 6 GeV era—most notably a massive, multi-hall effort in July 2000—the procedure remains a foundational concept today, even though formal, dedicated spin dance results have not yet been published for the upgraded 12 GeV machine [66, 121, 162].

4.6 Optics Solution for SBS Experiments

Across the SBS program, the magnet optics employed all four quadrupoles and the dipole magnet (Section 5.3). The GEN and GEN-RP experiments read out all eight photomultiplier tubes (PMTs) on the Møller detector, whereas GEp, at higher beam energy, used only the upper four PMTs (Section 5.4). The magnet settings are selected to maximize the simulated Møller coincidence rates while simultaneously minimizing the Levchuk correction to the analyzing power A_{zz} , keeping A_{zz} itself as high and as flat as possible [83]. The correction applied to A_{zz} accounts for the Levchuk effect, which reduces the effective analyzing power due to the intrinsic momentum of bound target electrons [135]. Because the extracted beam polarization is inversely proportional to the effective analyzing power, precise knowledge and minimization of this correction are essential for reducing systematic uncertainties in the final polarization result [163].

Figure 19 shows the simulated analyzing power and coincidence-rate scan for quadrupole Q2 at 4.3 GeV, with the remaining magnet settings listed in Table 1.

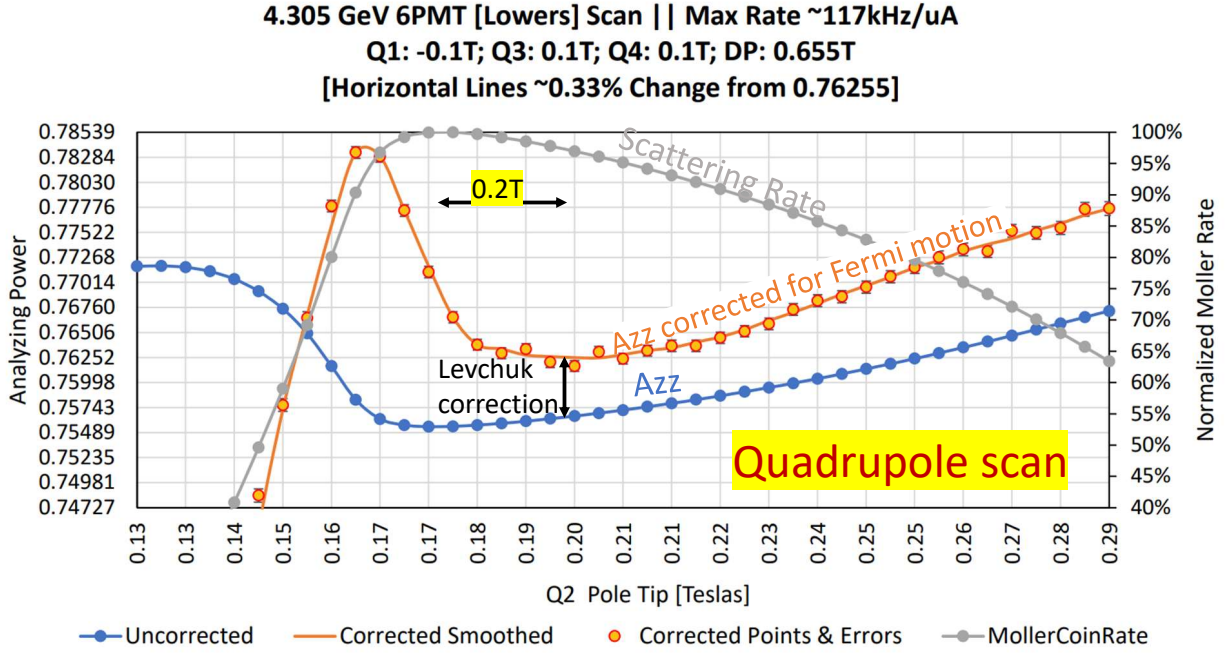


Figure 19: Simulated analyzing power and coincidence-rate scan for quadrupole Q2 at 4.3 GeV [164]. The optimal setting is at $0.2T$, where the corrected analyzing power curve is flat and Levchuk correction has the minimal effect.

Magnet scans are performed at all operational energy levels to determine the optimal working region that provides both a stable analyzing power and a high coincidence rate under realistic measurement conditions. The ideal operating point corresponds to a narrow region where the corrected A_{zz} curve remains relatively flat and high, in the region where the Levchuk correction is smallest, while the simulated Møller coincidence rate is near its maximum. A flat region in the analyzing power curve is particularly important because it reduces sensitivity to small fluctuations in the magnet current or field strength during data taking. Selecting this region minimizes the Levchuk correction and therefore reduces the associated systematic uncertainty in the analyzing power. At the same time, higher coincidence rates improve the statistical precision of the polarization measurement by increasing the number of detected Møller events within a given acquisition time.

Energy (GeV)	Experiments	Q1 (A)	Q2 (A)	Q3 (A)	Q4 (A)	Dip (A)	f_Q	f_{DP}	A_{zz}
2.10	GE _n , GE _n -RP	100.00	23.57	0.00	41.37	120	1.17	1.00	0.75420
4.29	GE _n , GE _n -RP	-41.08	97.33	42.10	41.37	242	1.14	1.00	0.76250
4.36	GE _p	-41.08	97.00	42.10	41.37	214	1.15	1.00	0.75935
6.37	GE _n , GE _n -RP	-183.30	60.50	84.40	93.39	314	1.63	1.05	0.76930
6.48	GE _p	-183.29	58.00	88.66	93.39	313	1.91	1.05	0.77178
8.45	GE _n , GE _n -RP	-183.30	-223.70	186.00	187.40	414	1.25	1.06	0.77390
8.59	GE _p	-183.29	-223.73	200.00	187.39	425	—	1.06	0.77511
10.70	GE _p	-228.60	-275.49	172.00	245.27	471	1.85	—	0.77429

Table 1: Optimized magnet settings for each beam energy used across all SBS experiments. Values in bold red indicate the quadrupole current selected from the optics scan at each energy. f_Q and f_{DP} are scale factors used for scaling max-rate magnet currents to reach lowest Levchuk correction region, for the selected quadrupole and the dipole, respectively.

Table 1 summarizes the optimized magnet settings for each beam energy used across all SBS experiments. For every energy configuration, the corresponding quadrupole and dipole currents were selected based on the rate and analyzing power scans described above, ensuring operation within the region of stable corrected A_{zz} and high Møller coincidence rates. Although some beam energies appear very similar, the values correspond to distinct accelerator configurations. During the SBS program, small adjustments to the CE-BAF energy setup were made to accommodate various experimental requirements and pass configurations. As a result, nominally close energy values represent separate machine tunings, each requiring its own dedicated optics optimization and magnet calibration. Consequently, each entry in Table 1 reflects the specific accelerator conditions under which the corresponding experiment was conducted.

CHAPTER 5

The Hall A Møller Polarimeter — Apparatus and Software

The Hall A Møller polarimeter’s experimental setup consists of hardware and software components that work together to precisely measure electron beam polarization. Additionally, a Monte Carlo simulation software models the behavior of the Møller polarimeter, supporting polarimetry research and helping to reduce systematic uncertainties.

The overall design follows the standard principles of Møller polarimetry, wherein the electron beam’s longitudinal polarization is determined from the spin-dependent asymmetry observed in polarized electron-electron scattering. This measurable asymmetry depends directly on the product of the beam and target polarizations and the analyzing power of the scattering event. Because the extracted beam polarization is inversely proportional to both the target polarization and the analyzing power, a fractional error in the magnetic field, detector acceptance, or kinematic selection propagates almost directly into the beam polarization; the apparatus described in this chapter is therefore designed to fix each of these quantities at the percent level, and the resulting contributions are quantified in the systematic budget of Chapter 7. More detailed information about the Hall A Møller polarimeter and its capabilities is available in [19, 145].

5.1 Møller Polarimeter Overview

General description

The Hall A Møller Polarimeter is a system of devices used for beam polarimetry at Jefferson Lab’s Hall A. How the beam polarization enters the SBS measurements depends on the observable: in the asymmetry-based GEN-II extraction it is a direct multiplicative factor, so its fractional uncertainty propagates into the extracted form factor and sets a working target on the polarimetry at the $\sim 1\%$ level, whereas in the ratio-based GEN-RP and GEp measurements it cancels to leading order and is needed only for the absolute calibration of the individual polarization components and to bound any drift. Maintaining sub-1% polarimetry, and continuously refining it, meets the most demanding of these needs and supports the broader high-precision program at Jefferson Lab [165, 166].

At the Thomas Jefferson National Accelerator Facility (Jefferson Lab), many experiments require knowledge of beam polarization at or below the 1% level, including parity-violation experiments and asymmetry-based form-factor measurements. As discussed in [145] and later performance studies, systematic effects such as target heating, magnetic field alignment, detector acceptance, and background subtraction must be carefully controlled to achieve this precision. Each of these effects is treated as a separate line of the systematic budget in Chapter 7 (Table 16); the remainder of this chapter describes the hardware and procedures that bound them. The Møller polarimeter in Hall A operates intermittently during experimental running to provide independent polarization measurements that are used to calibrate or cross-check other polarimetry systems, such as Compton polarimeters, when available.

Figure 20 is a diagram of the Møller polarimeter. It shows the main components of the

polarimeter. The scale, shape, and size ratios of the components do not reflect the actual polarimeter, and the diagram is intended only for basic visualization.

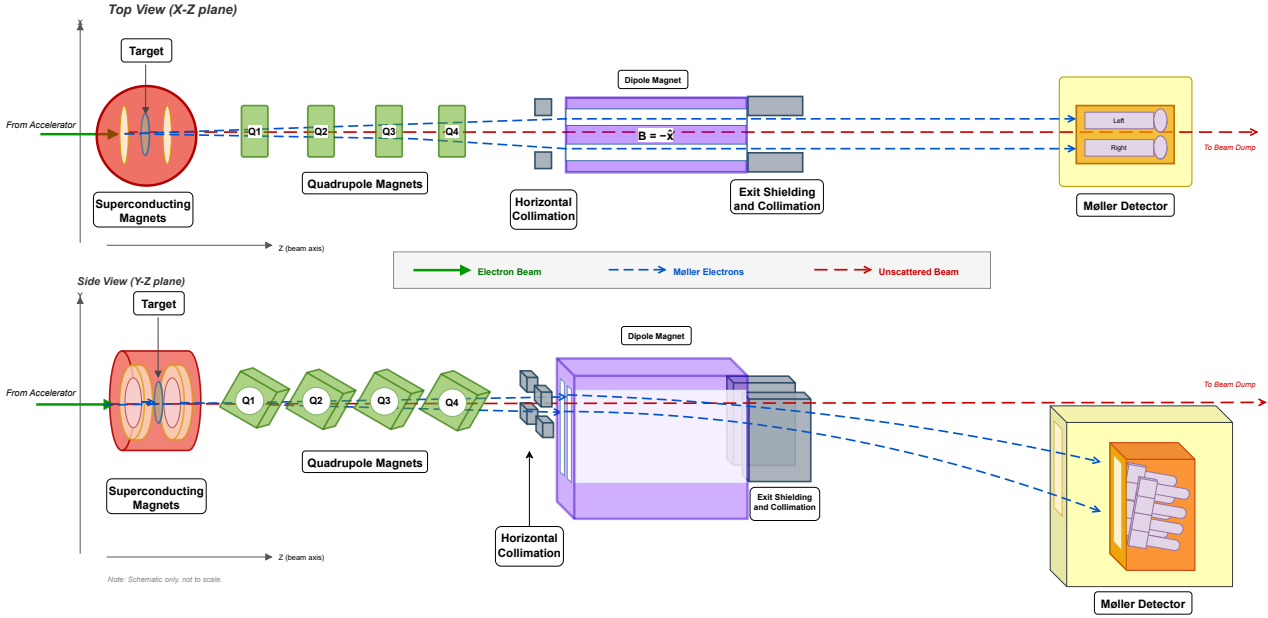


Figure 20: Diagram of the Hall A Møller polarimeter showing the main components. The scale, shape, and size ratios do not reflect the actual polarimeter and are intended only for basic visualization.

The polarimeter sits upstream of the main Hall A spectrometers. During a polarization measurement, the electron beam is aimed at a thin, magnetized ferromagnetic foil target. A magnetic spectrometer system analyzes the momentum of scattered Møller electrons, and the detector registers them in coincidence to suppress single-arm Coulomb/Mott backgrounds. The single-arm Mott cross section exceeds the Møller cross section by orders of magnitude, so without the coincidence requirement the polarization-sensitive signal would be buried; because Møller electrons emerge back-to-back in the center-of-mass frame and reach the two arms simultaneously, requiring a coincidence isolates them from the uncorrelated single-arm background. Accidental coincidences are analytically subtracted, see

Section 6.4.

The primary hardware components of the Møller polarimeter consist of the Møller target, superconducting Helmholtz coils surrounding the target, four quadrupole magnets, a dipole magnet, collimators positioned at the dipole magnet’s entrance and exit, the Møller calorimeter detector, and the data acquisition system.

The software components of the polarimeter include data acquisition software, a data processing and analysis software stack, and a simulation software package built on the CERN GEANT4 platform, which operates separately from the other software and hardware elements.

These systems work together to control the movement of scattered electrons from the target to the detector, ensuring precise acceptance and reducing background noise. Separating the hardware operation from the GEANT4-based simulation framework allows for independent validation of optics settings and systematic studies, all without disrupting live data collection.

The Møller polarimeter’s hardware components measure a helicity-induced asymmetry in the electron beam. Essential requirements include an incident electron beam, a polarized electron target, and a detector to capture scattered electrons. Magnets positioned between the target and detector eliminate unscattered beam electrons, low-energy background electrons, and other unwanted particles.

The magnet configuration is designed to direct symmetric Møller pairs produced at specific center-of-mass angles while suppressing elastic electron-nucleus scattering and other backgrounds. The configuration targets center-of-mass angles near 90° , where the longitudinal analyzing power A_{zz} reaches its maximum of $7/9$ and is locally flat (Section 7.7); selecting this region maximizes the analyzing power while minimizing its sensitivity to ac-

ceptance and to the Levchuk effect. Consequently, the optics solution directly determines the effective analyzing power, which in turn sets the analyzing-power contribution to the systematic budget.

5.2 The Polarized Target System

Møller Target

Jefferson Lab conducts fixed-target particle physics experiments. In the case of the Møller polarimeter, there are four target foils: three ferromagnetic foils used for polarization measurements and one weakly diamagnetic foil. During the SBS experiments, the three ferromagnetic foils are 10- μm , 10- μm , and 4- μm iron (Fe) foils [167, 168]. Because the target enters the polarization extraction as P_{target} , the choice of foil material and thickness directly sets one of the largest systematic contributions; iron is used because its polarization can be computed from first principles rather than measured run-by-run.

Iron's unpaired electrons in its 3d shell experience parallel spin alignment because the exchange interaction energetically favors this configuration, resulting in iron's ferromagnetic behavior, which is what allows the electron polarization to be obtained from the known saturation magnetization rather than from a separate per-run measurement. Under a strong magnetic field, the foil magnetizes and the spins of the polarized electron fraction align parallel or antiparallel to the beam direction. This property makes iron an effective target for double-polarized Møller scattering, enabling the measurement of asymmetries between the two spin orientations [129].

The effective target electron polarization is derived from the saturation magnetization after removing the orbital contribution to the iron moment, accounting for the temperature

dependence of the saturation magnetization and the high-field forced-magnetization slope, and dividing by the number of electrons in iron [129]. Only the two 3d electrons per iron atom are polarized, giving a target electron polarization of order 8%; because this value is fixed by the saturation magnetization to better than half a percent, it becomes the dominant and largely irreducible systematic of the measurement (the target-polarization term in Table 16). For thin iron foils exposed to high magnetic fields, the foil is driven to full magnetic saturation, which fixes the target electron polarization at its known saturation value. Using multiple foil thicknesses helps optimize the balance between statistical accuracy and target heating effects at different beam currents. A thinner foil deposits less beam power and therefore suffers less heating-induced demagnetization, at the cost of a lower coincidence rate; the 4- μm foil is used at the highest beam currents, where heating would otherwise pull the magnetization below saturation, while the thicker foils recover statistics at lower currents.

The remaining target is an 11- μm copper foil. Copper is weakly diamagnetic, meaning its electrons do not align with the magnetic field and exhibit negligible net electron spin polarization, so it produces no helicity-correlated asymmetry. When the electron beam passes through this copper foil, there is no helicity-induced asymmetry, making it useful for measuring false asymmetries caused by instrumental effects that are unrelated to beam polarization. Measurements on the copper foil provide an important systematic cross-check, particularly for helicity-correlated beam properties such as position, angle, or intensity variations. Because any asymmetry measured on the unpolarized copper target is by construction instrumental, these measurements set the null-asymmetry contribution to the systematic budget.



Figure 21: Image of the Møller target ladder [169].

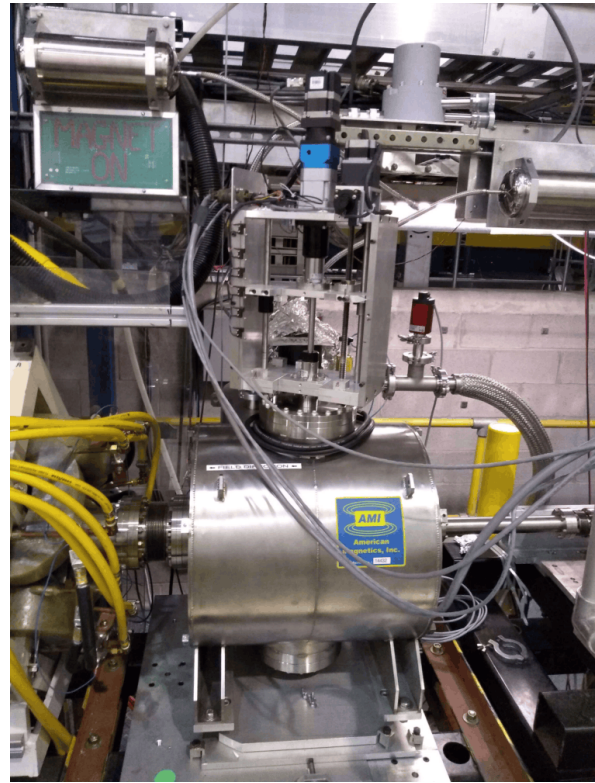


Figure 22: Image of the Møller superconducting target magnet system [170].

Superconducting Magnet

There are two approaches to polarize the target. The first, commonly called the classic technique, employs a relatively weak external magnetic field and does not reach full magnetization saturation. The second, known as the “brute force” method, uses a strong external magnetic field to fully magnetize the target [171, 172]. The brute-force approach is adopted because full saturation pins the magnetization at its known value M_{sat} , removing the dependence on a field calibration that the unsaturated classic method must carry; this is what converts the target polarization from a measured quantity into a calculable constant.

To bring the Møller ferromagnetic target to full polarization saturation, the target is placed

perpendicular to a strong magnetic field of approximately 4 T generated by a superconducting Helmholtz magnet. The magnet coil windings must be held below their superconducting critical temperature so they can carry the large persistent current needed for a multi-tesla field without resistive heating [83]. The Helmholtz configuration is selected to ensure a uniform magnetic field across the target foil's spatial extent, reducing gradients that could cause incomplete or uneven magnetization. Because the beam polarization varies inversely with the target polarization, any non-uniformity that left part of the beam-sampled foil below saturation would bias the result by the same fraction; the Helmholtz geometry holds the field uniform across the foil so that the entire illuminated region remains at M_{sat} .

This magnet requires a training procedure after a thermal cycle to room temperature or after a quench. Section 6.1 explains this training process. While in use, the magnet demands a high current, and a key operational constraint is the time needed to ramp up to the target magnetic field, which can take about an hour. Careful monitoring of cryogenic conditions and current stability is required during ramping to prevent quenches and to maintain safe operation. This ramp time, together with the invasiveness of the measurement to ongoing Hall A running, is part of why direct Møller measurements are performed only periodically, and why some production periods rely on polarization inferred from neighboring measurements (Chapter 7).

5.3 The Magnetic Spectrometer

Figure 23 shows an ideal dipole magnetic field and an ideal quadrupole magnetic field.

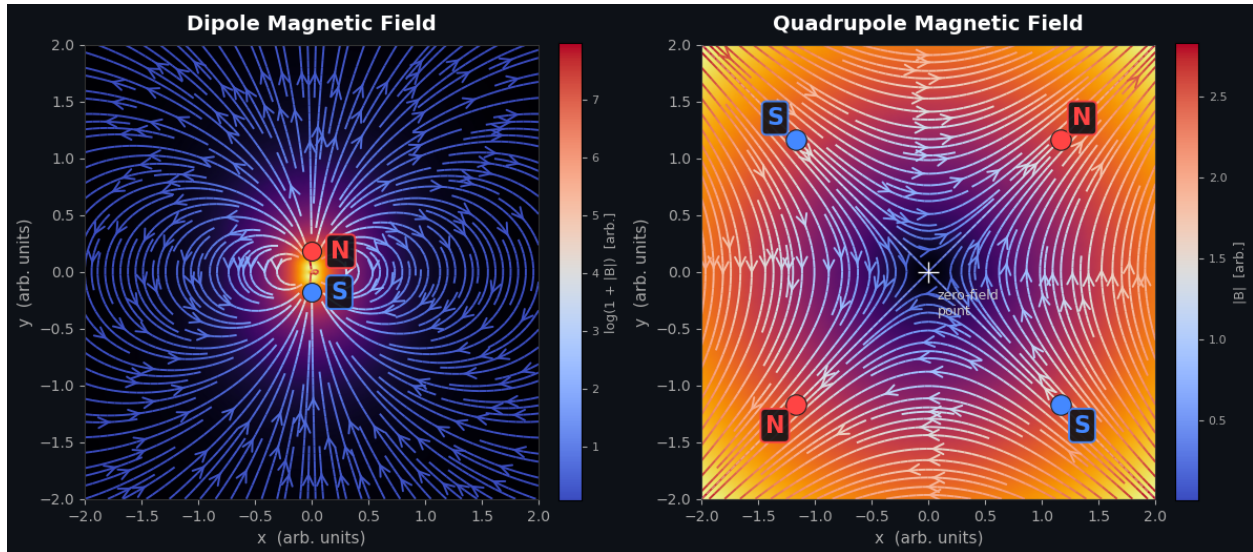


Figure 23: Vector diagrams of ideal dipole and quadrupole magnetic fields.

Quadrupoles

Four quadrupole magnets, each capable of producing a magnetic field of approximately 0.6 T at the pole tip, guide the Møller electrons toward the dipole magnet [83]. These quadrupoles can focus or defocus the electron trajectories horizontally or vertically, enabling the scattered electrons to be moved away from the beam axis and directed toward the dipole.

The specific currents required for each quadrupole magnet, together with the dipole current, define the optics solution for a given experiment. Large-scale simulations determine the optimal magnetic field settings before commissioning.

Because quadrupoles focus in one transverse plane while defocusing in the orthogonal plane, their relative strengths and positions must be carefully tuned to maintain symmetric transport of the two Møller electrons. The quadrupole settings fix which center-of-mass scattering angles are transported to the detector, and therefore set the effective analyzing power averaged over the acceptance; symmetric transport keeps that acceptance centered

on the flat region of A_{zz} , which limits both the optics systematic and the Levchuk sensitivity. Small deviations in current can shift the acceptance and introduce systematic effects in the measured asymmetry. Additional studies during the polarization measurement window improve the optical model and reduce the optics-related contributions to the systematic budget.

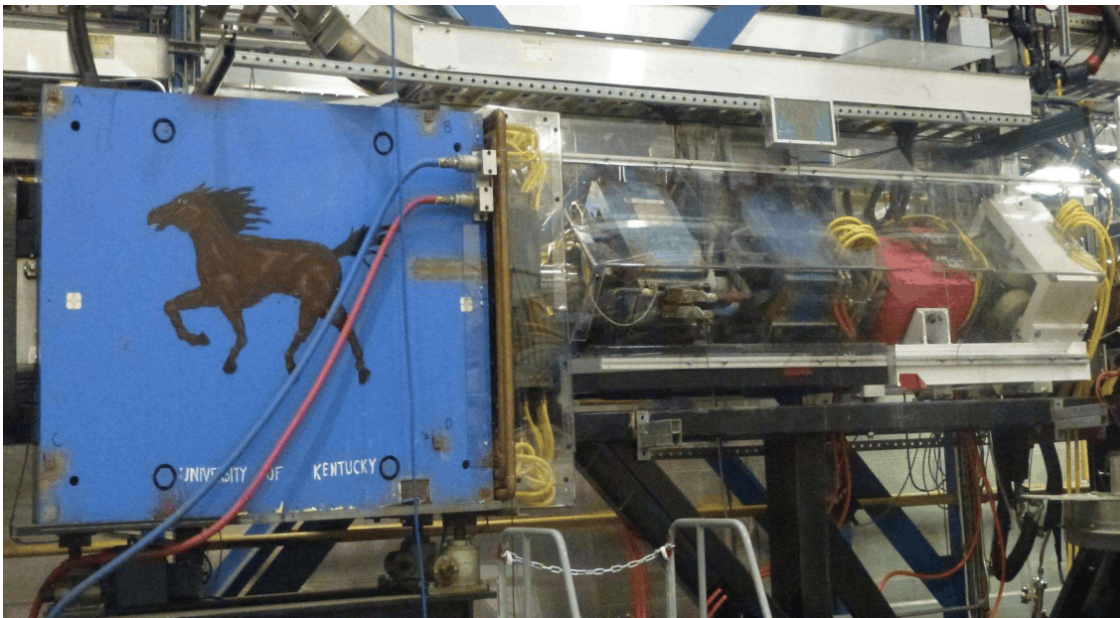


Figure 24: Image showing all Møller quadrupoles and the Møller dipole magnet inside Hall A [170].

Dipole

Møller electrons passing through the horizontal collimator enter the dipole vacuum chamber without a beam pipe [173]. These electrons then encounter a magnetic field that deflects their paths downward. The dipole exit windows are sealed with a 100- μm -thick titanium window. Electrons pass through this window and then travel through air toward the Møller detector [83]. Both the dipole magnetic field strength and the beam energy influence how much each electron bends. The strength of the dipole magnetic field and the current are determined by analyzing simulated data. Like with quadrupoles, specialized

studies during the measurement phase verify the best settings and help minimize systematic uncertainties. The dipole also enables momentum selection by spatially separating electrons based on their bending radius. This separation helps reduce low-energy background electrons and determines the detector system’s geometric acceptance. An error in the dipole field map would shift the reconstructed bend angle, and hence the accepted center-of-mass angle, moving the effective A_{zz} ; the field map is therefore characterized carefully so that this effect stays within the optics systematic.

Collimators

The Møller polarimeter’s collimation system features an adjustable horizontal collimator at the dipole’s entrance and lead shielding at the dipole box’s exit. Extra lead shielding surrounds the Møller detector and other areas to minimize background noise and gamma radiation from particle interactions with surrounding materials.

The horizontal collimator defines the geometric acceptance of Møller electrons entering the dipole. Its aperture controls the accepted azimuthal angular range, with narrower settings reducing backgrounds at the cost of counting rate and wider settings doing the reverse. The aperture is therefore set to keep the azimuthal acceptance symmetric, which bounds the Levchuk contribution to the analyzing power.

The lead shielding at the dipole exit and around the detector reduces secondary electrons and bremsstrahlung photons produced when primary electrons interact with vacuum windows, air, or structural materials. This shielding is especially important because electromagnetic showers can generate low-energy particles that imitate valid detector signals if not properly blocked.

5.4 The Møller Detector System

The Møller detector is a spaghetti calorimeter made of lead and scintillating fibers [174], along with an aperture detector. The whole system is housed in a lead-shielding enclosure. The calorimeter has two arms, each fitted with four vertically aligned scintillating fibers and photomultiplier tubes (PMTs) at the back to detect signals. The two-arm geometry exists to enforce the coincidence that defines a Møller event: a genuine pair deposits energy in both arms simultaneously, whereas single-arm backgrounds do not, so the calorimeter's segmentation is what makes the background rejection physical rather than statistical.

Incident electrons interact with the lead to produce an electromagnetic shower. As charged particles pass through the scintillating fibers, they excite the material, generating photons. These photons travel along the fibers to the photomultiplier tubes, where they release electrons at the photocathode through the photoelectric effect. The electrons are amplified along the dynode chain by successive secondary-electron emission. The total multiplication factor is the PMT gain. Since the system operates over a much larger dynamic range than a single photoelectron signal, the PMTs need to be set to appropriate high-voltage levels. Each photomultiplier tube is carefully characterized and calibrated prior to use. During gain matching, the high voltage for each PMT is adjusted so that all channels respond similarly to the same deposited energy. This simplifies data analysis and improves the effectiveness of asymmetry extraction. If the two arms or the two helicity states responded with different effective gain, that difference would appear as a beam-independent asymmetry, so gain matching suppresses an instrumental effect that is otherwise bounded only by the copper-foil null measurement. The output from each PMT is an analog signal.

The two-arm setup enables the coincidence detection of Møller electron pairs, greatly reducing background noise from single-particle events. Additional lead shielding helps min-

imize interference from ambient radiation and secondary particles generated upstream. Regular calibration and monitoring of detector response keep helicity-correlated electronic effects — the same effects bounded by the null-asymmetry systematic — from skewing the measured asymmetry.

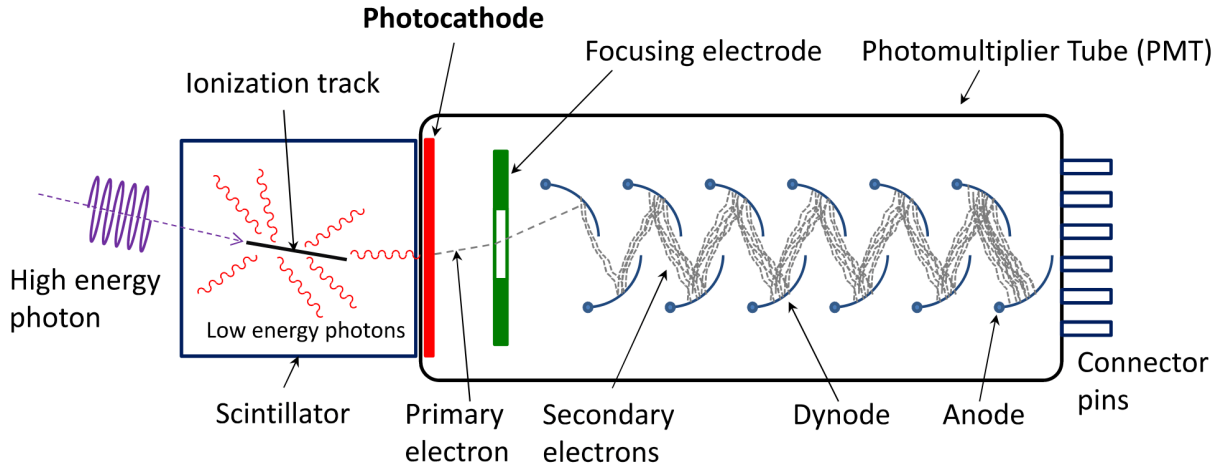


Figure 25: Diagram illustrating a high-energy photon striking a scintillating fiber, causing it to emit low-energy photons that are subsequently converted into photoelectrons and amplified within the photomultiplier tube (PMT) [175].

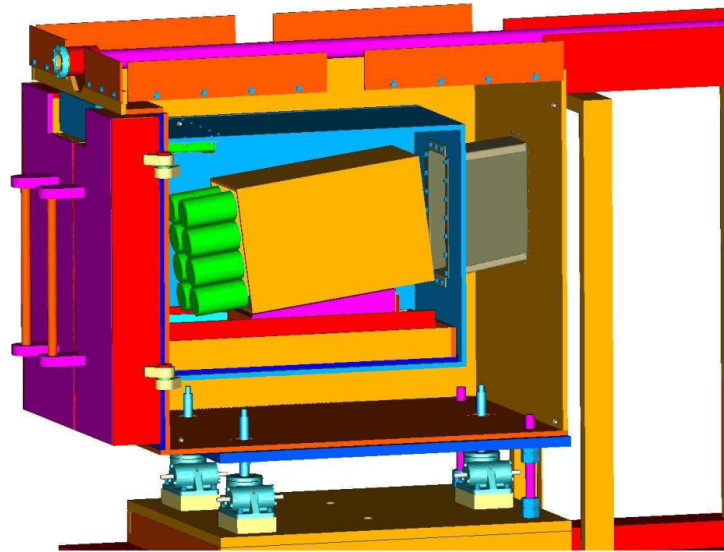


Figure 26: Møller detector inside the shielding box.

5.5 Data Acquisition and Software

The data acquisition (DAQ) hardware system consists of multiple electronic crates responsible for different tasks, including triggers, timing discrimination, coincidence logic, data handling, the high-voltage system, helicity-flipping trigger, scaler readout, analog-to-digital converters (ADCs), time-to-digital converters (TDCs), and Programmable Logic Unit (PLU) boards [176]. The DAQ converts analog PMT signals to digital form and transmits the right singles rate, left singles rate, and coincidence rate to storage for analysis. The timing discriminator enables setting a threshold below which signals are ignored, helping to remove electronic noise and low-energy background [83]. The helicity-flipping trigger sorts events based on the beam's helicity pattern. Section 2.3 details the electron beam's helicity pattern at Jefferson Lab. The PLUs classify event types such as left-arm singles, right-arm singles, and true coincidences. The scaler readout tracks how many events occur within a specific time window. The ADC modules digitize the integrated charge of each PMT pulse, providing information proportional to the deposited energy, while the TDC modules measure the relative timing between signals from the two detector arms. The coincidence timing window sets the accidental-coincidence rate that must later be subtracted (Section 6.4), and the discriminator threshold sets the trigger deadtime; both feed dedicated lines of the systematic budget, so their settings are fixed and logged for each run.

CODA Software

The CODA software handles data collection, event building, and storage before passing data to processing and analysis programs.

A run is a distinct period during which data are collected under stable conditions and is regarded as a single data set point. Runs can last from a few minutes to over an hour. In a complete Møller measurement, typically 20 to 50 runs are recorded. Each run receives

a unique number, is labeled, and stored. For the Møller polarimeter, data are saved in binary format. Different measurement types require different run configurations, including runs with and without the LED pulser. Runs with the LED on and the beam off are used to test the DAQ system and detector response, since the LED frequency is known and stable. These tests can be performed independently of accelerator operations and are useful for tuning thresholds, verifying timing alignment, and checking channel stability.

Because CODA interfaces directly with hardware triggers and readout modules, it guarantees synchronization among detector signals, helicity information, and beam-related scalers. A mismatch between the helicity assignment and the recorded detector data would either flip the sign of the extracted asymmetry or average it toward zero, so run configuration is verified before each measurement.

High Voltage (HV) Control & PLU Control software

Another part of the software stack manages the high-voltage supplies connected to the detector PMTs. This program allows remote control and monitoring of individual high-voltage channels.

Before operation, the PMTs undergo a gain-matching procedure because their gain depends on applied voltage and can drift over time due to aging or changes in operating conditions. Gain matching often requires applying different voltages to individual PMTs to achieve uniform response across all channels.

There is also software specifically for managing PLU configurations, which define the trigger logic for each channel. Part of the DAQ software's role includes programming hardware, like setting discriminator thresholds and configuring PLU logic.

Gain drift that correlates with the helicity flip would mimic a physics asymmetry, so high-

voltage stability is monitored continuously. Automated logging of the HV settings and trigger configurations ensures traceability, enables reproducibility of specific measurement conditions during later analysis, and supports the current-dependence and slit-dependence checks that bound this class of effect (the corresponding terms in Table 16).

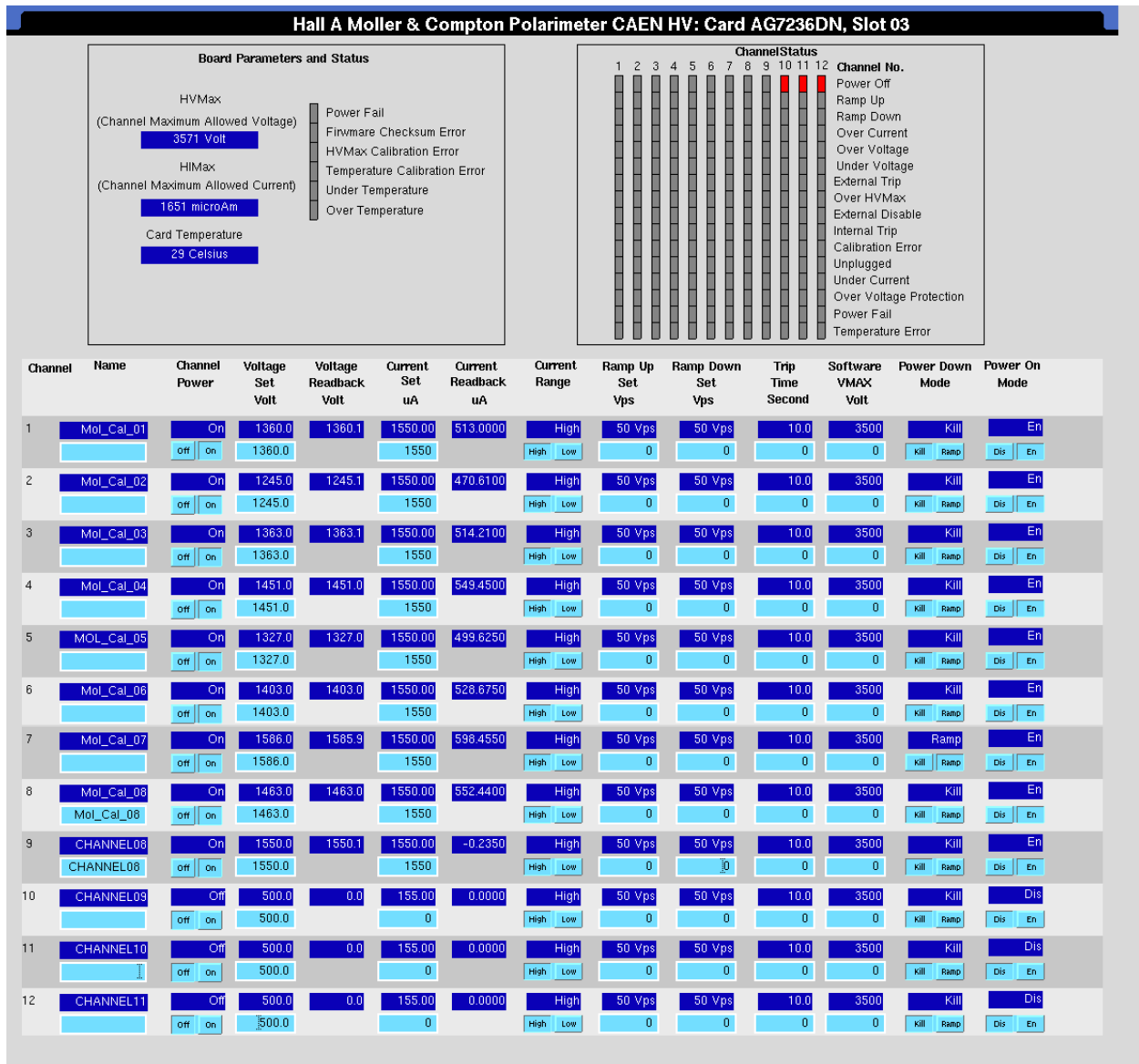


Figure 27: High-voltage control panel for the Møller detector photomultiplier tubes.

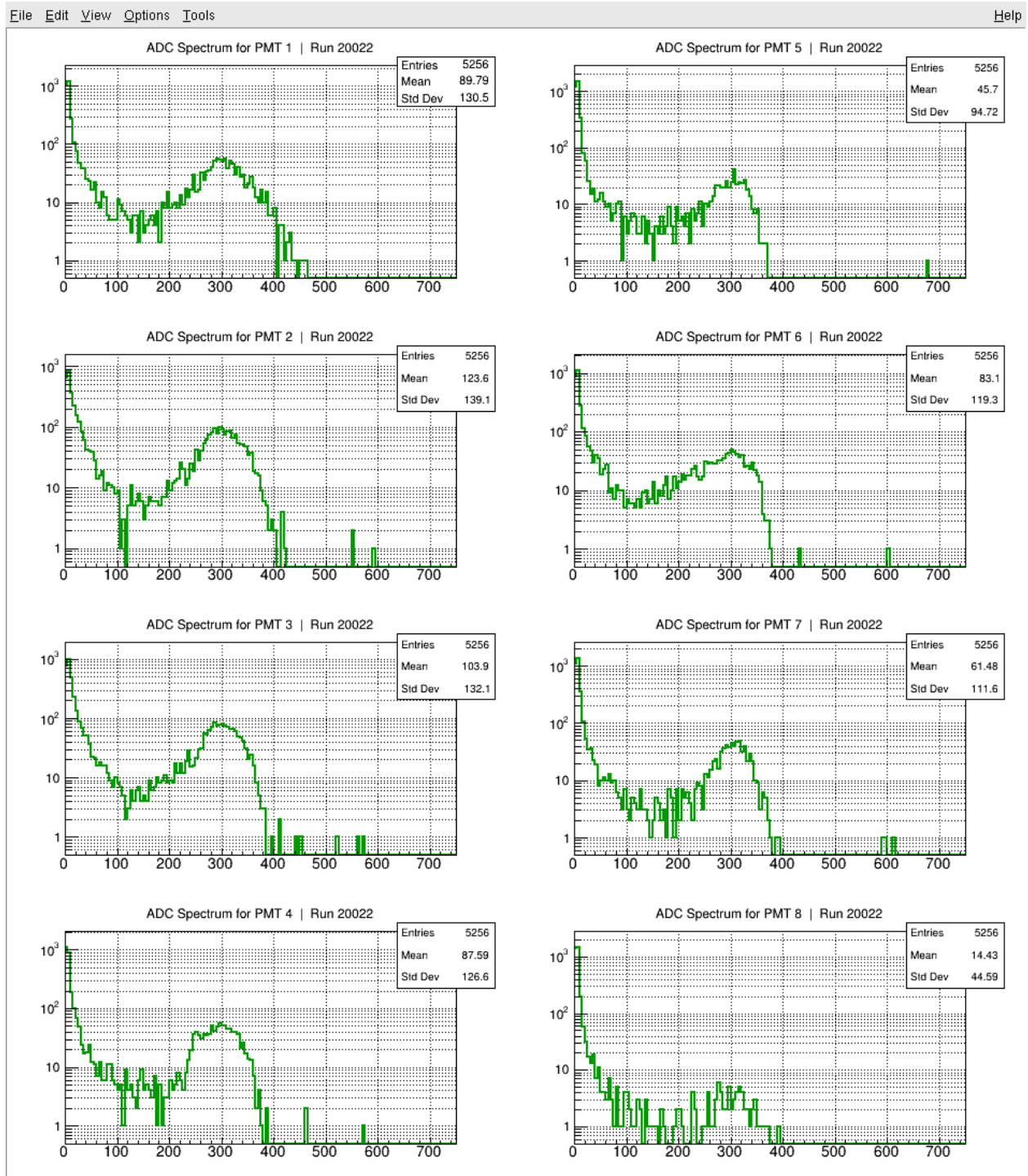


Figure 28: ADC readouts for the gain-matched photomultiplier tube channels.

Data Processing and Analysis Software

Once the data are collected according to the configured trigger logic, they are stored in binary format and must be converted into a structured format, such as CERN ROOT files. This conversion transforms raw electronic information into quantities that are more closely related to physical observables.

The primary Møller analysis script uses the recorded data files, along with additional inputs such as beam current monitor charge pedestal corrections, simulated analyzing power, and the Møller target polarization. The script outputs the left-arm singles rate, right-arm singles rate, coincidence rate, accidental rate, coincidence asymmetry, and charge asymmetry. From these quantities, the electron beam polarization is calculated using the known relationship between measured asymmetry, target polarization, beam polarization, and analyzing power (Equation 4.19). This step is where every element described in this chapter enters a single number: the target polarization from Section 5.2, the effective analyzing power from the simulation, and the measured asymmetry from the detector and DAQ combine through Equation 4.19 to yield the beam polarization.

The analysis also includes corrections for accidental coincidences, dead time, and beam charge normalization. Statistical uncertainties are propagated from counting statistics detailed in Section 7.1, while systematic contributions are evaluated separately. Each of these corrections appears as a named line in the systematic budget of Chapter 7 — accidentals, deadtime, and charge normalization — so the procedures described here map directly onto Table 16.

A secondary analysis script groups together runs of a similar type and purpose that occur within a short time span, ensuring that time-dependent beam variations are negligible. Combining these runs increases the effective sample size and reduces statistical uncer-

tainty. Grouping runs in this way is also what justifies the inverse-variance combination and the minimum-sample requirements used in Section 7.1. Additional cuts, such as removing beam trips or current discontinuities, are applied to produce a cleaner data set.

The outputs of these scripts are stored in a database for future reference, eliminating the need to reprocess raw data for each study. Storing processed results alongside configuration metadata enables consistent cross-checks and long-term monitoring of polarimeter performance over multiple experimental campaigns.

Møller Polarimeter Simulation Software

The Møller polarimeter simulation software is built on the CERN GEANT4 platform. GEANT4 is a toolkit for simulating particle interactions with matter using Monte Carlo methods. It includes detailed models of electromagnetic processes across a wide energy range and allows extraction of simulated detector observables.

The Møller Polarimeter Simulation package, often referred to as MolPol, incorporates all hardware components of the polarimeter at realistic geometric scales. It includes specialized event generators, such as a Møller scattering generator, which produces scattering events at the target based on the probability of interaction between the incident beam electron and polarized atomic electrons within a thin iron foil.

The generator accounts for the kinematics of electron-electron scattering and assigns appropriate weights corresponding to the spin-dependent cross section, enabling direct calculation of the effective analyzing power within the accepted phase space. The simulation provides capabilities that are impractical during live beam operation. Because Møller measurements are invasive to ongoing Hall A experiments and limited in available beam time, systematic studies cannot be performed indefinitely under real conditions. The simulation allows controlled variation of parameters such as magnetic field strengths, collimator aper-

tures, target thickness, and beam position. The output is written in the CERN ROOT format, which simplifies comparison with experimental data. In its current form, the simulation includes virtual detectors positioned before and after major hardware components, enabling detailed studies of particle transport and acceptance effects.

These virtual detectors make it possible to isolate the contribution of individual elements, such as multiple scattering in the target or the titanium dipole exit windows, or acceptance clipping at the collimator, to the overall detector response. The simulation's central output is the effective analyzing power $\langle A_{zz} \rangle$ within the real acceptance, together with its uncertainty — a quantity that cannot be measured directly — which enters Equation 4.19 and sets the analyzing-power line of the systematic budget.

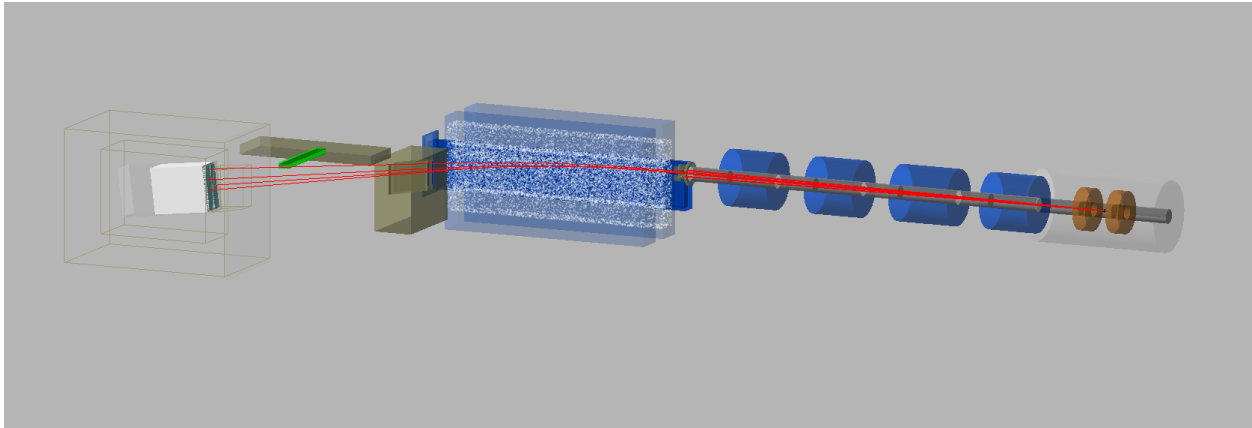


Figure 29: The Hall A Møller polarimeter in the MolPol simulation including Møller events passing through the spectrometer.

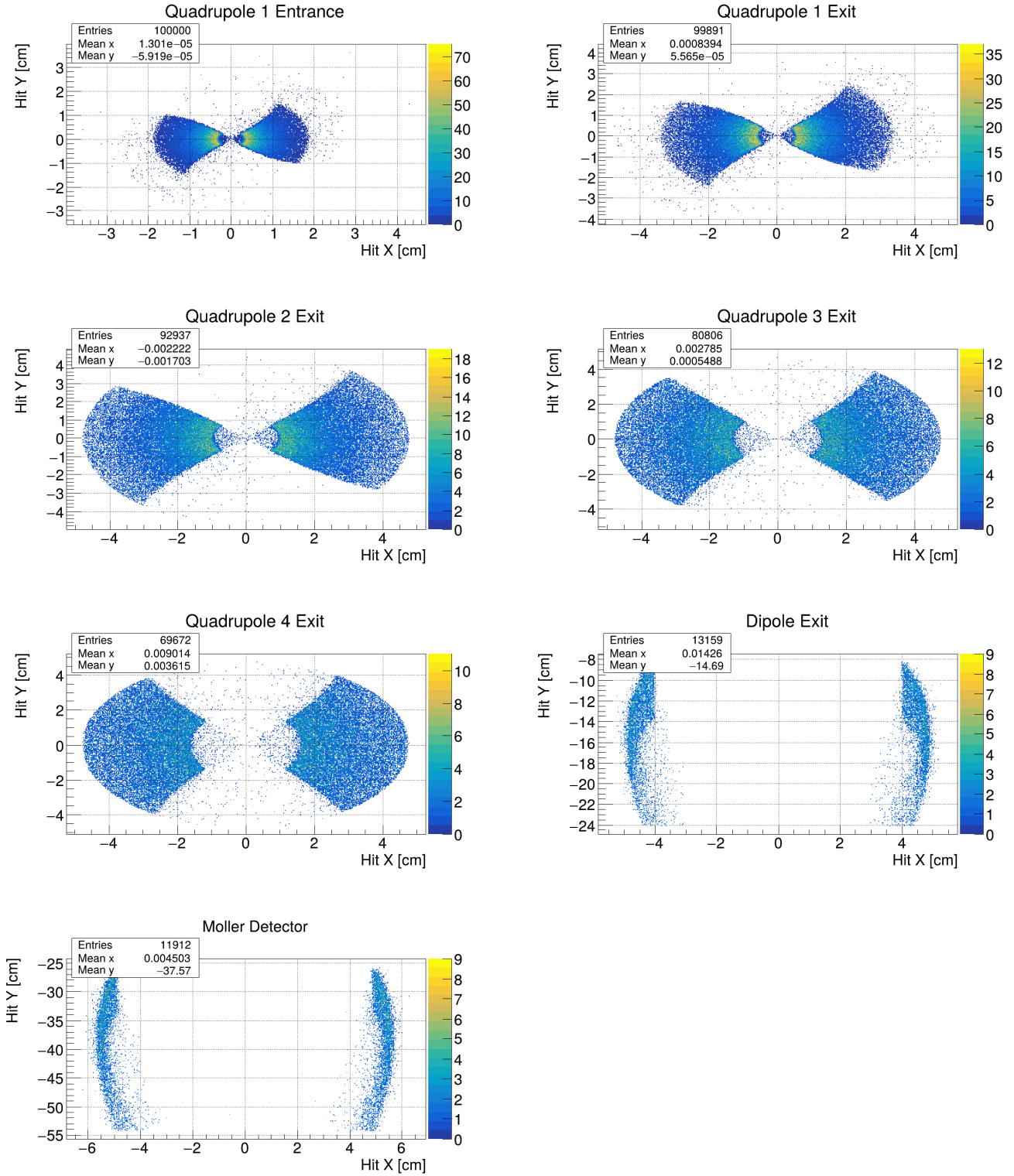


Figure 30: Hit x versus y at several detector planes throughout the spectrometer at 4.2 GeV SBS optics, illustrating the focusing, defocusing, and steering of Møller electrons onto the Møller detector.

CHAPTER 6

Data Analysis and Corrections

6.1 Measurement Protocol and Cadence

Pre-measurement procedure

The Møller measurement team completes a checklist the day before the measurement to ensure that all polarimeter components are functioning correctly. The checklist includes checking the target motion, the superconducting holding-field magnet power supply, and the cryogenic cooling system for the Møller target, as well as the detector high voltages and the CODA software, which is tested by running an LED test. The status of each system is recorded and documented.

If the superconducting Helmholtz magnet has been inactive for an extended period, it must undergo a training procedure prior to use. This process involves incrementally ramping the magnet through both positive and negative currents, allowing the superconducting coils to stabilize at each step. This cycle is repeated until the magnet consistently reaches the target field of approximately 4 T without quenching. Because training can take several hours, it should be completed well in advance of the scheduled measurement window to prevent operational delays.

This pre-measurement check is crucial because the Møller measurement window is usually limited in duration, and hardware or software issues can cause significant delays in beam

delivery. Detecting issues early, like unstable cryogenic conditions, high-voltage trips, or trigger misconfiguration, helps prevent wasting valuable accelerator time. The team records all checklist results to ensure traceability and to provide a reference for troubleshooting if anomalies appear in the data.

Møller measurement

At the start of the measurement, the protocol begins by instructing the Machine Control Center (MCC) to initiate the Møller measurement process. MCC requests the pre-calculated optimal optics settings that maximize the coincidence rate while minimizing systematic uncertainties such as the Levchuk effect [135]. As MCC fine-tunes the accelerator magnets and monitoring systems to produce a low-current beam for the hall, they simultaneously power and calibrate the Møller polarimeter magnets. This process can take over an hour. Many of the settings used during the Møller measurement differ from those in the main physics experiment. For instance, the beam raster patterns needed for high-current operation are usually disabled or adjusted during Møller measurements to maintain controlled beam conditions at the thin iron foil. Meanwhile, the Møller measurement team prepares the monitoring systems to ensure stable operating conditions. Later, MCC sends a message to ramp up the superconducting Helmholtz coils of the Møller target to approximately 4 T, a process that takes roughly one hour. During this ramp-up period, other subsystems are prepared: detector high voltages are activated, target motion and position are verified, magnet monitoring interfaces are launched, CODA and PLU control software are tested, and predefined run configurations are loaded.

Once MCC provides a stable, low-current beam, a harp scan is performed to measure the beam size and position within the beam pipe. This step prevents possible damage to the thin Møller target and ensures proper steering downstream. When the beam position and size are confirmed to be within acceptable limits, the detector PMT gain-matching pro-

cedure begins. Different high voltages are applied to each PMT, and the corresponding ADC signals, which are proportional to gain, are monitored. When all PMTs reach comparable gain values, the configuration is logged for future reference. At this stage, the Møller polarimeter is fully operational, and polarization measurements can begin. During the measurement, the measurement team requests that MCC interrupt the beam for a bleedthrough run. This run assesses the amount of unintended beam current entering the hall from other experimental setups at Jefferson Lab. If the leakage exceeds acceptable levels, MCC can adjust upstream slits to decrease it. Polarization measurements are then performed for each insertable Half-Wave Plate (iHWP) configuration. Copper foil measurements are also conducted for each HWP setting to detect false asymmetries unrelated to beam polarization. Collecting data at both HWP states offers an internal consistency check, because reversing the HWP changes the sign of the measured physics asymmetry while many instrumental asymmetries stay the same. Therefore, agreement between opposite HWP states increases confidence in the extracted beam polarization.

The crew may also conduct systematic studies to better constrain uncertainties. Magnet rate scans, as discussed in Section 6.7, are performed as needed to verify optimal optics settings and assess the impact of small current deviations on performance. Other systematic checks may include rotating the Møller target to study polarization alignment, examining how the direction of the holding magnetic field affects magnetization saturation, and evaluating correlations between beam orbit parameters and measured asymmetry. These controlled variations allow estimation of systematic uncertainties associated with optics tuning, target alignment, and magnetic field stability, which are later incorporated into the total polarization uncertainty budget.

After the measurement

After the measurement is completed, all beamline and polarimeter settings are restored to the main experiment's default configuration. The crew logs all changes before MCC reconfigures accelerator settings. The collected data, along with detailed electronic log entries, including a shift summary and run list, are entered into a database. Once the recorded information is reviewed and validated by the polarimeter team, primary and secondary analyses are performed. Final polarization results, along with associated statistical and systematic uncertainties, are archived and communicated to the main experimental collaboration for use in physics analyses.

Frequency of Measurements

The frequency of Møller measurements is determined by the scientists leading the primary physics experiment in the hall. Møller polarimetry is invasive because it requires beam conditions and magnet configurations that differ from those of the main experiment, and the polarimeter is installed directly on the beamline. The measurement process can require several hours under stable conditions. This creates a practical balance between minimizing interruptions to data collection and ensuring accurate knowledge of the beam polarization.

For experiments such as those conducted by the SBS collaboration, where beam polarization uncertainty does not dominate the total physics uncertainty, measurements may be performed approximately once per month or whenever significant beam parameters, such as the Wien filter angle or beam energy, change. More frequent measurements might be necessary if beam conditions are unstable or if the physics asymmetry is very sensitive to polarization uncertainty. Conversely, if polarization remains stable over time, the measurement interval can be lengthened.

It is also advisable to perform polarization measurements at the beginning and end of an

experimental “season” to reduce systematic uncertainty associated with temporal extrapolation of the beam polarization. Since physics data are collected continuously while direct polarization measurements are periodic, interpolation between measurement points is necessary. Performing measurements at well-defined time boundaries improves confidence in this interpolation and reduces long-term systematic uncertainty.

6.2 Charge Normalization and Pedestal Subtraction

The goal of Møller polarimetry is to determine the beam polarization, which is not measured directly but extracted from the measured scattering asymmetry using the standard Møller polarimetry formalism. The polarimeter directly measures the helicity-dependent coincidence asymmetry of scattered electrons, expressed as

$$A_{\text{raw}} = \frac{c_+ - c_-}{c_+ + c_-} \quad (6.1)$$

where c_+ and c_- are the recorded coincidence counts for the two electron beam helicity states.

One of the primary corrections accounts for differences in beam charge. The beam charge is measured by Beam Current Monitors (BCMs), and the average charge for each helicity state is denoted as Q_+ and Q_- . The charge asymmetry is defined as

$$A_Q = \frac{Q_+ - Q_-}{Q_+ + Q_-} \quad (6.2)$$

Since BCM signals are scaled electronic readouts rather than absolute charges, each helicity state needs normalization accordingly. Furthermore, the BCM output remains non-zero even when there is no beam current. This baseline, known as the BCM charge pedestal,

must be subtracted from the raw data. Consequently, the charge for each helicity state, after pedestal correction, is expressed as

$$Q_{\pm} = \bar{Q}(1 \pm A_Q) - P \quad (6.3)$$

where $\bar{Q}$ is the average beam charge and P is the measured pedestal. Figure 31 shows the BCM charge pedestals for each Møller measurement date.

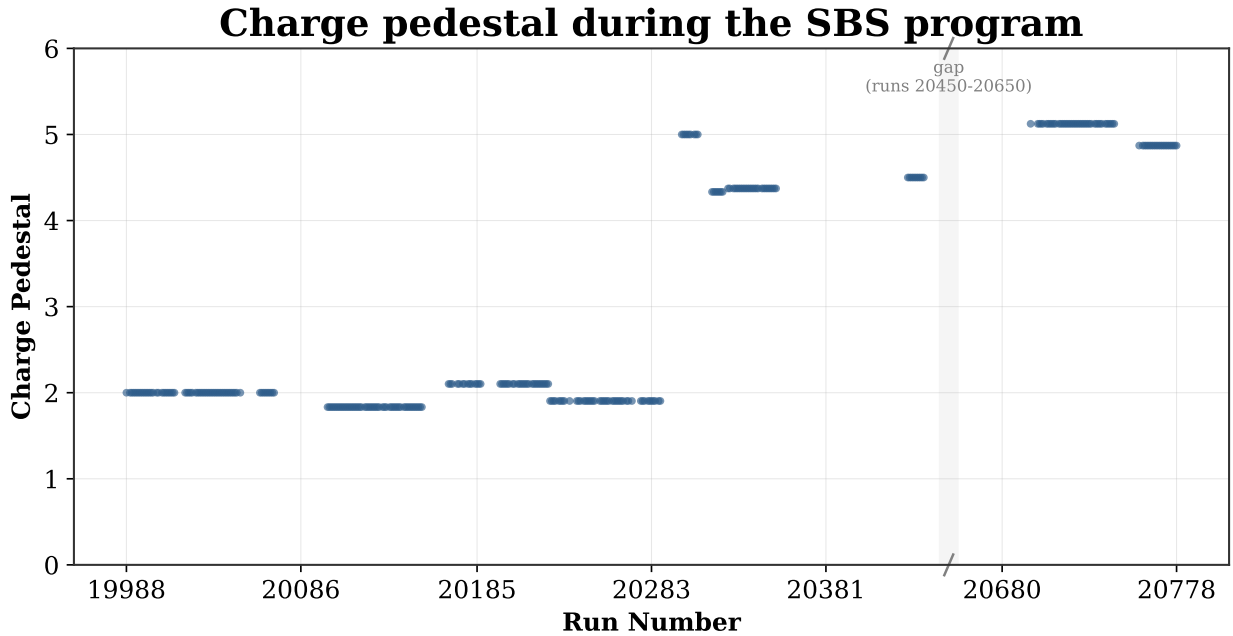


Figure 31: BCM charge pedestals for each Møller run during SBS.

Helicity-correlated charge asymmetries can generate false physics asymmetries if not corrected, since the measured coincidence rate scales linearly with beam current. Similar charge normalization procedures are discussed in [18], where helicity-correlated beam properties are tightly controlled.

An estimate of the size of this correction to the measured asymmetry arising from charge normalization and pedestal subtraction is given by the following expression, as discussed in

Eric King's thesis [83]:

$$\frac{A_{QP\text{corr}} - A_{\text{raw}}}{A_{\text{raw}}} \approx -A_Q \left(1 + \frac{P}{\bar{Q}}\right) A_{\text{raw}}^{-1} \quad (6.4)$$

where A_Q is the charge asymmetry, P is the charge pedestal value, and $\bar{Q}$ is the average charge. As with the other corrections, we apply it at the per-pattern level and report both per-run and group-pooled values; statistical errors are obtained from the SEM prescription of Sec. 7.1. The systematic uncertainty Δ_{sys} for the per-run numerical values is assigned as 20% of the correction magnitude. The corrections and their uncertainties are shown in Figure 32.

Notes for 2022-10-15.

During this date, a malfunction/mismatch in the BCM readout rendered the reported beam charge value unreliable. Consequently, the charge normalization and pedestal subtraction correction has been set to zero, and the $(1 + f_{\text{qped}}^{(i)})$ factor in $A_{\text{corr}}^{(i)}$ is replaced by unity for runs on this date. The methodology and assigned systematic uncertainty are discussed in Section 7.2.

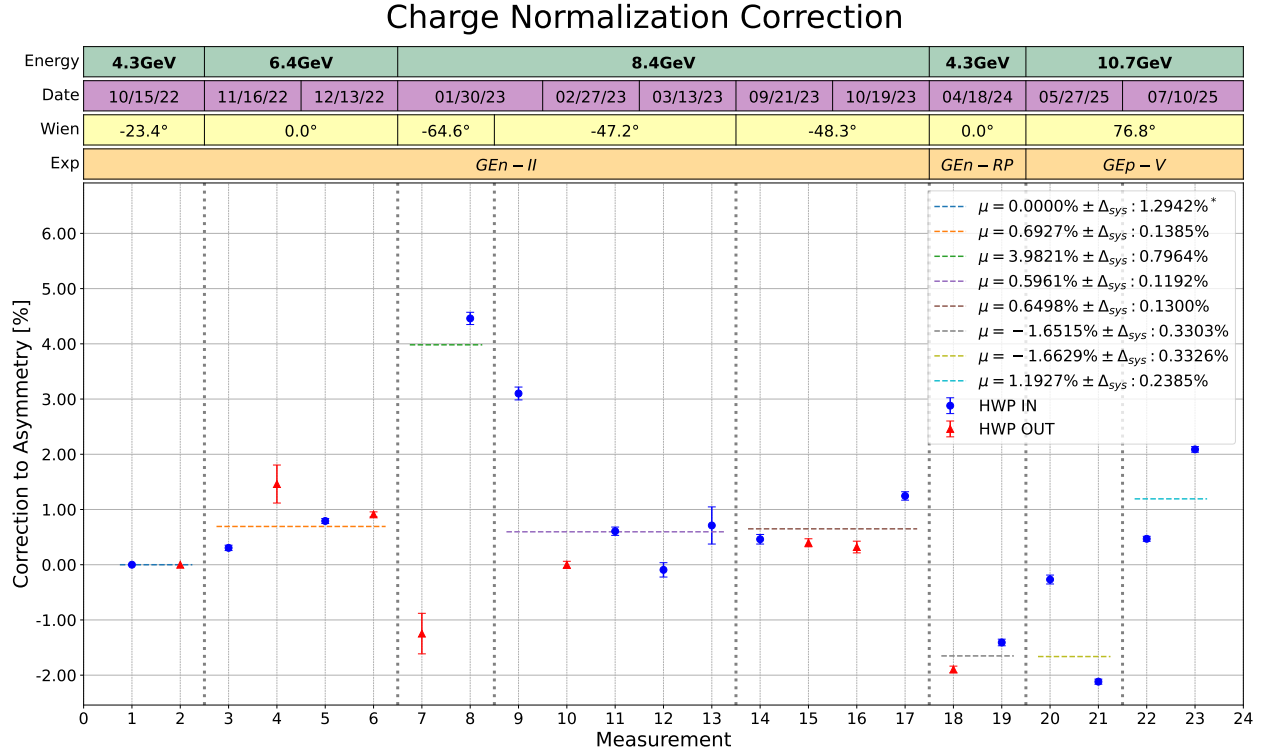


Figure 32: Group-pooled fractional charge normalization and pedestal subtraction corrections and uncertainties for all Møller run groups.

6.3 Deadtime Correction

The dead-time correction accounts for coincidence events lost when the DAQ is busy following a preceding event. At high rates, dead-time can reduce the measured coincidence rate and bias the extracted asymmetry. Various methods for addressing dead-time losses in coincidence systems are discussed in the literature [177, 178, 179, 180, 181].

We apply this correction and report both per-run and group-pooled values. The fractional dead-time correction implemented in our analysis follows a technote [182] on accidental and dead-time corrections for Hall A Møller polarimetry (April 2026); full credit for the derivation and the emulator-based determination of the time constants belongs to that

document.

Two distinct dead-time constants are used: τ_{ds} for single-arm events preceding a coincidence and τ_{dc} for coincidence events preceding a coincidence. We use the empirically optimized values $\tau_{ds} = 18.1$ ns and $\tau_{dc} = 30.5$ ns. The fractional correction is evaluated as:

$$\frac{A_{\text{mol}}^{\text{DT corr}} - A_{\text{raw}}}{A_{\text{raw}}} \approx R_L^s \tau_{ds} \frac{A_L^s}{A_{\text{raw}}} + R_R^s \tau_{ds} \frac{A_R^s}{A_{\text{raw}}} + 2R_C \tau_{dc}, \quad (6.5)$$

where the single-arm-only rates are $R_L^s = R_L - R_C$ and $R_R^s = R_R - R_C$, and the single-arm-only asymmetries A_L^s , A_R^s are defined in Sec. 7.1.

The systematic uncertainty Δ_{sys} is taken to be 50% of the correction magnitude. The group-pooled fractional dead-time correction for each Møller run is shown in Figure 33.

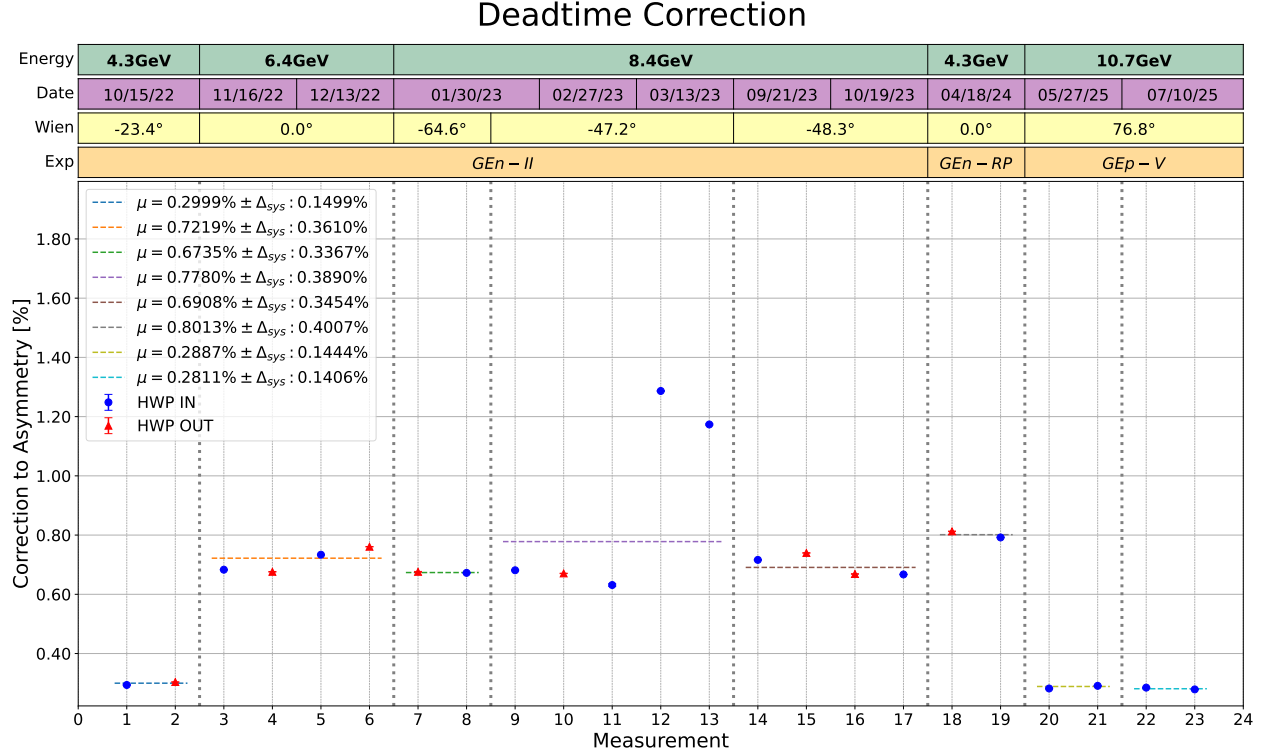


Figure 33: Group-pooled fractional dead-time correction to asymmetry for each Møller run group.

6.4 Accidental Correction

The accidental correction removes the contribution to the measured coincidence rate (and asymmetry) arising from random left–right coincidences between single-arm events. The fractional impact of accidentals on the measured asymmetry is typically small but can become non-negligible at high singles rates [183, 184, 185].

We apply the correction and quote both per-run values and group-pooled values. The fractional accidental correction we apply follows this technote [182] on accidental and dead-time corrections for Hall A Møller polarimetry (April 2026), to which the reader is referred for the full derivation. All equations and the empirically optimized coincidence window $2\tau_C = 22.6$ ns used below are taken from that document.

The fractional correction is given by:

$$\frac{A_{\text{mol}}^{\text{acc corr}} - A_{\text{raw}}}{A_{\text{raw}}} \approx \frac{R_{\text{acc}}}{R_C} \left(1 - \frac{A_{\text{acc}}}{A_{\text{raw}}} \right), \quad (6.6)$$

where the accidental rate and accidental asymmetry are evaluated as:

$$R_{\text{acc}} = 2\tau_C (R_L - R_C)(R_R - R_C), \quad (6.7)$$

$$A_{\text{acc}} = \frac{(R_R - R_C)(R_L A_L - R_C A_{\text{raw}}) + (R_L - R_C)(R_R A_R - R_C A_{\text{raw}})}{(R_R - R_C)(R_L - R_C)}. \quad (6.8)$$

The rates R_C , R_L , R_R and the asymmetries A_{raw} , A_L , A_R are defined in Sec. 7.1. For each group, per-pattern arrays from all member runs are concatenated and the correction is re-evaluated on the pooled sample. The systematic uncertainty Δ_{sys} for per-run values is taken to be 50% of the correction magnitude. The group-pooled accidental correction percentages for Møller measurements in SBS experiments are shown in Figure 34.

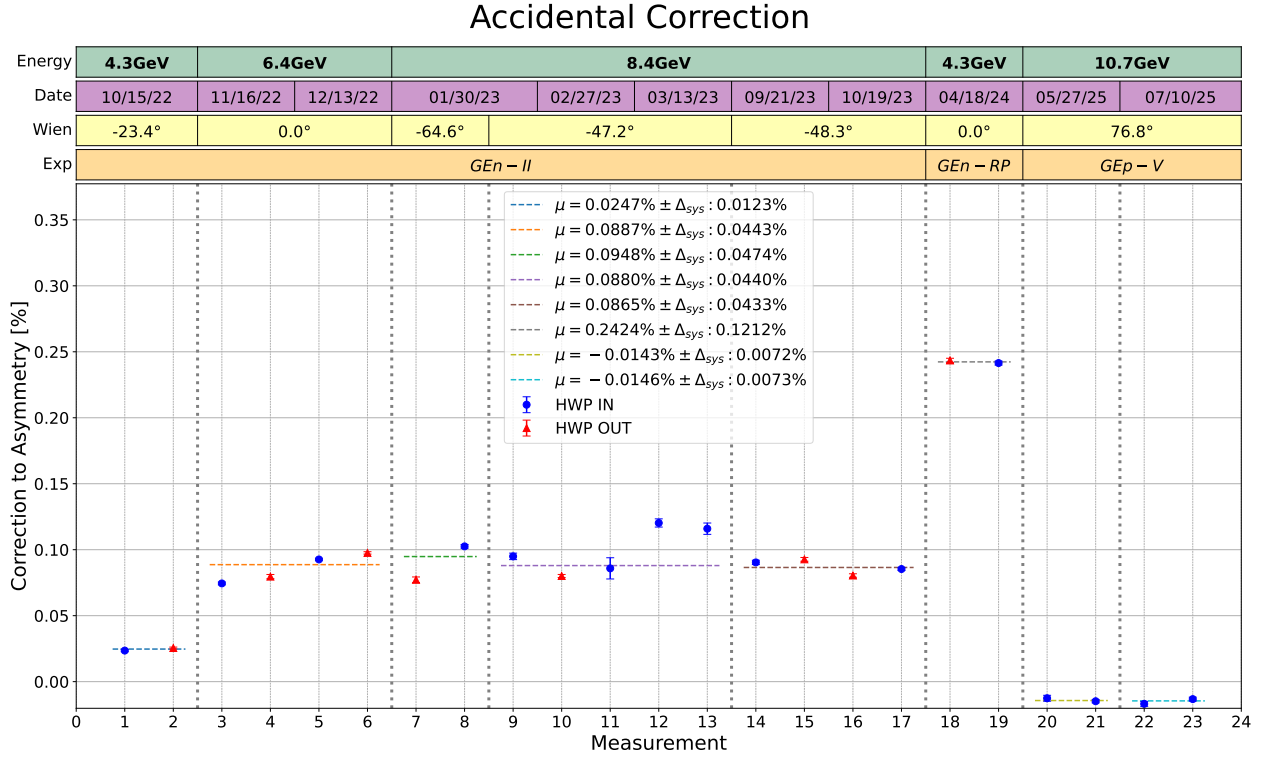


Figure 34: Group-pooled fractional accidental corrections for grouped Møller measurements in SBS experiments.

6.5 Null Asymmetry

In addition to polarization measurements on the iron foils, dedicated Møller runs are performed using a thin copper foil. Since copper is weakly diamagnetic and has no net spin polarization, the expected measured asymmetry is zero. Any non-zero asymmetry measured with copper, therefore, indicates false asymmetries caused by other beam properties or instrument effects. Null asymmetry measurements serve as an important systematic cross-check and are standard practice in precision Møller polarimetry, where controlling helicity-correlated systematics is crucial [18].

The insertion of the half-wave plate (HWP) reverses the sign of the measured asymme-

try. Therefore, for false asymmetry runs conducted with the HWP inserted, the measured asymmetry is multiplied by -1 to maintain a consistent physics sign convention. During GEn and GEn-RP runs, no Wien filter flips occurred within a given beam energy. However, changes in spin precession between different beam energies can cause an effective sign reversal. In the current configuration, with the target magnet field pointing downstream, the true physics asymmetry is expected to be positive. For each beam energy, the HWP state yielding a positive asymmetry is taken as the correct physics sign, and the opposite HWP state is multiplied by -1 . After applying these sign corrections, the null asymmetries are fit with a zeroth-degree polynomial. The results should be statistically consistent with zero. The fitted null asymmetries are shown in Figure 35, and the associated systematic uncertainties are discussed in Section 7.5.

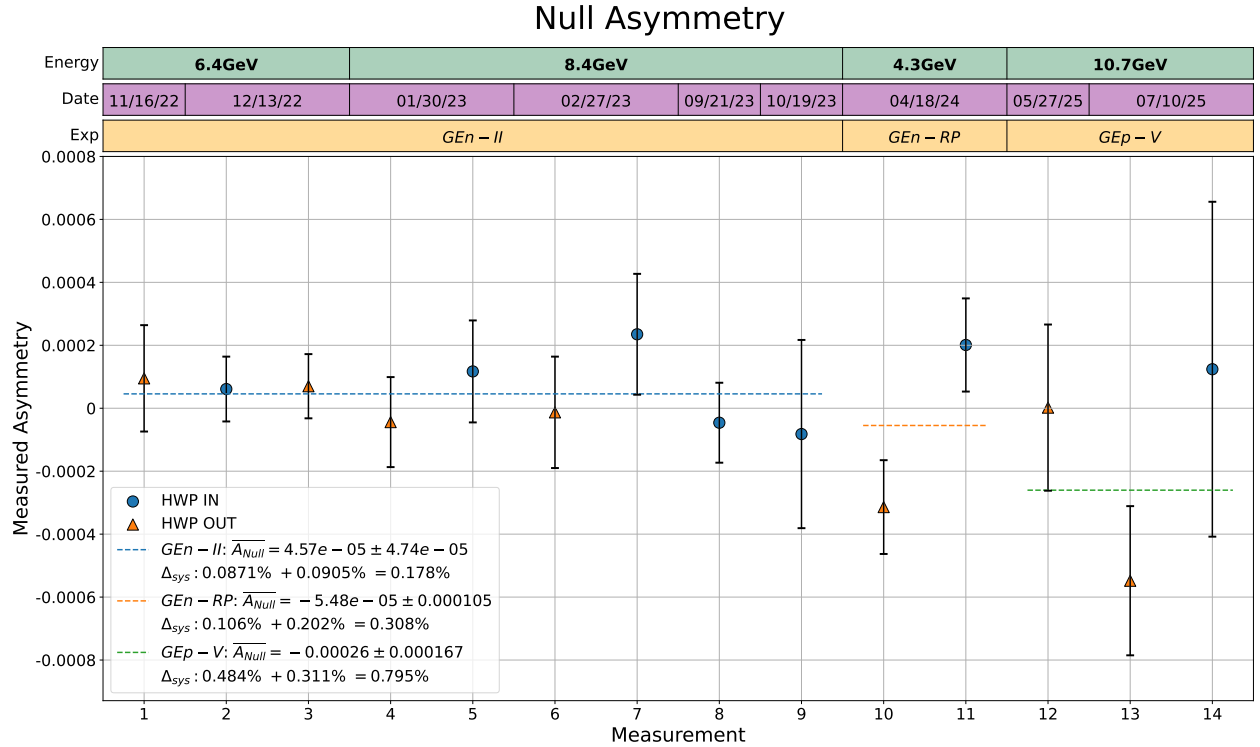


Figure 35: Copper foil false asymmetry measurements during the SBS program

6.6 Leakage Currents

During Møller measurements, other experimental halls, especially Halls B and C, may also be receiving beam. CEBAF delivers a single continuous wave beam to the experimental halls by mixing bunches for different halls at the photoinjector. Further downstream, RF separator cavities and a septum split the beam, directing each hall's bunches into its own beamline. This separation is never perfect. A small fraction of the current intended for the other halls can leak into Hall A and increase background coincidence rates through two distinct channels. First, bunches intended for other halls may not be fully separated, creating a beam "halo" that is spatially offset from the main Hall A trajectory. Second, imperfect turn-off of the drive lasers can deposit charge directly into the Hall A beam pulses. Because this charge is collinear with the main beam, collimators cannot reject it, establishing an irreducible floor on the coincidence-background rate. Hall C often operates at beam currents similar to Hall A, while Hall B's currents are considerably lower [67, 186].

To assess this effect, dedicated bleedthrough tests are conducted without intentionally delivering beam to Hall A. The resulting coincidence rate is then compared to rates from polarization runs. If leakage exceeds acceptable limits, the Machine Control Center (MCC) can adjust beamline slits to block the offset halo. However, the laser-induced leakage inside the main beam cannot be blocked this way. Because adjusting the slits successfully reduces the measured leakage, we know the offset halo—rather than the laser contamination—is the main source of background under these conditions. Close monitoring of leakage currents is vital to keep background contributions negligible compared to true Møller coincidence signals [41, 187].

The bleedthrough rate for each measurement is calculated using the following formula:

$$\text{Bleedthrough Rate} = \frac{R_C^i}{\overline{R}_C^{\text{Møller}}} \times 100\% \quad (6.9)$$

Figure 36 displays all bleedthrough measurements for the SBS experiments, and the associated systematic uncertainties are discussed further in Section 7.6.

Bleedthrough

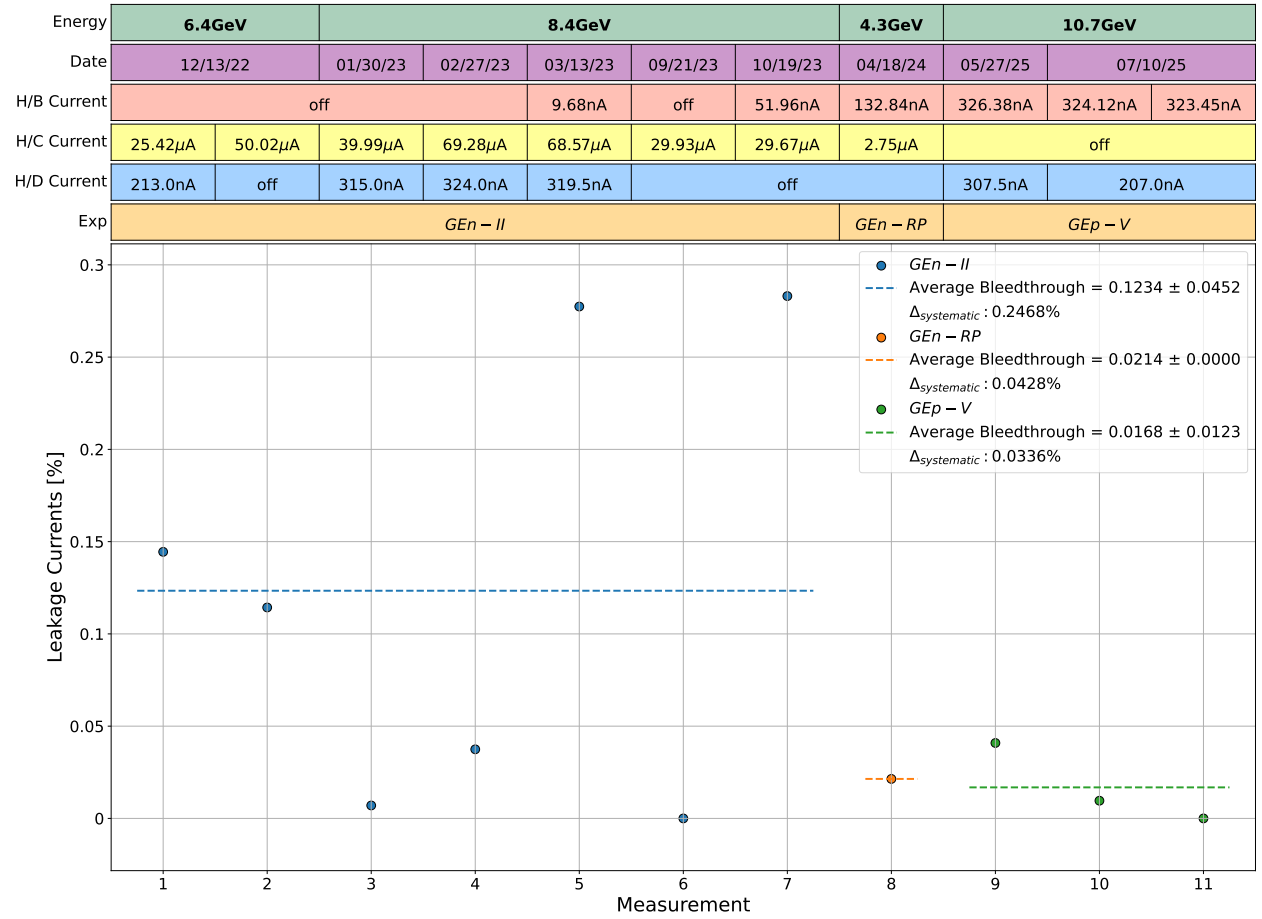


Figure 36: Beam-current bleedthrough measurements during the SBS program. Each point gives the fractional coincidence rate measured in Hall A when no beam was intentionally delivered to the hall, normalized to the average Møller coincidence rate on the same day (Equation 6.9). Beam energy, measurement date, SBS experiment, and currents in the other experimental halls are indicated for each run. Run-by-run values are listed in Table 8.

6.7 Magnet Rate Scans

The Møller measurement process involves performing magnet rate scans, usually on specific quadrupole or dipole magnets. The choice of magnets depends on the beam energy, with one or two magnets given priority. The coincidence rate is highly sensitive to the magnetic field configuration because these fields influence the acceptance and focusing of scattered electrons. During the scan, the magnet current is adjusted across a range around the predicted optimal value, and coincidence rates are recorded at each step. Smaller current increments are used near the expected maximum rate, based on earlier simulation studies.

Measured rate scans are compared extensively with GEANT-based simulations to study the polarimeter's acceptance. Discrepancies between data and simulation are used to refine magnetic field maps and alignment parameters, and to estimate systematic uncertainties in the calculated analyzing power A_{zz} .

6.8 Target Polarization

The extracted beam polarization is inversely proportional to the target polarization, so any uncertainty in the target magnetization propagates directly into the final polarimetry results. The target consists of thin iron foils magnetized to saturation by a strong external magnetic field [188]. For fully saturated iron at room temperature, an effective electron polarization of approximately 8% is achieved when averaged over all atomic electrons in the foil [129, 189]. Several factors can influence the target polarization in practice. Beam-induced heating is a key effect. As the beam current rises, more electrons transfer energy to the foil, raising its local temperature. According to the material's magnetization curve,

higher temperatures lead to decreased magnetization, which in turn reduces the effective electron polarization [129]. This effect depends not only on the beam current but also on the beam spot size. A smaller spot size results in a higher power density and therefore a larger local temperature rise. The beam spot size is measured using a harp scan. The harp scan provides the spatial distribution of the beam in the horizontal (x) and vertical (y) directions at the target location. From these distributions, the RMS widths in both directions are extracted. The effective beam spot radius is taken as the geometric mean of the two RMS widths, $R_{\text{spot}} = \sqrt{\sigma_x \sigma_y}$, which is the radius of a circular spot having the same effective area $\pi \sigma_x \sigma_y$ as the elliptical Gaussian beam. Once the beam spot radius and beam current are known, the steady-state temperature rise in the foil can be estimated from the balance between deposited beam power and heat removal by thermal conduction to the target frame (and radiative loss), using the thermal conductivity of iron; heat capacity governs only the transient time constant to reach equilibrium [190, 191]. The decrease in magnetization with temperature is based on empirical magnetization curves for iron, which allows calculation of the fractional reduction in target polarization relative to the standard 294 K value. The approach follows the method described in [129, 192], where temperature-dependent polarization effects in solid targets are detailed. These calculations are performed using analysis scripts [193], which use the measured beam current and harp-derived beam size to determine the fractional reduction in magnetization, thermal effects, and the resulting target polarization for each Møller run.

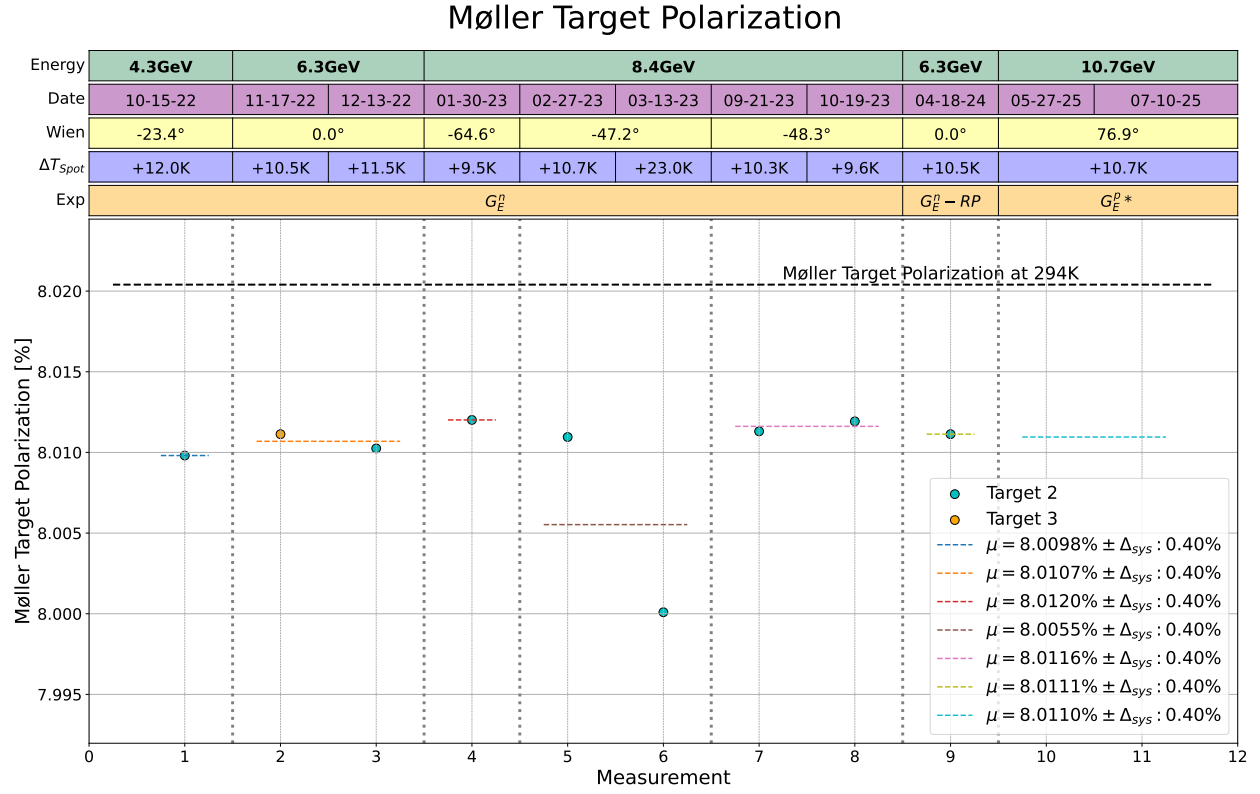


Figure 37: Møller target polarizations measured during the SBS experimental runs. The fit line shows the polarization for both foils; the GEp data point is omitted due to the Harp-Scanner being inoperative (see below).

Figure 37 shows the Møller target polarization for the 10-micron iron foils used during the GEN and GEN-RP experiments. The plot displays the polarization for each measurement day, along with the corresponding beam energies, estimated target temperature increases due to beam spot size and current, the associated experiment, and the beam Wien angle. The beam current was $1 \mu A$ for all measurements except for 2023-03-13, which shows a noticeable reduction in target polarization due to increased target temperature. A reference line indicates the polarization expected if the target temperature remained fixed at 294 K. The uncertainties associated with target polarization are discussed in Section 7.9.

Table 2 lists all target polarization values and their corresponding intermediate variables

for each measurement date.

Experiment	Date	I_{beam} (μ A)	E_{beam} (GeV)	σ_x (μ m)	σ_y (μ m)	R_{spot} (μ m)	ΔT (K)	P_{Target} (%)	$\Delta_{sys}^{Heating}$ (%)	$\Delta_{sys}^{Sat.}$ (%)	$\Delta_{sys}^{Inc. Sat.}$ (%)	Δ_{sys}^{Total} (%)
GEn	10/15/2022	1	4.3	191	76	120.48	12.0	8.010	0.02201	0.23721	0.25	0.34533
GEn	11/17/2022	1	6.3	304	131	199.56	10.5	8.011	0.01926	0.23717	0.25	0.38291
GEn	12/13/2022	1	6.3	138	158	147.66	11.5	8.010	0.02109	0.23720	0.25	0.37644
GEn	01/30/2023	1	8.4	395	200	281.07	9.5	8.012	0.01742	0.23714	0.25	0.37622
GEn	02/27/2023	1	8.4	413	92	194.93	10.7	8.011	0.01962	0.23718	0.25	0.37635
GEn	03/13/2023	2	8.4	276	96	162.78	23.0	8.000	0.04219	0.23750	0.25	0.37840
GEn	09/21/2023	1	8.4	314	153	219.18	10.3	8.011	0.01889	0.23716	0.25	0.37630
GEn	10/19/2023	1	8.4	350	218	276.22	9.6	8.012	0.01761	0.23715	0.25	0.37623
GEn-RP	04/18/2024	1	6.3	329	121	199.52	10.5	8.011	0.01926	0.23717	0.25	0.37637
GEp	05/27/2025	1	10.7	296.7	136.3	201.10	10.7	8.011	0.01962	0.23718	0.25	0.37633
GEp	07/10/2025	1	10.7	296.7	136.3	201.10	10.7	8.011	0.01962	0.23718	0.25	0.37635

Table 2: Target polarization systematic uncertainties for each SBS experimental run. The target polarization at $T = 294$ K is $P_{Target} = 8.0204\%$, with a saturation polarization uncertainty of $\delta P_{sat} = 0.00019$ out of $P_{sat} = 0.080204$ ($\sim 0.24\%$ fractional systematic uncertainty). For further details see Ref. [129].

During the GEp experiment, the harp scanner was not operational, so direct measurements of the beam spot size were unavailable. In this case, the average beam size from previous measurements was used as an estimate, while the exact beam current for each run was retained in the scaling calculation. This introduces an additional systematic uncertainty associated with possible variations in beam size that are not captured by the average value. The impact of this assumption is included explicitly in the target polarization error budget discussed in Section 7.9.

6.9 Spin Precession – $\delta p/p$

The spin precession correction should be evaluated separately for each production period. These periods are marked in Figure 38 by vertical dashed lines with timestamps. During each period, the beam energy, Wien filter angle, and optics configuration are typically stable, although small drifts in beam momentum can affect the final spin orientation at the target.

In the ultra-relativistic regime relevant for CEBAF energies, the total spin precession angle is determined by the beam energy and the cumulative magnetic bending angle in the accelerator arcs. As discussed in Section 4.5, the change in the spin precession angle can be expressed in terms of the fractional deviation in beam momentum, $\delta p/p$. Because beam energy and momentum are virtually interchangeable at these energies ($\delta E/E \approx \delta p/p$), multiplying the nominal spin precession angle by $\delta p/p$ yields the absolute deviation in the spin rotation angle [194, 195].

During a Møller measurement, the beam momentum deviation is measured with high precision, providing an accurate value of $\delta p/p$ at that specific time. Outside of dedicated Møller measurements, however, the fractional momentum deviation can vary due to small drifts in accelerator settings. These variations are the primary source of uncertainty in the spin precession calculation. For each production period, the applied correction accounts for the difference between the average fractional momentum deviation during production running and the deviation measured during the Møller polarization measurement. If multiple Møller measurements occur within the same production period, their $\delta p/p$ values are averaged.

Spin precession builds up incrementally each time the beam passes through an accelerator arc, and the contribution from each arc is proportional to the beam energy in that

arc, so the total accumulated spin rotation scales with the pass-summed energy (Eq. 6.10). Therefore, spin precession corrections must be evaluated separately for each specific energy setting and pass number, even if the Wien angle remains unchanged [196].

The total spin precession angle accumulated by the beam from the injector to the Hall A target is obtained by summing the precession through each 180° recirculation arc and the final Hall A transport arc:

$$\Delta\phi_{\text{tot}} = \frac{\theta}{440.65 \text{ MeV}} \left[\sum_{k=1}^n E_k^{\text{N}} + \sum_{k=1}^{n-1} E_k^{\text{S}} \right] + \frac{\phi}{440.65 \text{ MeV}} E_{\text{HallA}}, \quad (6.10)$$

where n is the pass number (1–5 at CEBAF), $\theta = 180^\circ$ is the bend angle of each recirculation arc, and $\phi = 37.5^\circ$ is the bend angle of the Hall A extraction arc. The beam energy in the north arc on pass k is $E_k^{\text{N}} = E_{\text{inj}} + k E_{\text{NL}} + (k - 1) E_{\text{SL}}$, and in the south arc on pass k it is $E_k^{\text{S}} = E_{\text{inj}} + k E_{\text{NL}} + k E_{\text{SL}}$, where E_{inj} is the injector energy and E_{NL} , E_{SL} are the per-pass energy gains in the north and south linacs, respectively. The final-pass beam extracted to Hall A has energy $E_{\text{HallA}} = E_{\text{inj}} + n E_{\text{NL}} + n E_{\text{SL}}$. The constant 440.65 MeV reflects the anomalous magnetic moment of the electron, $a_e = (g - 2)/2$, and converts the product of beam energy and bend angle into the $g-2$ spin precession angle relative to the momentum direction in the ultra-relativistic limit.

Further details regarding the uncertainty propagation and its contribution to the total polarization systematic error are provided in Section 7.10.

6.10 Polarization Estimation for Unmeasured Periods

Accelerator and beam data relevant to Hall A polarimetry were continuously monitored throughout the SBS program during active electron beam delivery. The data are segmented

into *production periods*, defined as intervals with a constant beam energy, Wien angle, and photoinjector laser spot. These periods may extend beyond the standard physics data collection windows (for specific SBS experiment data, see Section 7.11) [19].

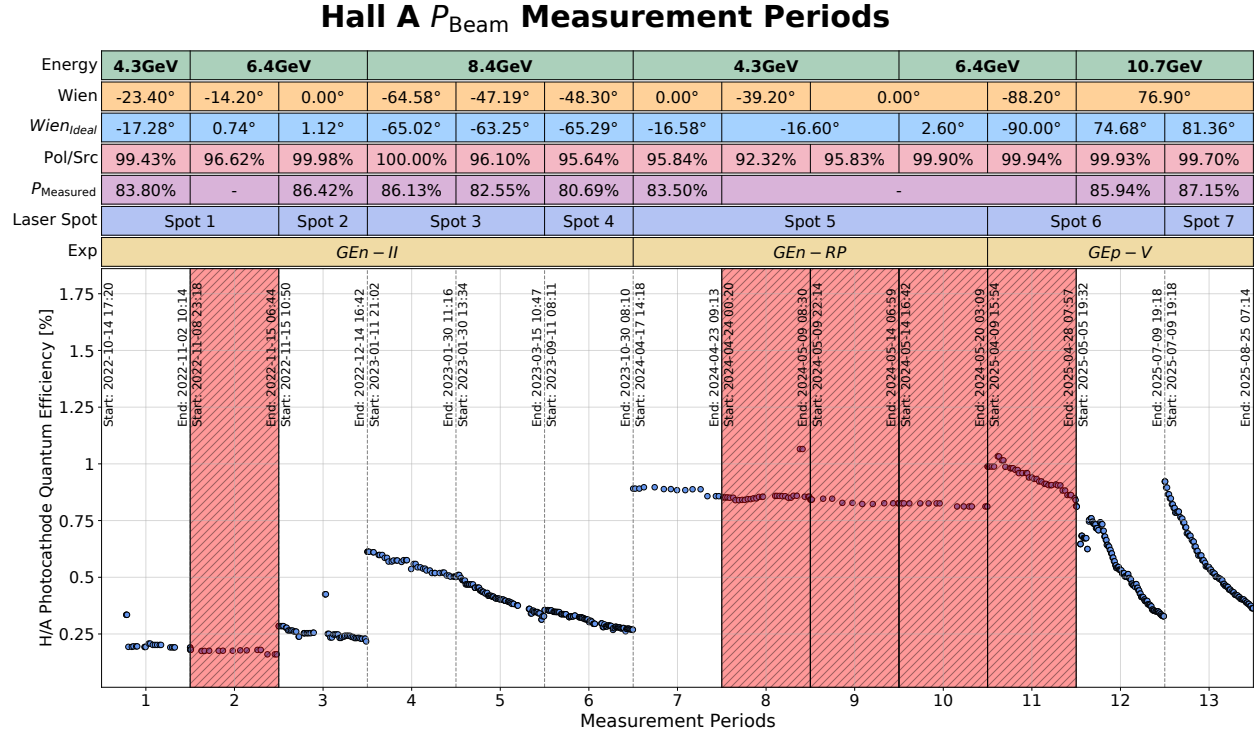


Figure 38: Hall A photocathode quantum efficiency across all production periods and beam polarimetry availability.

Figure 38 displays the measured Wien angle alongside the ideal Wien angle required to deliver maximum longitudinal polarization to Hall A. This ideal calculation accounts for the injector and linac energies, total spin precession, and radiated synchrotron energy. The *Pol/Src* subplot illustrates the maximum achievable fraction of the source polarization, or maximum longitudinal polarization assuming a 100% polarized source. Finally, the figure tracks laser spot variations at the photoinjector alongside all measured Hall A photocathode quantum efficiency values to visualize the spot changes. Table 3 displays the data pertaining to Figure 38.

Start Date	End Date	Experiment	E_{beam} (GeV)	θ_{Wien} (deg)	Møller?	$\theta_{\text{Wien}}^{\text{ideal}}$ (deg)	Pol/Src (%)	P_{Measured} (%)
2022-10-14 17:20:51	2022-11-02 10:14:15	GEn	4.29	-23.40	yes	-17.28	99.43	83.80
2022-11-08 23:18:37	2022-11-15 06:44:24	GEn	6.37	-14.20	no	0.74	96.62	-
2022-11-15 10:50:08	2022-12-14 16:42:20	GEn	6.37	0.00	yes	1.12	99.98	86.42
2023-01-11 21:02:26	2023-01-30 11:16:43	GEn	8.45	-64.58	yes	-65.02	100.0	86.13
2023-01-30 13:34:49	2023-03-15 10:47:48	GEn	8.44	-47.19	yes	-63.25	96.1	82.55
2023-09-11 08:11:13	2023-10-30 08:10:20	GEn	8.45	-48.30	yes	-65.29	95.64	80.69
2024-04-17 14:18:21	2024-04-23 09:13:58	GEn-RP	4.29	0.00	yes	-16.58	95.84	83.50
2024-04-24 00:20:05	2024-05-09 08:30:39	GEn-RP	4.29	-39.20	no	-16.6	92.32	-
2024-05-09 22:14:26	2024-05-14 06:59:36	GEn-RP	4.29	0.00	no	-16.6	95.83	-
2024-05-14 16:42:51	2024-05-20 03:09:55	GEn-RP	6.37	0.00	no	2.6	99.9	-
2025-04-09 15:54:28	2025-04-28 07:57:20	GEp	6.45	-88.20	no	-90.0	99.94	-
2025-05-05 19:32:12	2025-07-09 19:18:54	GEp	10.66	76.90	yes	74.68	99.93	85.94
2025-07-09 19:18:54	2025-08-25 07:14:57	GEp	10.66	76.90	yes	81.36	99.7	87.15

Table 3: Production periods and Møller measurement availability.

Using the connection between longitudinal beam polarization and the Wien angle, as shown in Equation 6.11, we can estimate the source polarization from the measured beam polarization. From there, we can determine the polarization at a different kinematic setting using that source polarization. This method yields more accurate results for periods close to those with known beam polarizations [125, 197].

$$P_{\text{Measured}} = P_{\text{source}} \cos(\theta_{\text{wien}} - \theta_{\text{wien}}^{\text{ideal}}) \quad (6.11)$$

The figures below display the estimated beam polarizations for all SBS experiments. Details on the error calculations and datasets are provided in 7.12.

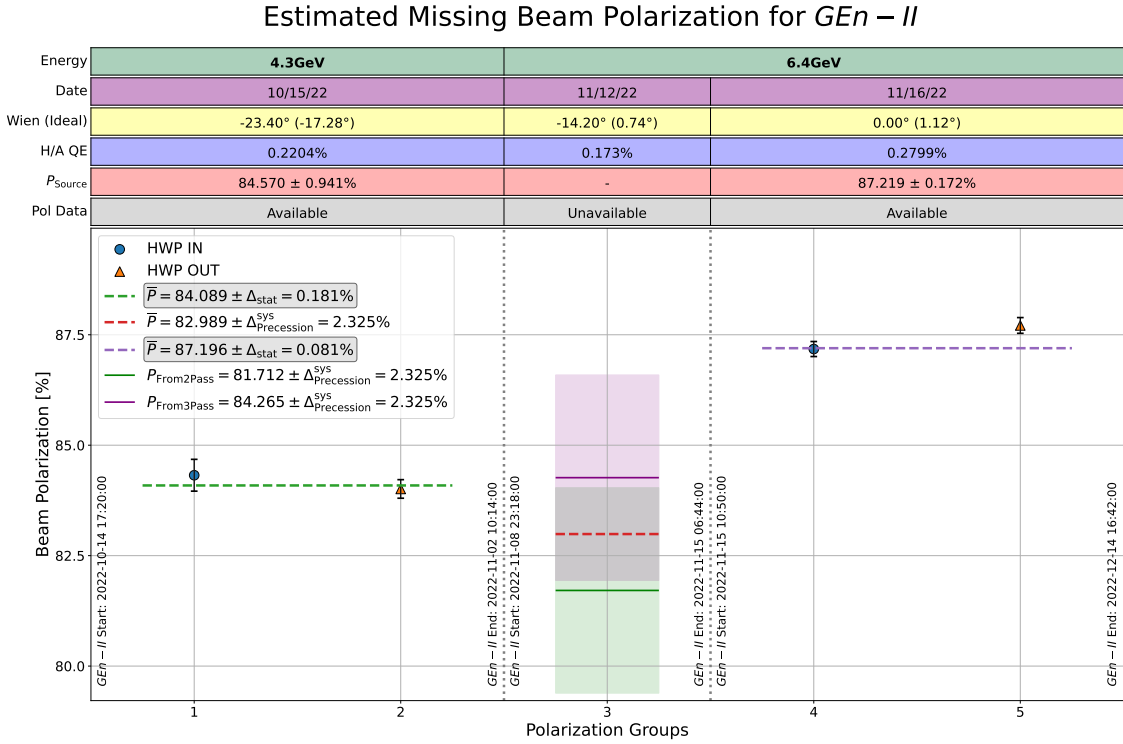


Figure 39: Forward-projected Hall A beam polarization for the unmeasured *GEn* interval at 6.37 GeV (2022-11-08 to 2022-11-15), obtained from the preceding Møller measurement through Equation 6.11. Legend highlights mark directly measured polarizations whose uncertainties are statistical only; the projected point carries a systematic uncertainty from the Wien-angle offset and spin-precession model.

The beam polarization for the period 2022-11-08 to 2022-11-15 was calculated by forward projecting from the prior production period. This was done by first extracting the source polarization from the earlier Møller measurement, then applying the Wien angle and ideal Wien angle of the new spot to determine what fraction of that source polarization Hall A could have received.

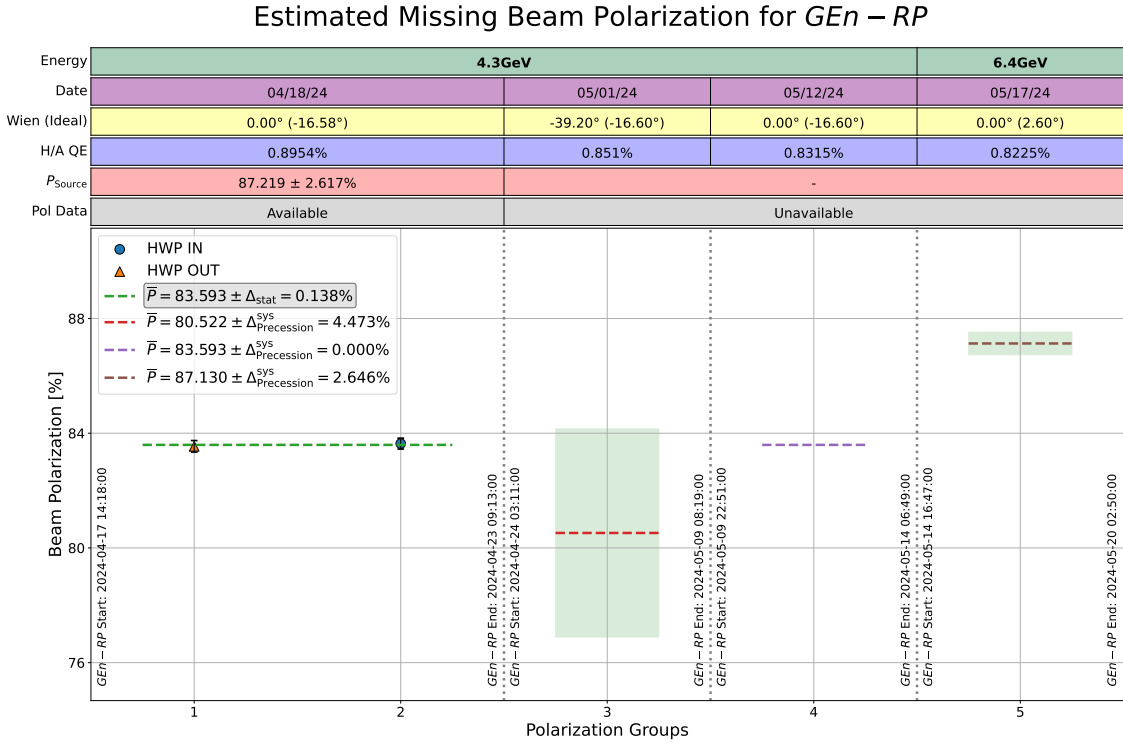


Figure 40: Beam polarization assignments for the three GEn-RP production periods without Møller coverage, extrapolated from neighboring measurements at 4.29 and 6.37 GeV using the Wien-angle offset from its ideal value. Highlighted legend entries reproduce Møller results and retain statistical uncertainties; extrapolated periods are assigned systematic uncertainties from the polarization-projection procedure.

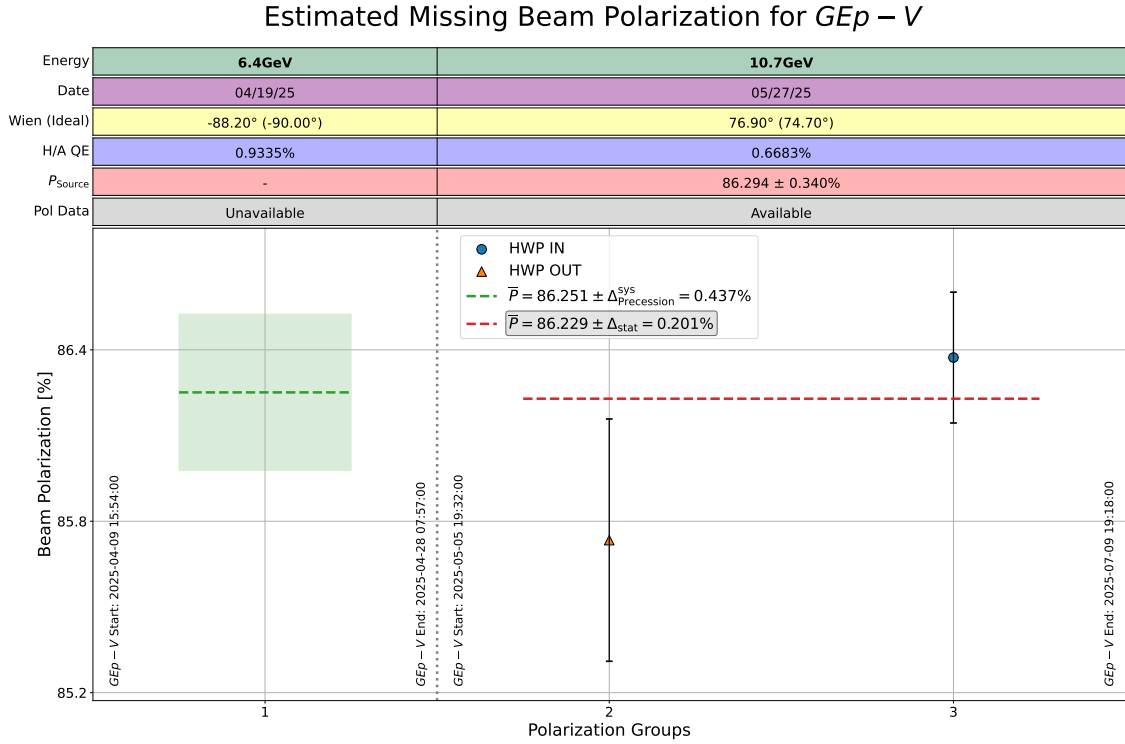


Figure 41: Inferred GEP longitudinal beam polarization for the April 2025 production window at 6.45 GeV prior to dedicated Møller polarimetry, based on the deviation of the set Wien angle from the ideal setting. Values highlighted in the legend are taken from measured polarizations and quoted with statistical errors only; the estimated interval inherits a systematic uncertainty from Wien-angle and spin-precession effects.

CHAPTER 7

Uncertainty Analysis

7.1 Statistical Uncertainties

Statistical Uncertainty of the Measured Asymmetry

Electron beam polarization in Hall-A-style Møller polarimetry is extracted from the measured counting-rate asymmetry between parallel and antiparallel spin configurations, $A_{\text{meas}} = (N^+ - N^-)/(N^+ + N^-)$, which is related to the beam polarization through the known analyzing power and target polarization. Under Poisson counting statistics, the dominant statistical uncertainty arises from the finite number of detected Møller events and propagates as

$$\delta A_{\text{meas}} = \sqrt{\frac{1 - A_{\text{meas}}^2}{N_{\text{tot}}}} \approx \frac{1}{\sqrt{N_{\text{tot}}}}, \quad (7.1)$$

where $N_{\text{tot}} = N^+ + N^-$, and the approximation holds because $A_{\text{meas}}^2 \sim 2.5 \times 10^{-3} \ll 1$ for the measured Møller asymmetries. Consequently, the relative statistical uncertainty on the extracted beam polarization scales inversely with the square root of accumulated Møller electrons, motivating operation at high rates and long integration times [173].

Detailed Hall A studies have demonstrated that sub-percent statistical precision can be achieved within minutes at modest beam currents, provided backgrounds and dead-time effects are controlled. In precision experiments, such as those involving parity-violation,

this well-understood statistical uncertainty from the Møller polarimeter propagates directly into the final physics asymmetry error budget, underscoring the importance of reliable statistical modeling in beam polarization measurements.

Statistical Uncertainties of the Corrections

Each of the three corrections discussed in the following subsections (accidental 7.4, charge-pedestal 7.2, and dead-time 7.3) is computed run-by-run from helicity-pattern scaler data. The statistical error we quote on every correction is the empirical *standard error of the mean (SEM)* of that correction across the helicity patterns of the run.

By treating each valid pattern as an independent measurement of the correction, the pattern-to-pattern spread divided by $\sqrt{N}$ naturally folds in counting statistics alongside any other random pattern-to-pattern variation. This is done automatically without requiring analytic error propagation from the underlying scaler counts. Systematic uncertainties arising from the time constants τ_{ds} , τ_{dc} , τ_C , the pedestal determination, and the scaler-to-rate conversions are not included here; they are assigned separately (see the relevant subsections).

Notation and inputs.

Let N be the number of valid patterns in a given run, indexed by $i = 1, \dots, N$. For each scaler branch $x \in \{c^\pm, \ell^\pm, r^\pm, b^\pm\}$ (coincidence, left arm, right arm, and beam-current monitor, with $\pm$ the helicity state), x_i is the per-pattern value and $\bar{x} = \frac{1}{N} \sum_i x_i$ is the run-averaged value. Counts are converted to rates using the scaler-to-rate factor $k = f_{\text{hel}}/(N_{\text{pat}}/2) = 15 \text{ Hz/count}$, where f_{hel} is the helicity flip frequency and $N_{\text{pat}} = 4$ for the quartet pattern used during the SBS program (Figure 6). Run-averaged rates are held

fixed as constants in the per-pattern formulas:

$$R_C = \frac{k}{2}(\bar{c}^+ + \bar{c}^-), \quad R_L = \frac{k}{2}(\bar{\ell}^+ + \bar{\ell}^-), \quad R_R = \frac{k}{2}(\bar{r}^+ + \bar{r}^-), \quad R_L^s \equiv R_L - R_C, \quad R_R^s \equiv R_R - R_C. \quad (7.2)$$

Per-pattern raw quantities.

The pattern-level raw asymmetries (coincidence, left, right, charge) all have the same structural form:

$$A_X^{(i)} = \frac{X_i^+ - X_i^-}{X_i^+ + X_i^-}, \quad X \in \{\text{raw} (\equiv c), L (\equiv \ell), R (\equiv r), q (\equiv b)\}. \quad (7.3)$$

The single-arm-only asymmetries, obtained by subtracting the coincidence contribution from the measured single-arm asymmetries, are:

$$A_{LS}^{(i)} = \frac{R_L A_L^{(i)} - R_C A_{\text{raw}}^{(i)}}{R_L^s}, \quad A_{RS}^{(i)} = \frac{R_R A_R^{(i)} - R_C A_{\text{raw}}^{(i)}}{R_R^s}. \quad (7.4)$$

Per-pattern corrections.

The fractional accidental, dead-time, and charge-pedestal corrections defined in the subsections below (Eqs. 6.6, 6.5, 6.4) are evaluated pattern-by-pattern by substituting the per-pattern asymmetries into the same expressions used at the run level:

$$f_{\text{acc}}^{(i)} = \frac{R_{\text{acc}}}{R_C} \left(1 - \frac{A_{\text{acc}}^{(i)}}{A_{\text{raw}}^{(i)}} \right), \quad f_{\text{dt}}^{(i)} = R_L^s \tau_{ds} \frac{A_{LS}^{(i)}}{A_{\text{raw}}^{(i)}} + R_R^s \tau_{ds} \frac{A_{RS}^{(i)}}{A_{\text{raw}}^{(i)}} + 2R_C \tau_{dc}, \quad f_{\text{qp}}^{(i)} = -\frac{A_q^{(i)}}{A_{\text{raw}}^{(i)}} \left(1 + \frac{Q_{\text{ped}}}{Q_{\text{avg}}} \right), \quad (7.5)$$

where $A_{\text{acc}}^{(i)}$ is the per-pattern evaluation of Eq. 6.8, R_{acc} is the run-level accidental rate from Eq. 6.7, and $Q_{\text{avg}} = (\bar{b}^+ + \bar{b}^-)/2$. The fully corrected per-pattern asymmetry uses the

same multiplicative chain as the run-level central value:

$$A_{\text{corr}}^{(i)} = A_{\text{raw}}^{(i)} \cdot (1 + f_{\text{qped}}^{(i)}) \cdot (1 + f_{\text{acc}}^{(i)}) \cdot (1 + f_{\text{dt}}^{(i)}), \quad (7.6)$$

with the qped factor replaced by unity for runs taken on 2022-10-15 (see Section 6.2).

Run-level error.

For any of $X \in \{A_{\text{raw}}, f_{\text{acc}}, f_{\text{dt}}, f_{\text{qped}}, A_{\text{corr}}\}$, the run-level statistical error is the sample standard deviation of the N per-pattern values divided by $\sqrt{N}$:

$$\sigma_X = \frac{s_X}{\sqrt{N}}, \quad s_X = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (X^{(i)} - \bar{X})^2}, \quad (7.7)$$

where s_X is the (Bessel-corrected) sample standard deviation of the N per-pattern values and σ_X is the corresponding standard error of the mean.

Group-level values and errors.

Runs sharing the same group number are combined by concatenating their per-pattern arrays into a single pooled sample of size $N_{\text{pool}} = \sum_{\text{runs}} N_{\text{run}}$.

The group central value X_{group} is obtained by evaluating the same formulas on the *pooled* run-averaged inputs; the group statistical error is the SEM of $X^{(i)}$ computed across the pooled patterns:

$$\sigma_{X,\text{group}} = \frac{s_{X,\text{pool}}}{\sqrt{N_{\text{pool}}}}, \quad s_{X,\text{pool}} = \sqrt{\frac{1}{N_{\text{pool}}-1} \sum_{i=1}^{N_{\text{pool}}} (X_{\text{pool}}^{(i)} - \bar{X}_{\text{pool}})^2}. \quad (7.8)$$

Runs with $N < 2$ have an undefined per-pattern sample variance and are excluded from the inverse-variance group combination rather than assigned $\sigma_X = 0$, which would other-

wise give such a run infinite weight. Runs are combined as

$$\bar{X} = \frac{\sum_j w_j X_j}{\sum_j w_j}, \quad w_j = \frac{1}{\sigma_{X,j}^2}. \quad (7.9)$$

7.2 Charge Normalization Systematics

Fractional corrections due to charge normalization and BCM charge pedestal subtractions are illustrated in Figure 32. These are averaged for each Wien angle, aligning with the main experiment’s beam polarization reporting method. Based on the well-understood charge-correction procedure, we assign a systematic uncertainty of 20% of the correction magnitude in each region.

The stability of the beam current monitor (BCM) pedestal is essential for precise charge normalization, especially at low currents where pedestal offsets can be similar to signal levels. On 2022-10-15, the BCM signal malfunctioned and was not detected during measurement. As a result, the data for that day was manually corrected, with a 0% correction indicated. To account for the hardware malfunction and ensure that errors in charge normalization and pedestal subtraction for that day are thoroughly considered, we allocated a systematic uncertainty of 120% of the average size of all other corrections. This approach covers potential errors and results in a systematic uncertainty of 1.29% for this period. This approach ensures that any unrecognized pedestal instability or normalization bias from that day is included in the systematic uncertainty, preventing unnoticed errors from impacting the final polarization results. The table below lists all values used to compute the charge normalization and BCM pedestal corrections for each measurement group.

Group	Date	Exp	QPed (ch.)	A_q	Q_{avg} (ch.)	$A_{\text{raw}}^{\text{Group}}$	$\delta A_{QP}^{\text{Group}}$ (% frac.)	$\sigma_{\delta A_{QP}^{\text{Group}}}$ (% frac.)	$\Delta_{\text{sys}}^{\text{Group}}$ (%)
3136	2022-10-15	GEn	0.000e+00	-5.662e-03	1.594e+01	-5.248e-02	0.000e+00	0.000e+00	1.2942*
3137	2022-10-15	GEn	0.000e+00	8.048e-06	2.253e+01	5.215e-02	0.000e+00	0.000e+00	1.2942*
3103	2022-11-17	GEn	2.000e+00	-1.052e-04	2.358e+02	5.316e-02	3.050e-03	4.849e-04	0.0610
3106	2022-11-17	GEn	2.000e+00	8.170e-04	2.365e+02	-5.288e-02	1.461e-02	3.441e-03	0.2922
3108	2022-12-13	GEn	2.000e+00	-4.131e-04	2.261e+02	5.265e-02	7.916e-03	4.490e-04	0.1583
3111	2022-12-13	GEn	2.000e+00	4.774e-04	2.349e+02	-5.279e-02	9.119e-03	4.477e-04	0.1824
3118	2023-01-30	GEn	1.833e+00	-7.250e-04	2.255e+02	-5.405e-02	-1.247e-02	3.671e-03	0.2494
3123	2023-01-30	GEn	1.833e+00	-2.267e-03	2.288e+02	5.123e-02	4.460e-02	1.110e-03	0.8920
3126	2023-01-30	GEn	1.833e+00	-1.537e-03	2.308e+02	4.996e-02	3.101e-02	1.163e-03	0.6202
3127	2023-02-27	GEn	2.102e+00	-2.249e-05	2.311e+02	-5.141e-02	-1.689e-05	6.374e-04	0.0003
3130	2023-02-27	GEn	2.102e+00	-4.272e-04	2.303e+02	5.084e-02	6.067e-03	7.534e-04	0.1213
3162	2023-03-14	GEn	1.904e+00	4.759e-05	5.062e+02	5.129e-02	-9.313e-04	1.299e-03	0.0186
3163	2023-03-14	GEn	1.904e+00	-3.605e-04	4.594e+02	5.093e-02	7.107e-03	3.363e-03	0.1421
3146	2023-09-21	GEn	5.000e+00	-2.250e-04	2.378e+02	4.982e-02	4.612e-03	8.679e-04	0.0922
3148	2023-09-21	GEn	5.000e+00	1.934e-04	2.448e+02	-4.969e-02	3.972e-03	7.259e-04	0.0794
3149	2023-10-19	GEn	4.333e+00	1.571e-04	2.224e+02	-5.002e-02	3.202e-03	1.060e-03	0.0640
3150	2023-10-19	GEn	4.333e+00	-5.715e-04	2.225e+02	4.955e-02	1.245e-02	7.629e-04	0.2491
3154	2023-11-21	GEn	4.373e+00	5.295e-04	2.166e+02	4.487e-02	-1.204e-02	1.071e-03	0.2408
3156	2023-11-21	GEn	4.373e+00	9.269e-04	2.316e+02	-4.318e-02	2.187e-02	1.025e-03	0.4375
3158	2024-04-18	GEn-RP	4.500e+00	9.729e-04	2.275e+02	5.232e-02	-1.896e-02	5.990e-04	0.3793
3161	2024-04-18	GEn-RP	4.500e+00	-7.188e-04	2.129e+02	-5.213e-02	-1.408e-02	5.973e-04	0.2816
3166	2025-05-27	GEp	5.124e+00	1.543e-04	2.611e+02	5.319e-02	-2.681e-03	8.066e-04	0.0536
3167	2025-05-27	GEp	5.124e+00	-1.112e-03	2.662e+02	-5.460e-02	-2.118e-02	4.606e-04	0.4235
3175	2025-07-10	GEp	4.871e+00	2.475e-04	2.342e+02	-5.385e-02	4.692e-03	4.531e-04	0.0938
3177	2025-07-10	GEp	4.871e+00	-1.088e-03	2.297e+02	5.319e-02	2.088e-02	5.041e-04	0.4177

Table 4: Group-pooled charge normalization and pedestal subtraction correction values. Each row corresponds to one group.

7.3 Deadtime Correction Systematics

The deadtime correction factors are shown in Figure 33, which includes all relevant information about the beam and the experiments. The percentage corrections are grouped by

the Wien angle, as beam polarizations are reported for SBS experiments in that manner.

Deadtime arises from finite detector and DAQ processing times. At high counting rates, this leads to measurable event loss that can bias the asymmetry if left uncorrected. In precision polarimetry, uncorrected deadtime can bias asymmetry measurements, especially if the rate differs between helicity states. To account for possible inaccuracies in the deadtime correction, we assign a systematic uncertainty of 50%, setting the uncertainty for each region equal to the correction magnitude. The deadtime correction applied to each polarization group is presented in the table below.

Group	Date	Exp	R_L^s (Hz)	R_R^s (Hz)	R_C (Hz)	A_L^s	A_R^s	$A_{\text{raw}}^{\text{Group}}$	τ_{DS} (s)	τ_{DC} (s)	$\delta A_{DT}^{\text{Group}}$ (% frac.)	$\sigma_{\delta A_{DT}^{\text{Group}}}$ (% frac.)	$\Delta_{\text{sys}}^{\text{Group}}$ (%)
3136	2022-10-15	GEn	4.34e+04	3.72e+04	3.92e+04	-2.022e-02	-1.942e-02	-5.248e-02	1.810e-08	3.050e-08	2.938e-03	6.698e-06	0.1469
3137	2022-10-15	GEn	4.42e+04	3.81e+04	4.00e+04	1.956e-02	1.932e-02	5.215e-02	1.810e-08	3.050e-08	3.038e-03	5.421e-06	0.1519
3103	2022-11-17	GEn	1.18e+05	1.11e+05	8.42e+04	2.047e-02	2.174e-02	5.316e-02	1.810e-08	3.050e-08	6.831e-03	6.360e-06	0.3415
3106	2022-11-17	GEn	1.17e+05	1.09e+05	8.43e+04	-1.983e-02	-2.084e-02	-5.288e-02	1.810e-08	3.050e-08	6.760e-03	9.515e-06	0.3380
3108	2022-12-13	GEn	1.20e+05	1.15e+05	9.52e+04	1.885e-02	1.894e-02	5.265e-02	1.810e-08	3.050e-08	7.336e-03	5.864e-06	0.3668
3111	2022-12-13	GEn	1.25e+05	1.19e+05	9.89e+04	-1.858e-02	-1.903e-02	-5.279e-02	1.810e-08	3.050e-08	7.606e-03	5.860e-06	0.3803
3118	2023-01-30	GEn	1.10e+05	1.14e+05	8.53e+04	-2.099e-02	-2.063e-02	-5.405e-02	1.810e-08	3.050e-08	6.760e-03	1.306e-05	0.3380
3123	2023-01-30	GEn	1.12e+05	1.16e+05	8.68e+04	1.802e-02	1.758e-02	5.123e-02	1.810e-08	3.050e-08	6.724e-03	8.651e-06	0.3362
3126	2023-01-30	GEn	1.10e+05	1.18e+05	8.74e+04	1.776e-02	1.809e-02	4.996e-02	1.810e-08	3.050e-08	6.812e-03	1.507e-05	0.3406
3127	2023-02-27	GEn	1.14e+05	1.14e+05	8.55e+04	-1.993e-02	-1.932e-02	-5.141e-02	1.810e-08	3.050e-08	6.706e-03	6.924e-06	0.3353
3130	2023-02-27	GEn	1.24e+05	1.29e+05	6.28e+04	1.914e-02	2.064e-02	5.084e-02	1.810e-08	3.050e-08	6.313e-03	4.510e-05	0.3157
3162	2023-03-14	GEn	2.30e+05	2.42e+05	1.51e+05	2.141e-02	2.248e-02	5.129e-02	1.810e-08	3.050e-08	1.287e-02	1.628e-05	0.6433
3163	2023-03-14	GEn	2.07e+05	2.22e+05	1.39e+05	2.096e-02	2.209e-02	5.093e-02	1.810e-08	3.050e-08	1.174e-02	2.196e-05	0.5868
3146	2023-09-21	GEn	1.01e+05	1.37e+05	9.13e+04	1.803e-02	1.869e-02	4.982e-02	1.810e-08	3.050e-08	7.162e-03	1.019e-05	0.3581
3148	2023-09-21	GEn	1.04e+05	1.42e+05	9.42e+04	-1.777e-02	-1.891e-02	-4.969e-02	1.810e-08	3.050e-08	7.397e-03	8.932e-06	0.3698
3149	2023-10-19	GEn	1.09e+05	1.13e+05	8.47e+04	-1.896e-02	-1.882e-02	-5.002e-02	1.810e-08	3.050e-08	6.687e-03	8.070e-06	0.3343
3150	2023-10-19	GEn	1.09e+05	1.12e+05	8.45e+04	1.862e-02	1.822e-02	4.955e-02	1.810e-08	3.050e-08	6.670e-03	6.786e-06	0.3335
3154	2023-11-21	GEn	1.20e+05	1.13e+05	9.05e+04	1.677e-02	1.712e-02	4.487e-02	1.810e-08	3.050e-08	7.106e-03	9.781e-06	0.3553
3156	2023-11-21	GEn	1.22e+05	1.15e+05	9.26e+04	-1.557e-02	-1.566e-02	-4.318e-02	1.810e-08	3.050e-08	7.201e-03	1.009e-05	0.3601
3158	2024-04-18	G-RP	1.69e+05	1.46e+05	1.08e+05	1.428e-02	1.336e-02	5.232e-02	1.810e-08	3.050e-08	8.126e-03	9.106e-06	0.4063
3161	2024-04-18	G-RP	1.69e+05	1.41e+05	1.06e+05	-1.422e-02	-1.321e-02	-5.213e-02	1.810e-08	3.050e-08	7.920e-03	8.286e-06	0.3960
3166	2025-05-27	GEp	3.58e+04	5.41e+04	3.12e+04	2.585e-02	3.150e-02	5.319e-02	1.810e-08	3.050e-08	2.820e-03	1.280e-05	0.1410
3167	2025-05-27	GEp	3.75e+04	5.58e+04	3.21e+04	-2.756e-02	-3.258e-02	-5.460e-02	1.810e-08	3.050e-08	2.909e-03	7.231e-06	0.1454
3175	2025-07-10	GEp	4.67e+04	4.54e+04	3.15e+04	-2.956e-02	-3.026e-02	-5.385e-02	1.810e-08	3.050e-08	2.849e-03	1.033e-05	0.1424
3177	2025-07-10	GEp	4.60e+04	4.52e+04	3.10e+04	2.851e-02	2.933e-02	5.319e-02	1.810e-08	3.050e-08	2.789e-03	7.916e-06	0.1395

Table 5: Group-pooled dead-time correction values. Each row corresponds to one group. G-RP abbreviates GEn-RP.

7.4 Accidental Correction Systematics

Accidental corrections, along with their corresponding beam parameters, dates, and experiments, are shown in Figure 34. The percentage corrections are categorized by Wien angle and fitted with a constant to determine the average correction for each reporting period, similar to charge normalization and deadtime adjustments.

Accidental coincidences result from uncorrelated background events within the coincidence timing window. Although generally small, fluctuations in beam current or background conditions can alter their contribution. While the accidental corrections are largely accurate, some data variations remain, and efforts are underway to improve the correction method. As a precaution, a relative systematic uncertainty of 50% has been assigned to each spot to cover the maximum possible error, though for all spots, this uncertainty stays below 0.25%. Applying a 50% uncertainty ensures that any residual modeling errors do not bias the measured asymmetry. Table 6 lists the input quantities in Equation 6.6 for the accidental correction in each measurement group.

Group	Date	Exp	R_{acc} (Hz)	R_C (Hz)	A_{acc}	$A_{\text{raw}}^{\text{Group}}$	$\delta A_{\text{Accid}}^{\text{Group}}$ (% frac.)	$\sigma_{\delta A_{\text{Accid}}^{\text{Group}}}$ (% frac.)	$\Delta_{\text{sys}}^{\text{Group}}$ (%)
3136	2022-10-15	GEn	3.65e+01	3.92e+04	-3.964e-02	-5.248e-02	2.354e-04	8.556e-06	0.0118
3137	2022-10-15	GEn	3.80e+01	4.00e+04	3.888e-02	5.215e-02	2.548e-04	7.074e-06	0.0127
3103	2022-11-17	GEn	2.95e+02	8.42e+04	4.221e-02	5.316e-02	7.445e-04	1.076e-05	0.0372
3106	2022-11-17	GEn	2.89e+02	8.43e+04	-4.067e-02	-5.288e-02	7.961e-04	1.608e-05	0.0398
3108	2022-12-13	GEn	3.12e+02	9.52e+04	3.779e-02	5.265e-02	9.261e-04	9.045e-06	0.0463
3111	2022-12-13	GEn	3.36e+02	9.89e+04	-3.761e-02	-5.279e-02	9.757e-04	9.012e-06	0.0488
3118	2023-01-30	GEn	2.83e+02	8.53e+04	-4.162e-02	-5.405e-02	7.721e-04	2.135e-05	0.0386
3123	2023-01-30	GEn	2.92e+02	8.68e+04	3.560e-02	5.123e-02	1.025e-03	1.414e-05	0.0513
3126	2023-01-30	GEn	2.94e+02	8.74e+04	3.586e-02	4.996e-02	9.494e-04	2.458e-05	0.0475
3127	2023-02-27	GEn	2.92e+02	8.55e+04	-3.925e-02	-5.141e-02	8.008e-04	1.149e-05	0.0400
3130	2023-02-27	GEn	3.62e+02	6.28e+04	3.977e-02	5.084e-02	8.586e-04	8.042e-05	0.0429
3162	2023-03-14	GEn	1.26e+03	1.51e+05	4.388e-02	5.129e-02	1.203e-03	3.111e-05	0.0601
3163	2023-03-14	GEn	1.04e+03	1.39e+05	4.305e-02	5.093e-02	1.159e-03	4.292e-05	0.0580
3146	2023-09-21	GEn	3.14e+02	9.13e+04	3.672e-02	4.982e-02	9.037e-04	1.634e-05	0.0452
3148	2023-09-21	GEn	3.33e+02	9.42e+04	-3.668e-02	-4.969e-02	9.268e-04	1.428e-05	0.0463
3149	2023-10-19	GEn	2.79e+02	8.47e+04	-3.778e-02	-5.002e-02	8.044e-04	1.320e-05	0.0402
3150	2023-10-19	GEn	2.77e+02	8.45e+04	3.684e-02	4.955e-02	8.535e-04	1.109e-05	0.0427
3154	2023-11-21	GEn	3.05e+02	9.05e+04	3.389e-02	4.487e-02	8.238e-04	1.568e-05	0.0412
3156	2023-11-21	GEn	3.18e+02	9.26e+04	-3.123e-02	-4.318e-02	9.505e-04	1.615e-05	0.0475
3158	2024-04-18	GEn-RP	5.59e+02	1.08e+05	2.764e-02	5.232e-02	2.434e-03	1.652e-05	0.1217
3161	2024-04-18	GEn-RP	5.38e+02	1.06e+05	-2.743e-02	-5.213e-02	2.415e-03	1.508e-05	0.1207
3166	2025-05-27	GEp	4.37e+01	3.12e+04	5.735e-02	5.319e-02	-1.259e-04	2.234e-05	0.0063
3167	2025-05-27	GEp	4.72e+01	3.21e+04	-6.014e-02	-5.460e-02	-1.484e-04	1.226e-05	0.0074
3175	2025-07-10	GEp	4.79e+01	3.15e+04	-5.981e-02	-5.385e-02	-1.683e-04	1.875e-05	0.0084
3177	2025-07-10	GEp	4.69e+01	3.10e+04	5.784e-02	5.319e-02	-1.323e-04	1.452e-05	0.0066

Table 6: Group-pooled accidental correction values. Each row corresponds to one group.

7.5 Null Asymmetry Systematics

All false asymmetry measurements from the SBS experiments are presented in Figure 35.

As detailed in Section 6.5, half-wave plate (HWP) states with a negative Möller asymme-

try are sign-corrected ($f_i = \pm 1$). This ensures all values reflect the true positive asymmetry measured by the Møller detector and reference a consistent physical polarization orientation prior to averaging. The weighted averages, constant fits, and associated uncertainties for each experiment are also displayed. The weighted null asymmetry $\bar{A}_{null}$ and its contribution to the systematic uncertainty Δ_{sys}^{Null} are calculated as:

$$\bar{A}_{null} \pm \delta \bar{A}_{null} = \frac{\sum_i f_i A_{null,i} / (\delta A_i)^2}{\sum_i 1 / (\delta A_i)^2} \pm \frac{1}{\sqrt{\sum_i 1 / (\delta A_i)^2}} \quad (7.10)$$

$$\Delta_{sys}^{Null} = \frac{|\bar{A}_{null}| + \delta \bar{A}_{null}}{|\bar{A}_{\text{Møller}}|} \quad (7.11)$$

where f_i is the sign correction factor (± 1) for each asymmetry measurement, $\delta \bar{A}_{null}$ is the uncertainty from the constant fit, and $\bar{A}_{\text{Møller}}$ is the average of the absolute values of measured Møller asymmetries across all experiments. Equation (7.11) normalizes the fitted offset and its uncertainty to provide a conservative estimate of the maximum fractional deviation from sources unrelated to true beam polarization (e.g., instrumental asymmetries or residual helicity-correlated beam properties). Because this systematic uncertainty remains below 1%, false asymmetries do not significantly affect the overall polarization uncertainty. See Table 7 for details of null asymmetry runs.

Group	Date	Experiment	HWP	Energy (GeV)	$A_{measured}$	$\delta A_{measured}$	$\bar{A}_{M\ddot{o}ller}$	Sign	Correction
3105	2022-11-16	GEn	OUT	6.4	-9.80e-05	1.19e-04	0.05260	—	
3109	2022-12-13	GEn	IN	6.4	6.10e-05	1.03e-04	0.05260	+	
3110	2022-12-13	GEn	OUT	6.4	-7.00e-05	1.02e-04	0.05260	—	
3120	2023-01-30	GEn	OUT	8.4	4.40e-05	1.43e-04	0.05260	—	
3124	2023-01-30	GEn	IN	8.4	1.17e-04	1.62e-04	0.05260	+	
3128	2023-02-27	GEn	OUT	8.4	1.30e-05	1.77e-04	0.05260	—	
3131	2023-02-27	GEn	IN	8.4	2.35e-04	1.92e-04	0.05260	+	
3147	2023-09-21	GEn	IN	8.4	-4.60e-05	1.27e-04	0.05260	+	
3152	2023-10-19	GEn	IN	8.4	-8.20e-05	2.99e-04	0.05260	+	
3160	2024-04-18	GEn-RP	IN	4.3	-2.01e-04	1.48e-04	0.05260	—	
3159	2024-04-18	GEn-RP	OUT	4.3	-3.14e-04	1.49e-04	0.05260	+	
3168	2025-05-27	GEP	OUT	10.7	-3.90e-05	2.37e-04	0.05260	—	
3173	2025-07-10	GEP	OUT	10.7	5.48e-04	2.37e-04	0.05260	—	
3176	2025-07-10	GEP	IN	10.7	1.24e-04	5.32e-04	0.05260	+	

Table 7: Grouped false asymmetry measurements using a copper foil during the SBS program.

7.6 Leakage Currents Systematics

All bleedthrough measurements are displayed in Figure 36. Beam energy, measurement dates, related experiments, and currents of other experimental halls are included throughout both the bleedthrough and Møller measurements [121].

For each experiment, the individual data points are fitted with a constant, yielding a single bleedthrough value per experiment using the average ratio of measured coincidence rate to the average Møller coincidence rate:

$$\Delta_{sys} = \frac{2}{N} \sum_{i=1}^N \frac{R_C^i}{\bar{R}_C^{\text{Møller}}} \times 100\% \quad (7.12)$$

where R_C^i is the measured coincidence rate for run i , $\bar{R}_C^{\text{Møller}}$ is the average Møller coincidence rate of the day, and N is the total number of bleedthrough runs taken during the experiment.

CEBAF uses a single polarized injector with a separate drive laser per hall with Hall C drive-laser helicity configured to be opposite of Hall A's. Since the leaked beam carries opposite helicity, even minor contamination can dilute or partially cancel the measured asymmetry. To address this, we conservatively use double the average leakage currents as the total systematic uncertainty in each experiment [18]. This conservative approach ensures that the maximum potential polarization bias from injector cross-talk or incomplete Pockels cell isolation is effectively bounded [128].

Run	Date	Exp	Energy (GeV)	$I_{Hall\ B}$ (nA)	$I_{Hall\ C}$ (μ A)	$I_{Hall\ D}$ (nA)	Coin (Hz)	Møller Coin (Hz)	BleedRate (%)	Δ_{sys} (%)
20065	2022-12-13	GEn	6.4	0.0	50.02	0.0	110.0	96223	0.114318	0.2468
20052	2022-12-13	GEn	6.4	0.0	25.42	213.0	139.0	96223	0.144456	0.2468
20119	2023-01-30	GEn	8.4	0.0	39.99	315.0	6.0	85430	0.007023	0.2468
20181	2023-02-27	GEn	8.4	0.0	69.28	324.0	29.0	77438	0.037449	0.2468
20226	2023-03-13	GEn	8.4	9.68	68.57	319.5	399.0	143832	0.277406	0.2468
20307	2023-09-21	GEn	8.4	0.0	29.93	0.0	0.0	91890	0.000000	0.2468
20322	2023-10-19	GEn	8.4	51.96	29.67	0.0	238.0	84077	0.283074	0.2468
20432	2024-04-18	GEn-RP	4.3	132.84	2.75	0.0	23.0	107450	0.021405	0.0428
20732	2025-05-27	GEp	10.7	326.38	0.0	307.5	13.0	31802	0.040877	0.0336
20770	2025-07-10	GEp	10.7	324.12	0.0	207.0	3.0	31276	0.009592	0.0336
20771	2025-07-10	GEp	10.7	323.45	0.0	207.0	0.0	31276	0.000000	0.0336

Table 8: All bleedthrough runs and their relevant information, which are shown in Figure 36.

The systematic uncertainty due to beam current bleedthrough is 0.2468% for GEN, 0.0428% for GEN-RP, and 0.0336% for GEP

7.7 Analyzing Power and Optics Systematics

Systematic uncertainties on the analyzing power are estimated from the sensitivity of the simulated analyzing power to variations in the relevant parameters of the Møller polarimeter simulation. These parameters include the magnetic field strengths of the four quadrupole magnets and the dipole, radiative Corrections and the Levchuk effect, acceptance, and the intrinsic error of the Monte Carlo simulation. In Møller scattering, the analyzing power A_{zz} is calculable by QED and depends on the scattering angle in the center-of-mass frame; thus, any factor that changes the angular acceptance or the reconstructed kinematics directly affects the polarization measurement. For high-precision polarimetry, meticulous control of optics and detector acceptance is essential.

Regarding the magnet systematics, simulations were performed under ideal optics for each energy level detailed in Section 4.6. Additional simulations with tailored settings were conducted to assess the sensitivity of the analyzing power to each component. For each of the four quadrupole magnets and the dipole, the magnetic fields were individually altered by 1% in both directions, and the resulting percentage change in the corrected analyzing power was calculated. Furthermore, the dipole entrance lead jaws were closed by 5mm. Acceptance effects are especially crucial because the Møller analyzing power varies rapidly near the kinematic edges of the acceptance window. As a result, small geometric shifts can cause non-linear variations in the averaged A_{zz} . Similar systematic considerations are discussed in [159]. In separate tests, the Levchuk effect was disabled, then radiative corrections were enabled. Radiative corrections modify the effective analyzing power

both through genuine QED corrections to the polarized cross section and through their effect on the reconstructed kinematics and acceptance [141, 198], whereas the Levchuk effect arises because the electrons carrying the iron magnetization — predominantly the 3d electrons — are weakly bound valence electrons with a narrower intra-atomic momentum distribution than the tightly bound core (inner-shell) electrons, which make up most of the 26 atomic electrons and dominate the high-momentum part of the target. Folded with the finite acceptance and the angular dependence of A_{zz} , this difference between the polarized and unpolarized momentum distributions shifts the acceptance-averaged effective analyzing power. Lastly, the field of the target superconducting magnet was varied by 2% in each direction. All simulations using the provided optics solutions were run with event numbers such that the coincidence rate of the specific settings allowed for a 0.1% statistical error.

The percent change in analyzing power A_{zz} was calculated as:

$$\Delta = \frac{A_{zz}^{\text{base}} - A_{zz}^{\text{modified}}}{A_{zz}^{\text{base}}} \times 100\% \quad (7.13)$$

When the four quadrupole contributions were combined, the percent changes at each energy setting were minimal; the highest combined error across all energies was rounded up, and a conservative total of 0.1% was assigned to the quadrupoles as a group. The same conservative approach was applied to the dipole: the combined contribution from its $\pm 1\%$ field variations was rounded up to a total of 0.1% as well.

The total systematic error was then obtained by combining all individual contributions in quadrature:

$$\delta_{\text{total}} = \sqrt{\delta_1^2 + \delta_2^2 + \cdots + \delta_n^2} \quad (7.14)$$

Settings ↓ Energy(GeV) →	4.2904	4.3600	6.3729	6.4480	8.4484	8.5900	10.662
	(%)	(%)	(%)	(%)	(%)	(%)	(%)
All Quadrupoles	0.10000	0.10000	0.10000	0.10000	0.10000	0.10000	0.10000
Dipole	0.10000	0.10000	0.10000	0.10000	0.10000	0.10000	0.10000
Dipole Lead Jaws (ϕ)	0.10000	0.10000	0.10000	0.10000	0.10000	0.10000	0.10000
Levchuk Effect	0.19800	0.24000	0.19800	0.28200	0.30000	0.27900	0.02100
Radiative Corrections	0.10000	0.10000	0.10000	0.10000	0.10000	0.10000	0.10000
Target Solenoid	0.10000	0.10000	0.10000	0.10000	0.10000	0.10000	0.10000
Monte Carlo Stat	0.10000	0.10000	0.10000	0.10000	0.10000	0.10000	0.10000
Total	0.31497	0.34293	0.31497	0.37353	0.38730	0.37127	0.24585

Table 9: Analyzing power systematic error summary. The two crossed-out energy columns correspond to settings that GEp never utilized.

7.8 High Current and Slit Dependence Systematics

High Current dependence and slit dependence were not directly measured during the SBS experiments, but both remain potential sources of systematic uncertainty. Because the main SBS experiments operate at much higher beam currents than Møller polarimetry, we must account for possible differences in helicity-correlated beam properties—such as photocathode charge-limit effects at the source—that may not be captured at the lower currents used for polarization measurements. Based on a 2007 study [199], we applied a conservative systematic uncertainty of 0.5% for all SBS experiments for current dependence. Previous injector and source studies at Jefferson Lab have shown that slit dependence effects can be limited to below the sub-percent level through careful tuning [200], assigning a maximum systematic uncertainty of 0.1% to all SBS experiments. Using these historical upper limits provides a conservative estimate in the absence of a dedicated SBS-era study.

7.9 Target Polarization Systematics

Møller target polarizations and their associated systematic errors are shown in Figure 37. One source of systematic uncertainty comes from how well we understand the saturation spin polarization. The saturation magnetization of iron foils determines the maximum magnetization the material can achieve. As explained in [129], the room temperature target polarization for a fully saturated iron foil is 0.08020, with an uncertainty of 0.00019. To estimate the percentage systematic error caused by saturation polarization, this error is compared to the temperature-adjusted target polarization, which is often lower than the quoted value above [201].

Because beam polarizations are reported for each SBS experiment at various Wien angles, the polarization systematic error should be calculated individually for each case. While saturation polarization values differ among sections, the systematic errors are generally around 0.24% or less. Furthermore, a small systematic error arises from the target's heating correction, which minimally affects the total systematic error since errors are combined in quadrature.

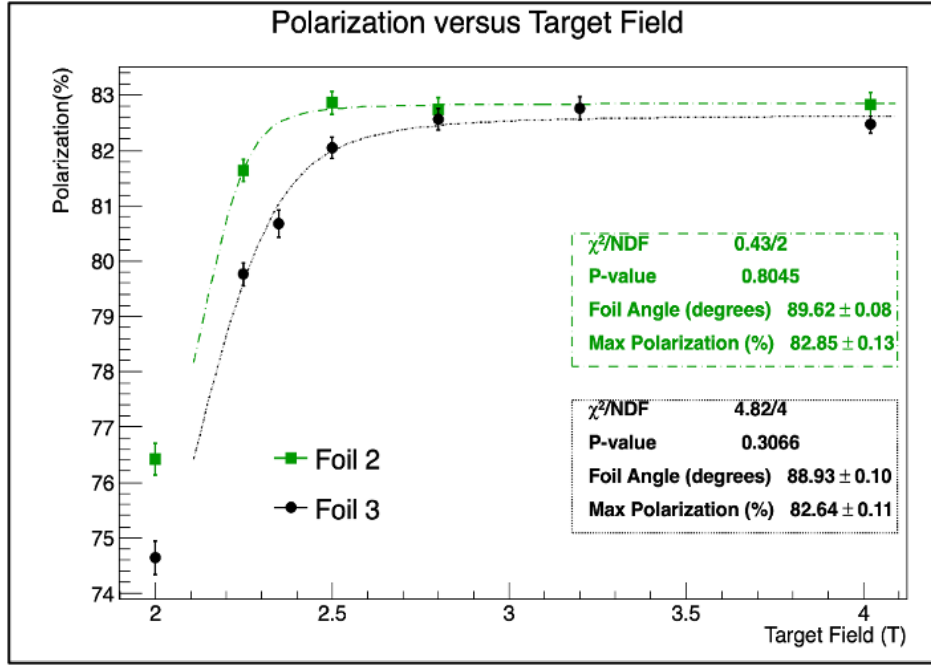


Figure 42: Stoner–Wohlfarth fits to the Møller target polarization for targets 2 and 3 as a function of the applied magnetic field near the target.

Figure 42 displays polarization measurements at different magnetic fields near the target, fitted using the Stoner-Wohlfarth model for Møller targets 2 and 3, both composed of 10-micron iron foils [202]. Although both targets achieve full saturation, a polarization difference of 0.25% remains, which is incorporated into the systematic uncertainties and combined in quadrature. The Stoner-Wohlfarth model offers a single-domain explanation for magnetization reversal and provides a useful single-domain approximation for the approach to saturation of thin ferromagnetic foils under strong applied fields [203, 204, 205, 206]. The observed 0.25% difference between targets probably results from minor differences in foil thickness, strain, or mounting conditions. Accounting for this variation as an independent factor ensures that any target-to-target discrepancies are thoroughly addressed.

As noted in Section 6.8, during the GEp experiment, the precise beam spot size was not

measured due to instrument limitations, and consequently, the target polarizations were estimated using the known beam current and the average of all prior beam spot sizes. The combined systematic uncertainties are approximately 0.35% for most measurement points. However, other factors might affect this figure, and more detailed studies are needed to verify that the total systematic uncertainty remains below 0.35%. To adopt a cautious approach, we assign a systematic error of 0.40% to all measurements to account for potential issues such as foil misalignment or wrinkles. Foil alignment relative to the beam axis affects the effective projection of magnetization along the beam direction. Even slight angular misalignments can decrease the longitudinal part of the target polarization. Adding an extra margin accounts for mechanical imperfections that aren't explicitly modeled, ensuring they are included within the quoted uncertainty, such as when targeting polarizations were theoretically estimated.

7.10 Spin Precession Systematics – $\delta p/p$

Variations in the fractional beam momentum ($\delta p/p$) between the dedicated Møller measurements and standard production runs induce direct variability in the longitudinal polarization at the target. The absolute shift in the spin orientation angle, $\delta\phi_{\text{tot}}$, is determined by evaluating the arc sums of Equation 6.10 in closed form and scaling the result by the fractional momentum mismatch, $\Delta(\delta p/p) = (\delta p/p)_{\text{prod}} - (\delta p/p)_{\text{Møller}}$:

$$\delta\phi_{\text{tot}} = \frac{\theta \Delta(\delta p/p)}{440.65 \text{ MeV}} \left[(2n-1) E_{\text{inj}} + n^2 E_{\text{NL}} + n(n-1) E_{\text{SL}} \right] + \frac{\phi \Delta(\delta p/p)}{440.65 \text{ MeV}} \left[E_{\text{inj}} + n E_{\text{NL}} + n E_{\text{SL}} \right] \quad (7.15)$$

where the bracketed terms represent the closed-form energy summations for the north

and south arcs and the Hall A arc, respectively. The variables and their standard physical units are explicitly defined as follows: n is the pass number (dimensionless), E_{inj} is the injector energy (MeV), E_{NL} and E_{SL} are the per-pass linac energy gains (MeV), $\theta = \pi$ rad (180°) is the recirculation-arc bend angle, and $\phi \approx 0.6545$ rad (37.5°) is the Hall A bend angle.

Once this total precession deviation is calculated, it propagates into the systematic uncertainty of the analyzing power. The systematic error, Δ_{sys} (dimensionless), is quantified by evaluating the fractional change in the longitudinal projection of the spin vector due to this angular shift:

$$\Delta_{\text{sys}} = \frac{\cos(\theta_{\text{offset}} - \delta\phi_{\text{tot}}) - \cos(\theta_{\text{offset}})}{\cos(\theta_{\text{offset}})} \quad (7.16)$$

where θ_{offset} (rad) is the angular difference between the empirically measured Wien angle and the ideal Wien angle that yields maximum longitudinal polarization.

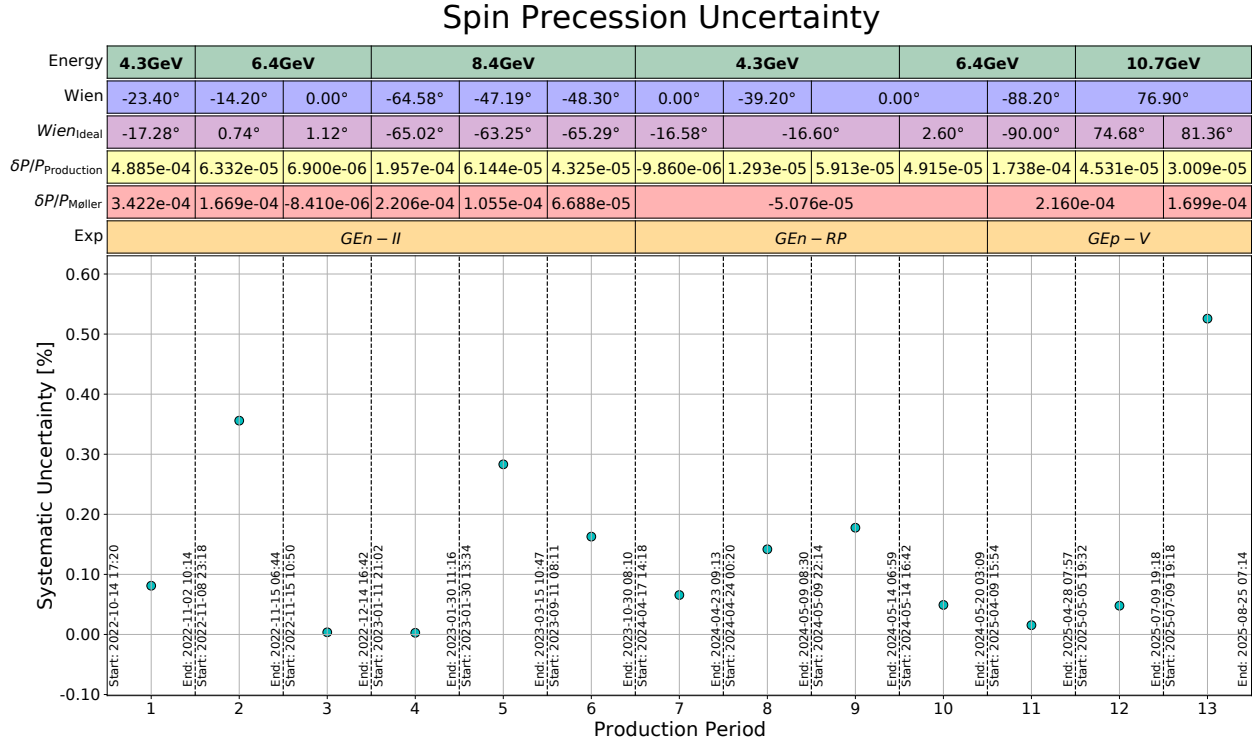


Figure 43: Systematic uncertainties from spin precession effects for Møller polarimetry across all production periods of the SBS program. The data points correspond to the values listed in Table 10.

Figure 43 presents the calculated systematic errors in spin precession across all production periods. The magnitude of these errors varies significantly with beam energy. At higher energies the per-arc spin precession increases, rendering the final longitudinal polarization component highly sensitive to minor momentum fluctuations. This increased sensitivity, coupled with larger $\Delta(\delta p/p)$ deviations frequently observed between Møller and production periods, accounts for the amplified systematic uncertainties at the highest SBS energies. Table 10 displays the spin precession systematics from the plot above.

Start Date	End Date	Experiment	$\delta p/p_{\text{prod}}$	$\delta p/p_{\text{Møller}}$	E_{beam} (GeV)	θ_{wien} (deg)	$\theta_{\text{wien}}^{\text{ideal}}$ (deg)	$\delta\phi_{\text{total}}$ (rad)	Δ_{sys} (%)
2022-10-14	2022-11-02	<i>GEN – II</i>	4.88e−4	3.42e−4	4.3	-23.3968	-17.28	0.0079	0.0811
2022-11-08	2022-11-15	<i>GEN – II</i>	6.33e−5	1.67e−4	6.4	-14.1985	0.74	-0.0130	0.3559
2022-11-15	2022-12-14	<i>GEN – II</i>	6.90e−6	−8.41e−6	6.4	0.0	1.12	0.0019	0.0036
2023-01-11	2023-01-30	<i>GEN – II</i>	1.96e−4	2.21e−4	8.4	-64.5836	-65.02	-0.0057	0.0027
2023-01-30	2023-03-15	<i>GEN – II</i>	6.14e−5	1.05e−4	8.4	-47.1942	-63.25	-0.0100	0.2832
2023-09-11	2023-10-30	<i>GEN – II</i>	4.33e−5	6.69e−5	8.4	-48.3	-65.29	-0.0054	0.1628
2024-04-17	2024-04-23	<i>GEN – RP</i>	−9.86e−6	−5.08e−5	4.3	0.0	-16.58	0.0022	0.0656
2024-04-24	2024-05-09	<i>GEN – RP</i>	1.29e−5	−5.08e−5	4.3	-39.2	-16.6	0.0034	0.1418
2024-05-09	2024-05-14	<i>GEN – RP</i>	5.91e−5	−5.08e−5	4.3	0.0	-16.6	0.0059	0.1777
2024-05-14	2024-05-20	<i>GEN – RP</i>	4.92e−5	−5.08e−5	6.4	0.0	2.6	0.0126	0.0491
2025-04-09	2025-04-28	<i>GEP – V</i>	1.74e−4	2.16e−4	6.4	-88.2	-90.0	-0.0054	0.0154
2025-05-05	2025-07-09	<i>GEP – V</i>	4.53e−5	2.16e−4	10.7	76.9	74.68	-0.0621	0.0478
2025-07-09	2025-08-25	<i>GEP – V</i>	3.01e−5	1.70e−4	10.7	76.9	81.36	-0.0508	0.5257

Table 10: $\delta p/p$ variation systematics

7.11 Polarization Drift Systematics

Beam polarization is not strictly static; it can drift over extended intervals between direct Møller measurements due to evolving conditions at the electron source. Factors such as quantum efficiency degradation at the photoinjector photocathode—and the subsequent laser spot adjustments required to maintain beam current—can introduce small variations in the delivered polarization.

To account for the increased uncertainty when projecting known polarization values across these measured production periods, an additional time-dependent systematic uncertainty (Δ_{sys}) is applied. Specifically, an absolute systematic error of 0.5% is added for every two weeks of polarization projection required. This scaling penalty is applied to any production data collected either prior to the first available Møller measurement or after the final

measurement of the experimental run. The specific assigned projection uncertainties for each relevant data collection period are detailed in Table 11.

Beam Start	Beam End	SBS Start	SBS End	Møller Start	Møller End	Δ_{sys} (%)	$\theta_{Wien}(\delta)$ (deg)	Exp-P
22-10-14 17:22	22-10-30 09:10	22-10-17 20:21	22-10-30 09:12	22-10-14 23:00	22-10-15 13:00 ⁺	0.50	-23.40(-6)	$G_E^n 2$
22-11-08 23:18	22-11-15 06:43	22-11-11 09:02	22-11-15 04:44				-14.20(-15)	$G_E^n 3$
22-11-16 02:58	22-12-14 16:41	22-11-16 09:19	22-12-14 16:41	22-11-17 08:00	22-12-13 23:30	0.00	0.00(-1)	$G_E^n 3$
23-01-11 21:03	23-01-30 11:15	23-01-15 17:36	23-01-30 04:14	⁺ 23-01-30 04:10	23-01-30 04:14	0.50	-64.58(2)	$G_E^n 4$
23-01-30 13:35	23-03-15 07:33	23-01-31 12:00	23-03-12 03:00	23-01-30 04:14	23-03-12 03:00	0.00	-47.19(16)	$G_E^n 4$
23-09-13 01:18	23-10-30 08:08	23-09-16 03:09	23-10-30 06:30	⁺ 23-09-21 08:00	23-10-19 07:45 ⁺	0.50	-48.30(17)	$G_E^n 4$
24-04-17 14:18	24-04-23 09:13	24-04-18 15:28	24-04-23 08:55	24-04-18 04:25	24-04-18 08:25 ⁺	0.25	0.00(17)	$G_E^{nRP} 2$
24-04-24 00:19	24-05-09 08:29	24-04-24 03:50	24-05-09 08:29				-39.20(-23)	$G_E^{nRP} 2$
24-05-09 22:14	24-05-14 06:58	24-05-09 23:46	24-05-14 06:58				0.00(17)	$G_E^{nRP} 2$
24-05-14 16:43	24-05-20 03:09	24-05-14 19:55	24-05-19 13:54				0.00(-3)	$G_E^{nRP} 3$
25-04-09 15:54	25-04-28 07:57	25-04-11 21:11	25-04-27 13:11				-88.20(2)	$G_E^p 3$
25-05-05 19:32	25-07-09 13:00	25-05-09 21:45	25-07-09 05:18	⁺ 25-05-27 08:35	25-05-27 16:20 ⁺	2.00	76.90(2)	$G_E^p 5$
25-07-09 15:00	25-08-25 07:13	25-07-09 19:54	25-08-25 07:13	25-07-10 08:25	25-07-10 14:10 ⁺	1.50	76.90(-5)	$G_E^p 5$

Table 11: Polarization drift systematics during periods with Møller measurements. A ⁺ symbol in the Møller Start and End columns indicates that beam polarization must be projected beyond the Møller measurement period, as production extended beyond the available Møller data.

7.12 Unmeasured Polarization Systematics

The table immediately below shows this forward-projected result. Below that is a second table showing the same polarization estimated by backward projecting from the production period immediately following the unmeasured interval. For both methods, an assumption was made that the Wien angle was not known to better than 5° of its intended value; this assumption was used to calculate two separate errors: one associated with the extracted source polarization, and one associated with the projected beam polarization at the period with missing data (referred to as the projected polarization in the tables). The projected and source errors were added in quadrature for both methods. The weighted

average of the two polarization estimates was then assigned to the spot, along with the average of their respective total errors [207]. This is not the conventional way of combining errors, but since both values are fairly conservative, this approach is reasonable. Further in this section, the calculation of these errors will be discussed in detail.

Start	End	E_{beam} (GeV)	θ_{Wien} (deg)	$\theta_{\text{Wien}}^{\text{ideal}}$ (deg)	Møller?	Exp.	P_{Meas} (%)	P_{src} (%)	δP_{src} (%)	P_{proj} (%)	P_{final} (%)	δP_{proj} (%)	δP_{total} (%)
2022-10-14	2022-11-02	4.291	-23.40	-17.28	yes	GEn	84.09	84.57	0.94	-	-	-	-
2022-11-08	2022-11-15	6.373	-14.20	0.74	no	GEn	-	-	-	81.71	82.99	2.33	2.51

Table 12: Forward polarization projection from 4.29 GeV measurements for GEn production.

Start	End	E_{beam} (GeV)	θ_{Wien} (deg)	$\theta_{\text{Wien}}^{\text{ideal}}$ (deg)	Møller?	Exp.	P_{Meas} (%)	P_{src} (%)	δP_{src} (%)	P_{proj} (%)	P_{final} (%)	δP_{proj} (%)	δP_{total} (%)
2022-11-08	2022-11-15	6.373	-14.20	0.74	no	GEn	-	-	-	84.27	82.99	2.33	2.33
2022-11-15	2022-12-14	6.373	0.00	1.12	yes	GEn	87.20	87.21	0.17	-	-	-	-

Table 13: Backward polarization projection from 6.37 GeV measurements for GEn production.

The uncertainty in the extracted source polarization is computed by propagating a 5° uncertainty in the Wien angle through the cosine relationship. Denoting the angular offset as $\theta_{\text{offset}} = \theta_{\text{Wien}} - \theta_{\text{Wien}}^{\text{ideal}}$, the fractional error on the source polarization is:

$$\delta P_{\text{src}} = \frac{P_{\text{src}}^+ - P_{\text{src}}^-}{2P_{\text{src}}} * 100\% = \frac{\frac{P_{\text{meas}}}{\cos(\theta_{\text{offset}} + 5^\circ)} - \frac{P_{\text{meas}}}{\cos(\theta_{\text{offset}} - 5^\circ)}}{2 \frac{P_{\text{meas}}}{\cos(\theta_{\text{offset}})}} * 100\% \quad (7.17)$$

Similarly, the uncertainty in the projected polarization at the unmeasured period is:

$$\delta P_{\text{proj}} = \frac{P_{\text{proj}}^+ - P_{\text{proj}}^-}{2P_{\text{projected}}} * 100\% = \frac{P_{\text{src}} \cos(\theta_{\text{offset}} + 5^\circ) - P_{\text{src}} \cos(\theta_{\text{offset}} - 5^\circ)}{2 P_{\text{src}} \cos(\theta_{\text{offset}})} * 100\% \quad (7.18)$$

Start	End	E_{beam} (GeV)	θ_{Wien} (deg)	$\theta_{\text{Wien}}^{\text{ideal}}$ (deg)	Møller?	Exp.	P_{Meas} (%)	P_{src} (%)	δP_{src} (%)	P_{proj} (%)	δP_{proj} (%)	δP_{total} (%)
2024-04-17	2024-04-23	4.288	0.00	-16.58	yes	GEN-RP	83.59	87.22	2.62	-	-	-
2024-04-24	2024-05-09	4.288	-39.20	-16.60	no	GEN-RP	-	-	-	80.52	3.63	4.47
2024-05-09	2024-05-14	4.288	0.00	-16.60	no	GEN-RP	-	-	-	83.593	0	0.00
2024-05-14	2024-05-20	6.368	0.00	2.60	no	GEN-RP	-	-	-	87.13	0.40	2.65

Table 14: Polarization projections for GEN-RP production periods at 4.29 GeV and 6.37 GeV.

During the GEN-RP experiment from 2024-04-24 to 2024-05-09, the significant difference between the Wien angle and its ideal value causes errors to increase substantially, surpassing 4%. However, from 2024-05-09 to 2024-05-14, since identical kinematic settings as those used between 2024-04-17 and 2024-04-23 were employed, the systematic error related to beam polarization extrapolation is considered to be 0%.

Start	End	E_{beam} (GeV)	θ_{Wien} (deg)	$\theta_{\text{Wien}}^{\text{ideal}}$ (deg)	Møller?	Exp.	P_{Meas} (%)	P_{src} (%)	δP_{src} (%)	P_{proj} (%)	δP_{proj} (%)	δP_{total} (%)
2025-04-09	2025-04-28	6.448	-88.20	-90.00	no	GEp	-	-	-	86.25	0.27	0.44
2025-05-05	2025-07-09	10.662	76.90	74.68	yes	GEp	86.23	86.29	0.34	-	-	-

Table 15: Polarization projections and measurements for GEp production periods.

CHAPTER 8

Results

8.1 Hall A Møller Measurements

The results from the fully analyzed, corrected, and grouped beam polarization measurements at Jefferson Lab’s Hall A, obtained with the Hall A Møller polarimeter, are shown in the Figure 44. Measurements taken within a few minutes under the same settings are grouped together to minimize statistical error. This grouping of closely spaced measurements under stable beam and optics conditions reduces fluctuations while maintaining systematic accuracy [208].

Above each group, the respective beam energy, measurement date, Wien angle at the injector, photocathode quantum efficiency at the injector, and the experiment conducted are listed. The legend shows the weighted average beam polarization for each measurement period, along with the statistical uncertainty of the fit and the overall systematic uncertainty for that period. A measurement period is the time during which the Wien angle, beam energy, and laser-spot at the injector remain constant. Since the total beam precession and the percentage of longitudinal beam polarization remain consistent during such period, it is feasible to consolidate the beam polarizations in this manner. Start and end dates are indicated near dashed vertical lines. The left side of the vertical axis indicates the percentage of electron beam polarization, while the right side displays the measured asymmetry percentage, as described in Equation 4.19.

For each measurement period, a single combined polarization value along with its errors is reported to facilitate the analysis of physics asymmetries in the main experiment, as described in Section 1.4.

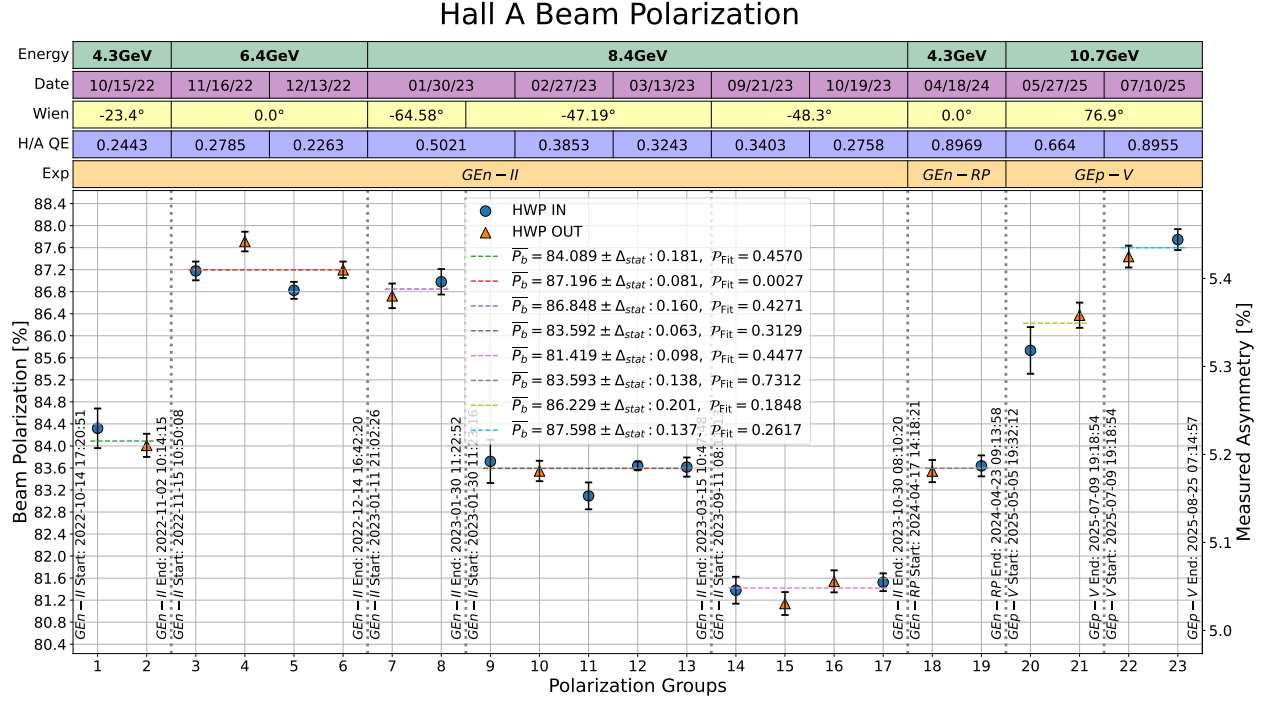


Figure 44: Beam polarization measurements at Hall A with grouped values and statistical uncertainties.

8.2 Hall C Møller Measurements During SBS

Beam polarization at Jefferson Lab was also measured in experimental Halls B and C for several periods. The results from the Hall C Møller polarimeter, compared to Hall A's measurements, are shown in Figure 45.

Longitudinal polarizations differ between halls due to variations in beam energy and accumulated spin precession. Although each hall receives a different component of the same source polarization, the measurements stay consistent. This agreement validates the con-

sistency of spin transport calculations and confirms that systematic effects associated with hall-dependent optics remain under control during SBS running.

Møller Polarimetry at Hall A & Hall C during the SBS Program

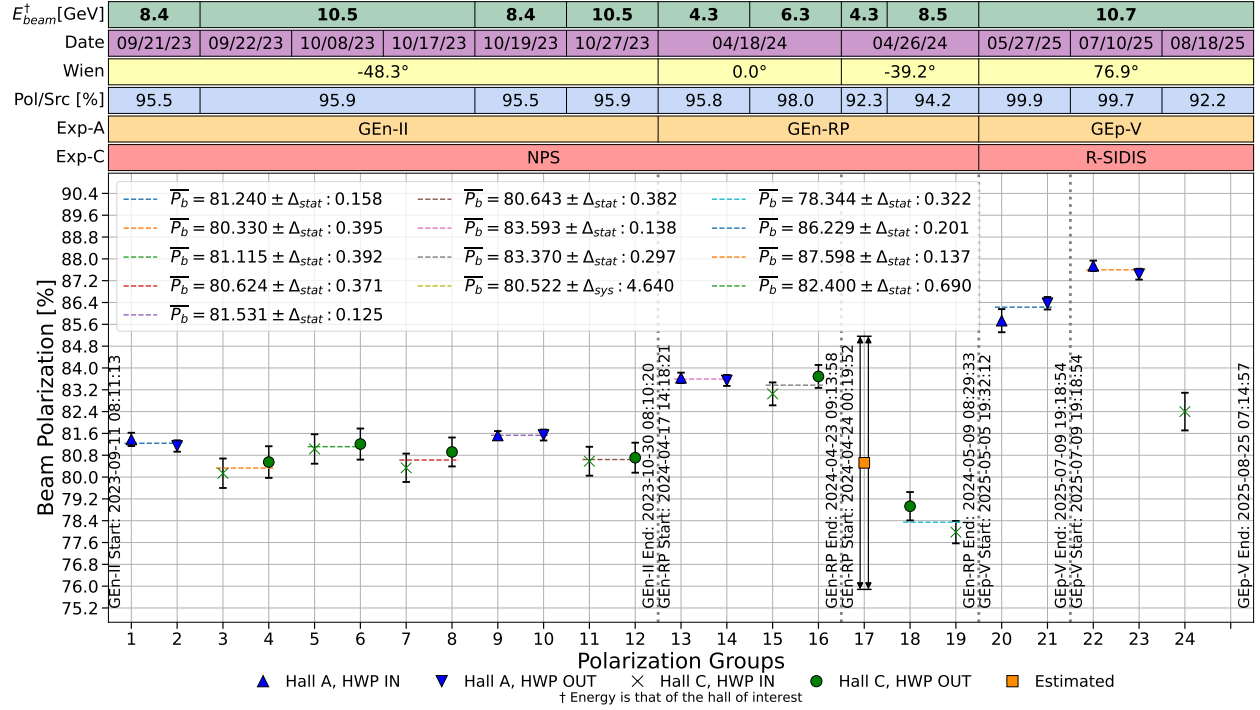


Figure 45: Comparison of Hall A and Hall C beam polarization during SBS experiments.

8.3 Estimated Missing Beam Polarizations

Figure 46 shows the estimated beam polarizations combined into one plot.

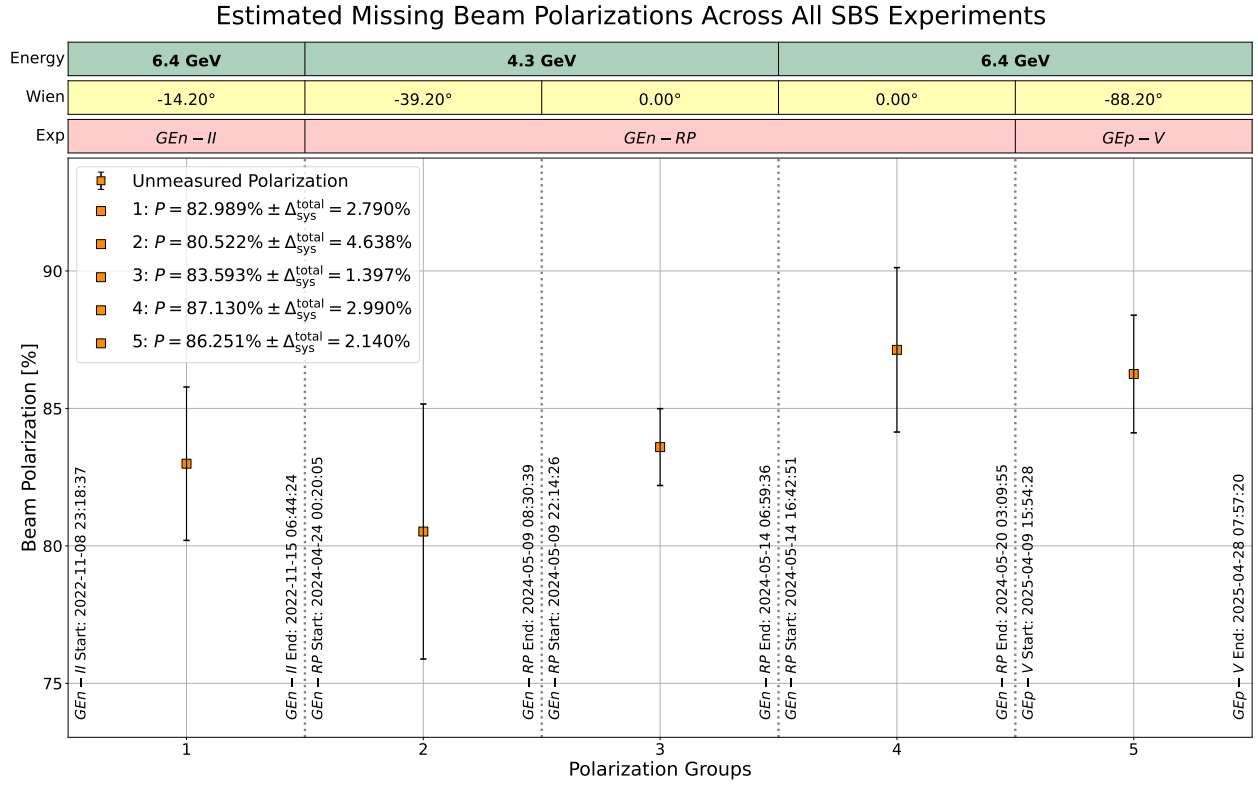


Figure 46: Estimated beam polarizations for all SBS production periods without direct Møller measurements, inferred from Wien-angle offsets and spin precession. A highlighted legend entry denotes a value taken from the beam polarization measurement plot; its uncertainty is the total systematic uncertainty from all factors. For all other points, the uncertainty is systematic and arises from estimating the beam polarization using Wien-angle offsets and spin precession.

8.4 Møller Polarimetry Systematic Uncertainty Summary

Table 16 summarizes the systematic uncertainties for each SBS production period. Each row gives the contribution from a specific correction or effect, in percent; the final row reports the quadrature sum used as the total systematic uncertainty for that period.

Pass Exp.	2 GEn	3* GEn	3 GEn	4 GEn	4 GEn	4 GEn	2 G-RP	2* G-RP	2* G-RP	3* G-RP	3* GEp	5 GEp	5 GEp
A_{zz}	0.32	0.34	0.34	0.39	0.39	0.39	0.32	0.32	0.32	0.34	0.32	0.25	0.25
Target	0.40	0.40	0.40	0.40	0.40	0.40	0.40	0.40	0.40	0.40	0.40	0.40	0.40
Deadtime	0.15	0.22	0.36	0.34	0.39	0.35	0.40	0.40	0.40	0.40	0.14	0.14	0.14
Chg. Norm.	1.29	0.74	0.14	0.80	0.12	0.13	0.33	0.33	0.33	0.33	0.33	0.33	0.24
Accidentals	0.01	0.02	0.04	0.05	0.04	0.04	0.12	0.12	0.12	0.12	0.01	0.01	0.01
Bleedthrough	0.25	0.25	0.25	0.25	0.25	0.25	0.04	0.04	0.04	0.04	0.03	0.03	0.03
Precession	0.08	0.36	0.00	0.00	0.28	0.16	0.07	0.14	0.18	0.05	0.02	0.05	0.53
Null Asym.	0.18	0.18	0.18	0.18	0.18	0.18	0.31	0.31	0.31	0.31	0.80	0.80	0.80
Pol. Drift	0.50	1.00	0.00	0.50	0.00	0.50	0.25	0.75	1.00	1.00	1.75	2.00	1.50
Pol. Est.	0.00	2.33	0.00	0.00	0.00	0.00	0.00	4.47	0.00	2.65	0.44	0.00	0.00
Curr. Dep.	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50
Slit Dep.	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.10
Stat. Transf.	0.00	0.18	0.00	0.00	0.00	0.00	0.00	0.14	0.14	0.14	0.20	0.00	0.00
Total	1.60	2.79	0.88	1.29	0.95	1.04	0.98	4.64	1.40	2.99	2.14	2.29	1.93

Table 16: Møller polarimetry systematic uncertainty summary for each SBS production period. All entries are in percent. Pass number denotes the CEBAF accelerator pass; an asterisk marks periods without a direct beam polarization measurement. G-RP abbreviates GEn-RP.

8.5 Beam Polarization Results Delivered to SBS Experiments

Table 17 compiles the polarization values delivered to the SBS physics analysis. For each run period it lists the weighted-average longitudinal beam polarization $\bar{P}_b$ together with the combined statistical and systematic uncertainties adopted for that interval.

Start	End	Exp	$\bar{P}_b$ (%)	Δ_{stat} (%)	Δ_{sys} (%)
2022-10-14 17:22	2022-11-02 10:14	GE _n	84.089	0.181	1.602
2022-11-08 23:18	2022-11-15 06:43	GE _n	82.898	0.000	2.790
2022-11-15 10:50	2022-12-14 16:42	GE _n	87.196	0.081	0.884
2023-01-11 21:03	2023-01-30 11:15	GE _n	86.848	0.160	1.289
2023-01-30 13:35	2023-03-15 07:33	GE _n	83.592	0.063	0.954
2023-09-13 01:18	2023-10-30 08:08	GE _n	81.419	0.098	1.038
2024-04-17 14:18	2024-04-23 09:13	GE _n -RP	83.593	0.138	0.983
2024-04-24 00:19	2024-05-09 08:29	GE _n -RP	80.522	0.000	4.638
2024-05-09 22:14	2024-05-14 06:58	GE _n -RP	83.593	0.000	1.397
2024-05-14 16:43	2024-05-20 03:09	GE _n -RP	87.130	0.000	2.990
2025-04-09 15:54	2025-04-28 07:57	GE _p	86.251	0.000	2.140
2025-05-05 19:32	2025-07-09 19:18	GE _p	86.229	0.201	2.291
2025-07-09 19:18	2025-08-25 07:13	GE _p	87.598	0.137	1.928

Table 17: Møller polarimetry for SBS final summary table.

8.6 Conclusions

In summary, the Møller Polarimetry Working Group successfully operated the Hall A Møller polarimeter throughout the SBS form-factor experiments. Together, we delivered calibrated longitudinal beam polarizations for all 13 production periods spanning from 2022 to 2025, with final polarization values ranging between 80.5%–87.6%.

To ensure the accuracy of these measurements, our group developed comprehensive period-by-period corrections that accounted for charge and pedestal effects, deadtime, and accidentals. We also established a rigorous systematic budget encompassing target effects, analyzing power, bleedthrough, false asymmetry, spin precession, and polarization drift. For the eight directly measured periods, we achieved systematic uncertainties between 0.88%–2.3%, alongside statistical uncertainties of $\leq 0.21\%$. For the remaining five inferred

periods, total systematic uncertainties ranged from $1.4\% - 4.6\%$, which were successfully derived utilizing the Wien and spin precession model.

The reliability of these results is further supported by overlapping measurements taken in Hall C, which confirm the consistency of our spin transport calculations. Ultimately, for GEN-II, the experiment most sensitive to polarimetry, our group maintained systematic uncertainties of $< 3\%$ across all periods, achieving an exceptional precision of 0.88% during our best measurement period. This achievement ensures that polarimetry uncertainties remain subdominant to other sources of error and fall perfectly in line with the rigorous experimental needs of the program.

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VITA

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Education

Syracuse University, Syracuse, NY

Ph.D. in Physics, 2026 (expected)

Dissertation research in experimental and computational particle physics with emphasis on instrumentation, data-driven analysis, and large-scale scientific systems.

City College of New York (CUNY), New York, NY

M.S. in Physics, 2019

Baruch College (CUNY), New York, NY

B.A. in Physics and Mathematics, 2016

Research Experience

Experimental and Computational Particle Physics

Syracuse University, 2021–Present

Advisor: Dr. Paul Souder

- Conduct research on medium-energy particle physics experiments, including Super Bigbite Spectrometer form factor measurements and Møller polarimetry for precision beam diagnostics.
- Develop and validate simulation and analysis pipelines using GEANT4, ROOT, C++, and Python to model detector response, optimize experimental configurations, and assess systematic uncertainties.
- Apply data-driven and machine learning methods to large experimental datasets for signal detection, particle tracking, detector tuning, and performance monitoring.
- Contribute to experiment operations, instrumentation studies, and collaborative analysis within a large, multi-institution scientific collaboration.
- Mentor junior students and collaborators in computational workflows, data analysis practices, and experimental concepts.

Computational Astrophysics

CityTech (CUNY), 2017–2019

Advisor: Dr. Viviana Acquaviva

- Studied Lyman Alpha Emitting galaxies to investigate galaxy formation and cosmic history.
- Applied machine learning methods including classification, regression, and support vector machines.
- Used Bayesian statistics to probabilistically classify emission-line galaxies.

Experimental Condensed Matter Physics

Meriles Group, City College of New York (CUNY), 2017–2018

- Studied nitrogen-vacancy and silicon-vacancy centers in diamond for long-term data storage applications.
- Utilized MATLAB to enhance the efficiency of calculations for fast-switching magnetic coils and to develop hardware control interfaces.
- Assisted with laboratory setup including solenoid construction, laser alignment, and designing a thermal management system.

Relevant Coursework

Graduate:

Computational Physics; Classical Mechanics; Advanced Electromagnetic Theory I & II; Modern Instrumentation; Thermodynamics and Statistical Mechanics; Quantum Mechanics I & II; Graduate Problem Solving; Analytical Dynamics; Curriculum Problems in Science Education

Undergraduate:

Linear Algebra and Matrix Analysis; Single and Multivariable Calculus; Algorithms and Programming (C++); Probability and Mathematical Statistics

Skills

Experimental & Hardware Systems:

Particle detector development, timing detectors, experimental instrumentation, detector

calibration, data acquisition support, hardware-software integration, laboratory setup and diagnostics

Programming & Scientific Computing:

Python, C++, C, SQL, Bash, Shell, Git, MATLAB, Mathematica

Modeling, Simulation & Data Analysis:

Numerical simulation, GEANT4, ROOT, statistical modeling, uncertainty analysis, machine learning

Computational Infrastructure:

Linux, HPC environments, Docker, Apache Airflow, Spark, reproducible data pipelines, SQL databases

Course & Curriculum Development:

Syllabus design, lecture development, problem sets, homework, laboratory exercises, exams, assessment design, learning outcomes alignment

Teaching Experience

Course Instructor

Syracuse University

AST 101: Our Corner of the Universe

Designed and delivered lectures for a general-education astronomy course emphasizing conceptual understanding, quantitative reasoning, and scientific literacy for non-majors.

Adjunct Lecturer

Baruch College (CUNY), 2016–2019

Fundamentals of Experimental Physics (Laboratory)

Fundamentals of Physics: Theory and Practice (Lecture)

Calculus I Preview

Beginning and Intermediate Algebra

Designed course materials, laboratory exercises, homework, and exams; held regular office hours and provided academic mentoring to a diverse student population.

Teaching Assistant

Syracuse University, 2019–2021

Major Concepts of Physics; Stars, Galaxies, and the Universe

Workshop Instructor and Teaching Assistant

Baruch College (CUNY), New York, NY, 2016–2018

Calculus I; Beginning and Intermediate Algebra

Tutoring Experience

Certified Advanced Tutor

Baruch College Student Academic Consulting Center, 2013–2019

Provided one-on-one and small-group tutoring in physics and calculus; supported exam preparation and long-term skill development.

Varsity Tutors, 2015–2017

Professional Memberships

American Physical Society (APS)

APS Division of Nuclear Physics

Biomed Society, Baruch College (former Vice President)

Work Experience

Research Assistant

Syracuse University, 2021–Present

Supported experimental operations, data analysis, and computational infrastructure for large-scale physics experiments.

Administrative Assistant (Part-Time)

Baruch College Student Academic Consulting Center

Supported academic services through record keeping, database maintenance, and student-facing operations.

Awards

Student Academic Consulting Center, Front Desk Staff of the Year, 2017

CUNY Math Challenge Finalist, 2016

Talks

Hall A Status Update, APS April Meeting, 2024

Hall A Møller Polarimetry Results for SBS G_E^n and G_E^n -RP, APS April Meeting, 2024

Møller Polarimetry Updates for G_E^n , Thomas Jefferson Lab Hall A Annual Summer Meeting, 2023

Preliminary Hall A Møller Polarimetry Results for SBS G_E^n , APS Division of Nuclear Physics, 2023

Using Tracking to Study the Levchuk Effect and Radiative Corrections, APS April Meeting, 2023

Separation of Lyman Alpha Emitting galaxies from contaminants in the HETDEX survey using machine learning, NYU/AMNH/CUNY AstroFest, 2018

Separation of Lyman Alpha Emitting galaxies from contaminants in the HETDEX survey using machine learning, Astronomical Society of New York, 50th Anniversary Meeting, 2018

Publications & Major Collaboration Projects

Super Bigbite Spectrometer (SBS) Collaboration, Jefferson Lab

Measurement of the Neutron Electromagnetic Form Factor Ratio G_E^n/G_M^n at High Q^2 .

Contribution: Experimental operations during data collection, physics data analysis, and beam polarimetry.

Measurement of Kaon Structure Function through Tagged Deep Inelastic Scattering (TDIS).

Contribution: Experimental operations support, data analysis, and beam polarization measurements.

Measurement of the Ratio G_E^n/G_M^n by the Double-Polarized $^2\text{H}(e, e'n)$ Reaction.

Contribution: Detector operation during production running, experimental data analysis, and beam polarimetry.

Polarization Transfer in Wide-Angle Charged Pion Photoproduction.

Contribution: Experimental running support, data analysis, and beam polarization studies.

Hall C Collaboration, Jefferson Lab

Detailed Studies of the Nuclear Dependence of F_2 in Light Nuclei.

Contribution: Experimental operations and data acquisition support during production running.

Inclusive Scattering from Nuclei at $x > 1$ in the Quasielastic and Deeply Inelastic Regimes.

Contribution: Detector and data-taking support during experimental operations.

Deuteron Electro-Disintegration at Very High Missing Momenta.

Contribution: Experimental operations support during data collection.

Neutral Particle Spectrometer (NPS) Collaboration, Jefferson Lab

Exclusive Deeply Virtual Compton and Neutral Pion Cross-Section Measurements in Hall C.

Contribution: Experimental operations support during data-taking periods.

Measurement of Semi-Inclusive π^0 Production as Validation of Factorization.

Contribution: Experimental operations support and detector monitoring.

Deeply Virtual Compton Scattering off the Neutron with the Neutral Particle Spectrometer in Hall C.

Contribution: Experimental operations support during data collection.

Measurements of the Ratio $R = \sigma_L/\sigma_T$, p/d Ratios, P_t Dependence, and Azimuthal Asymmetries in Semi-Inclusive DIS π^0 Production from Proton and Deuteron Targets using the NPS in Hall C.

Contribution: Experimental operations support during production running.

The manuscripts corresponding to these projects are in preparation within the respective collaborations.